Unit 2 BEEE (Part 2)

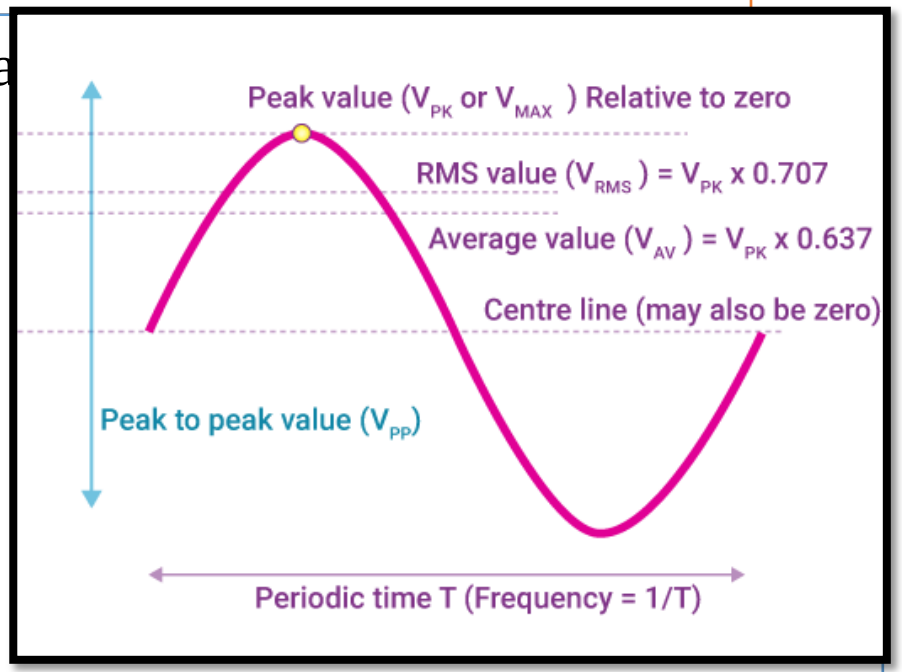
CONTENTS

1. Generation of alternating voltage
2. Basic definitions- average and RMS values, phasor and phase difference , sums on phasors
3. Single phase ac series and parallel consisting of – R,L,C,RL,RC, RLC ckts.
4. Definitions-real,reactive and apparent power
5. Admittance Y
6. Series and parallel resonance (only theory)
7. Generation of 3 phase voltages, voltage & current relationships in Star and Delta connections.

Nirmala K

Average and RMS values

- Peak /Maximum value (Vp or Vm) -The maximum value rea an alternating quantity in one cycle is known as the peak value.



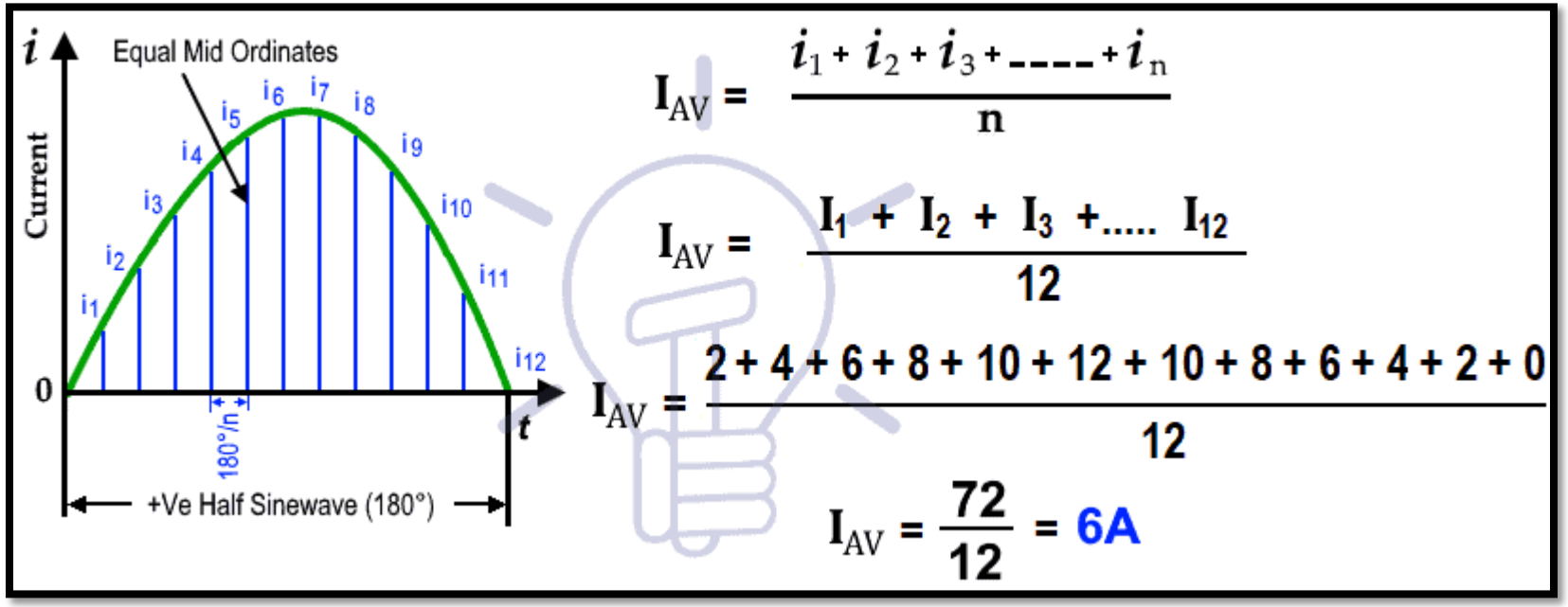
- RMS value of AC component-

RMS or root mean square current/voltage of the alternating current/voltage represents the d.c. current/voltage that dissipates the same amount of power as the average power dissipated by the alternating current/voltage. For sinusoidal oscillations, the RMS value equals peak value divided by the square root of 2.

$$\Rightarrow Irms = \sqrt{\frac{(i1)^2}{n} + \frac{(i2)^2}{n} + \cdots \frac{(in)^2}{n}} = \frac{Im}{\sqrt{2}}$$

$$V_{rms} = \frac{1}{\sqrt{2}} * V_p = 0.7071 * V_p$$

- AVERAGE Value- The average of all the instantaneous values of an alternating voltage and currents over one complete cycle gives the average Value.



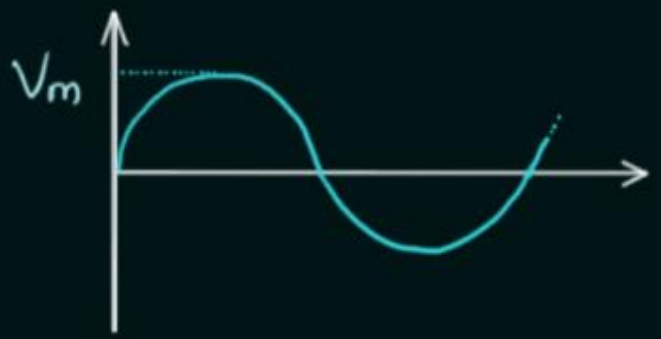
$$I_{AV} = \frac{i_1 + i_2 + i_3 + \text{----} + i_n}{n}$$

$$I_{AV} = \frac{I_1 + I_2 + I_3 + \ldots.. I_{12}}{12}$$

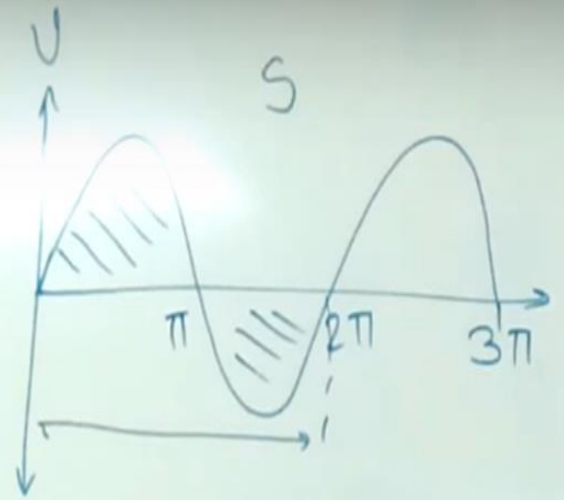
$$I_{AV} = \frac{2+4+6+8+10+12+10+8+6+4+2+0}{12}$$

$$I_{AV} = \frac{72}{12} = 6A$$

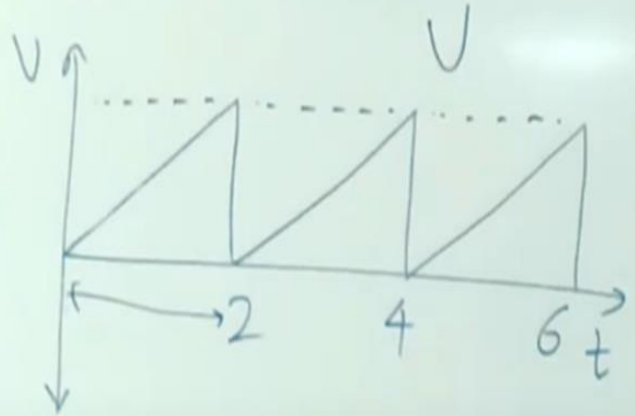
RMS & Average Values of Standard Waveforms

	V_{rms}	V_{avg}
	$\frac{V_m}{\sqrt{2}}$	0 {F.W.} $\frac{2V_m}{\pi}$ {H.W.}
	$\frac{V_m}{\sqrt{2}}$	$\frac{2V_m}{\pi}$
	$\frac{V_m}{2}$	$\frac{V_m}{\pi}$
	V_m	0 {F.W.} V_m {H.W.}
	$\frac{V_m}{\sqrt{2}}$	$\frac{V_m}{2}$
	$\frac{V_m}{\sqrt{3}}$	$\frac{V_m}{2}$
	$\frac{V_m}{\sqrt{3}}$	$\frac{V_m}{2}$

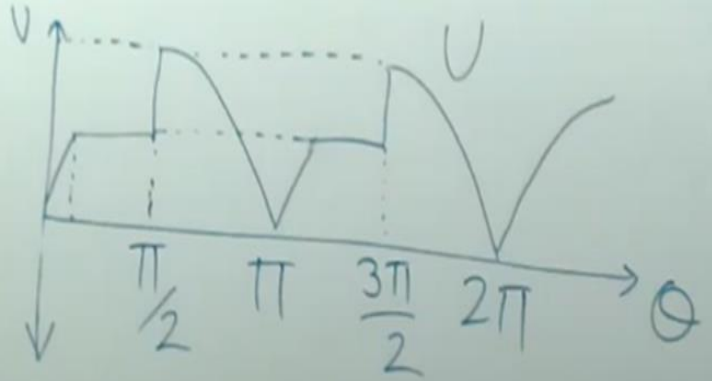
Activate Windows
Go to Settings to activate Windows.

① 

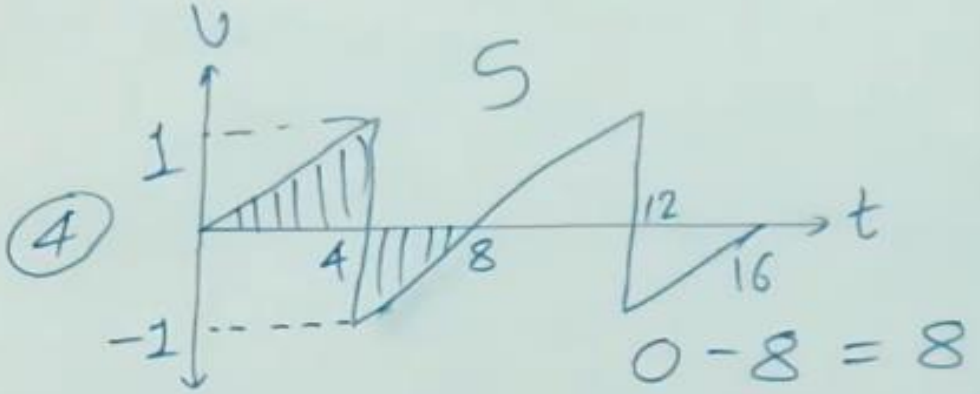
duration of cycle $= 2\pi$

② 

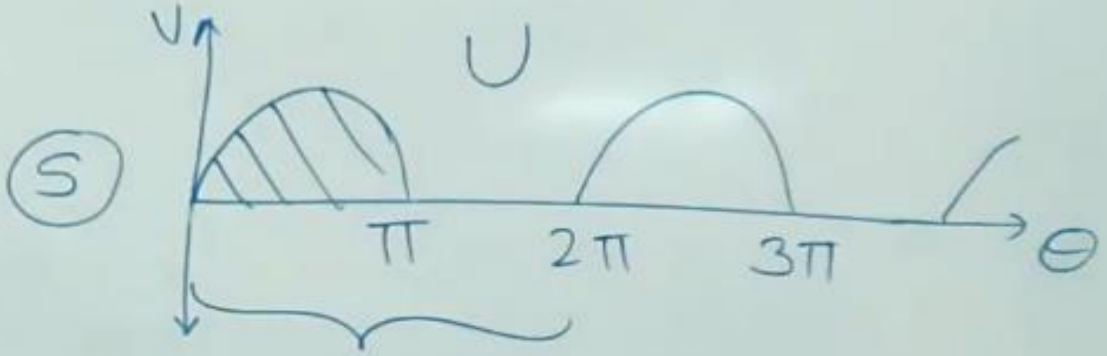
duration of cycle $= 0 - 2 = 2$

③ 

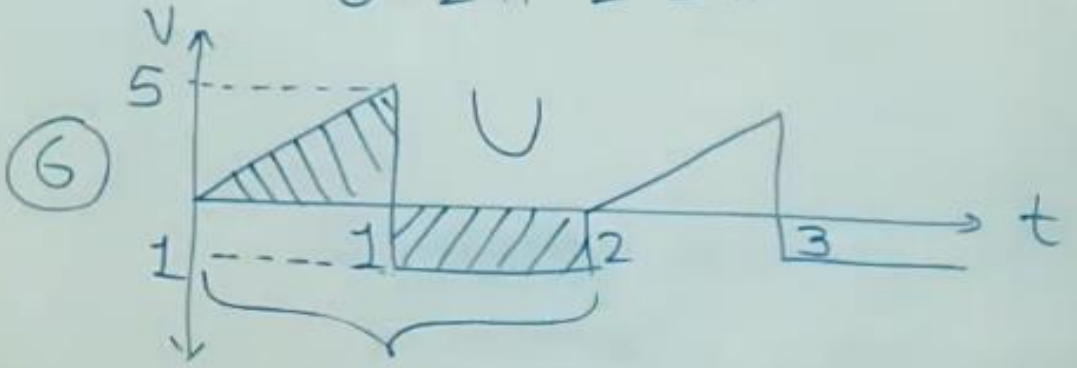
Duration of cycle $= 0 - \pi = \pi$

④ 

$0 - 8 = 8$

⑤ 

$0 - 2\pi = 2\pi$

⑥ 

$0 - 2 = 2$

Activate Windows

Average value of given waveform

Average value and RMS value

Symmetrical Waveform

- Avg value = $\frac{\text{Area of half cycle}}{\text{Base length of half cycle}}$
- RMS value = $\frac{\sqrt{\text{Area of half cycle of sq wave}}}{\text{Base length of half cycle}}$

Asymmetrical Waveform

- Average value = $\frac{\text{Area of full cycle}}{\text{Base length of full cycle}}$
- RMS value: $\frac{\sqrt{\text{Area of full cycle of square wave}}}{\text{Base length of full cycle}}$

1) Identify whether given waveform is symmet

2) Determine duration of a cycle

3) Write equation of curve for each interval in a cycle

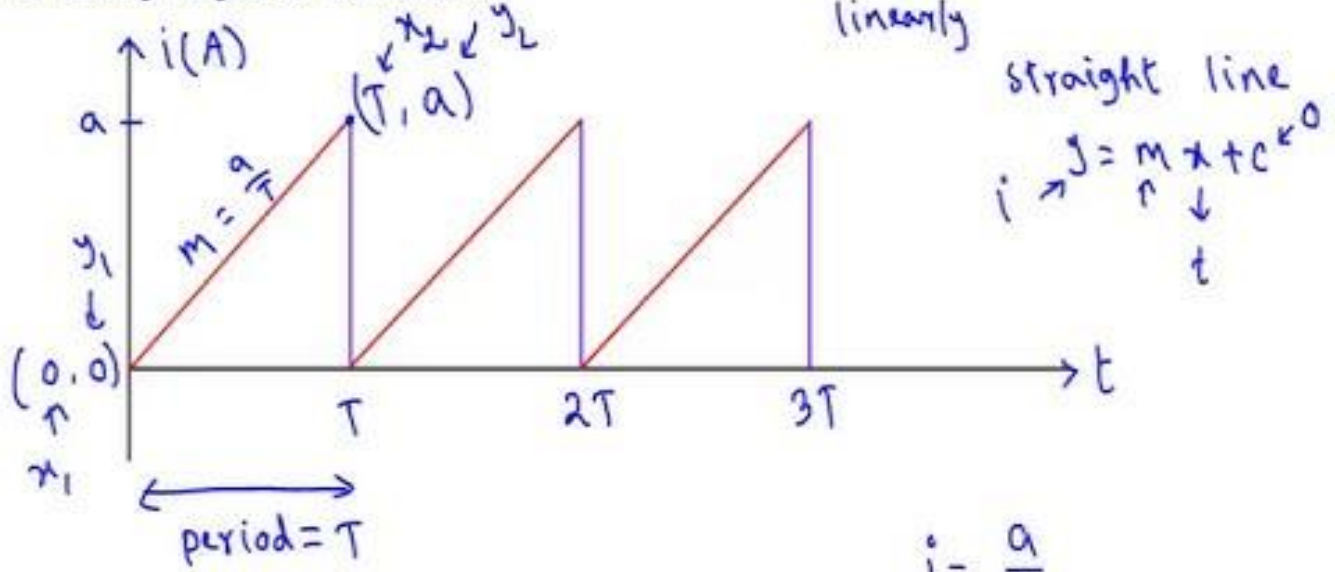
4) For Symmetry $Avg = \frac{\int_{\text{Half cycle}} \text{Equation of curve}}{\text{Duration of half cycle}}$

For Unsymmetric $Avg = \frac{\int_{\text{Full cycle}} \text{Equation of curve}}{\text{Duration of full cycle}}$

Activate Windows
Go to Settings to activate Windows.

RMS Value : Example 2

Determine the rms value of the following triangular current waveform.



Sawtooth waveform linearly

straight line

$y = mx + c$ ($c = 0$)

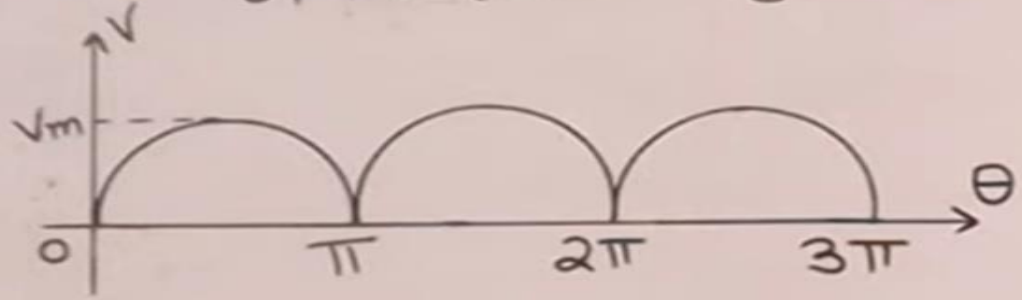
$i \rightarrow y$, $t \rightarrow x$

$i = \frac{a}{T} t$

$$\text{Slope, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$$

$$= \frac{a - 0}{T - 0} = \frac{a}{T}$$

=> find Average value and RMS value of the following waveform.



$V = V_m \sin\theta$ for 0 to π

$$V_{avg} = \frac{\int_0^{\pi} V \, d\theta}{\pi}$$

$$= \frac{\int_0^{\pi} V_m \sin\theta \, d\theta}{\pi}$$

$$= V_m \frac{\int_0^{\pi} \sin\theta \, d\theta}{\pi}$$

$$= 0.636 V_m$$

$$V_{rms} = \sqrt{\frac{\int_0^{\pi} V^2 \, d\theta}{\pi}}$$

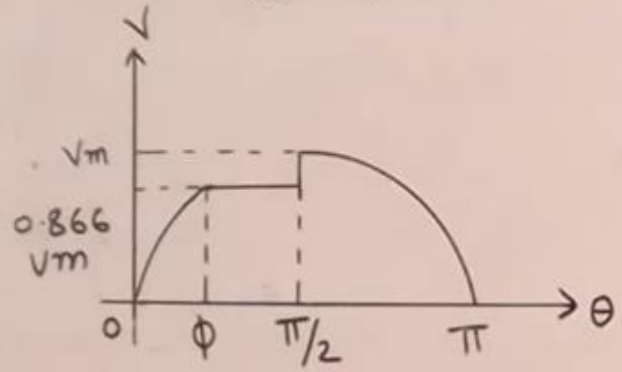
$$= \sqrt{\frac{\int_0^{\pi} V_m^2 \sin^2\theta \, d\theta}{\pi}}$$

$$= V_m \sqrt{\frac{\int_0^{\pi} \sin^2\theta \, d\theta}{\pi}}$$

$$= 0.707 V_m$$

PlanetOjas

⇒ find RMS value of given waveform



$V = V_m \sin\theta$

When $\theta = \phi$ $V = 0.866\, V_m$

$0.866 V_m = V_m \sin\phi$

$\phi = \sin^{-1} 0.866$

$= 60$

$\phi = \pi/3$

$V = V_m \sin\theta$ for 0 to $\pi/3$

$V = 0.866\, V_m$ for $\pi/3$ to $\pi/2$

$V = V_m \sin\theta$ for $\pi/2$ to π

$$V_{rms} = \sqrt{\frac{\int_0^{\pi} v^2 d\theta}{\pi}}$$

$$= \sqrt{\frac{\int_0^{\pi/3} V_m^2 \sin^2\theta d\theta + \int_{\pi/3}^{\pi/2} (0.866\, V_m)^2 d\theta + \int_{\pi/2}^{\pi} V_m^2 \sin^2\theta d\theta}{\pi}}$$

$$= V_m \sqrt{\frac{\int_0^{\pi/3} \sin^2\theta d\theta + \int_{\pi/3}^{\pi/2} 0.866^2 d\theta + \int_{\pi/2}^{\pi} \sin^2\theta d\theta}{\pi}}$$

$$= 0.687\, V_m$$

Average value and RMS value

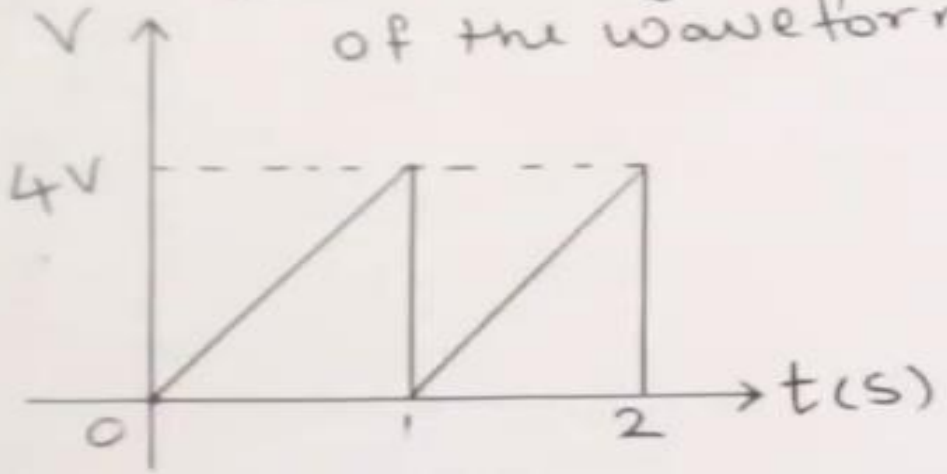
Symmetrical Waveform

- Avg value = $\frac{\text{Area of half cycle}}{\text{Base length of half cycle}}$
- RMS value = $\sqrt{\frac{\text{Area of half cycle of sq wave}}{\text{Base length of half cycle}}}$

Asymmetrical Waveform

- Average value = $\frac{\text{Area of full cycle}}{\text{Base length of full cycle}}$
- RMS value = $\sqrt{\frac{\text{Area of full cycle of square wave}}{\text{Base length of full cycle}}}$

=> find average value of the waveform



$$V_{avg} = \frac{\int_0^1 V\,dt}{}$$

$$= \int_0^1 4t\,dt$$

$$= 2V$$

Writing equation for 0 to 1 x_2 y_2

1, 4

0, 0

x_1 y_1

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y}{x_1 - x}$$

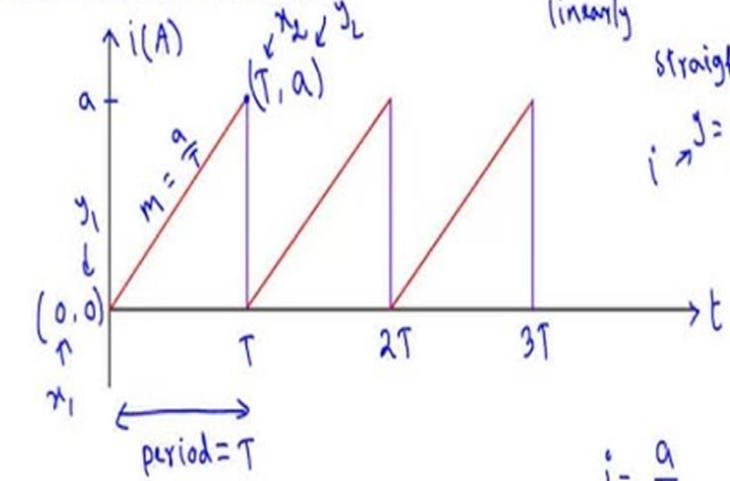
$$\frac{4}{1} = \frac{V}{t}$$

$$V = 4t \quad \text{for } 0 \text{ to } 1$$

RMS Value : Example 2

Determine the rms value of the following triangular current waveform

Sawtooth wavetooth linearly

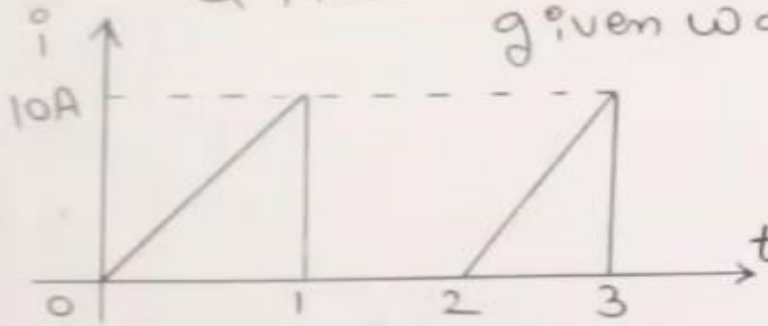


$i = \frac{a}{T}$

$$\text{Slope, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$$

$$= \frac{a - 0}{T - 0} = \frac{a}{T}$$

Q find Rms value of the given waveform



$$V_{rms} = \sqrt{\frac{\int_0^2 i^2\,dt}{2}}$$

$$= \sqrt{\frac{\int_0^1 10^2 t^2\,dt + \int_1^2 0\,dt}{2}}$$

$$= 4.082$$

Writing equation for 0 to 1

(0, 0) = (x_1, y_1), (1, 10) = (x_2, y_2)

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y}{x_1 - x}$$

$$\frac{10}{1} = \frac{i}{t}$$

$$i = 10t \quad \text{for } 0 \text{ to } 1$$

$$i = 0 \quad \text{for } 1 \text{ to } 2$$

2.2.1 Geometric Waveforms

UEx. 2.2.1 (MU - Q. 1(a), Dec. 18, 4 Marks) Find the RMS value of all waveform shown in Fig. Ex. 2.2.1.

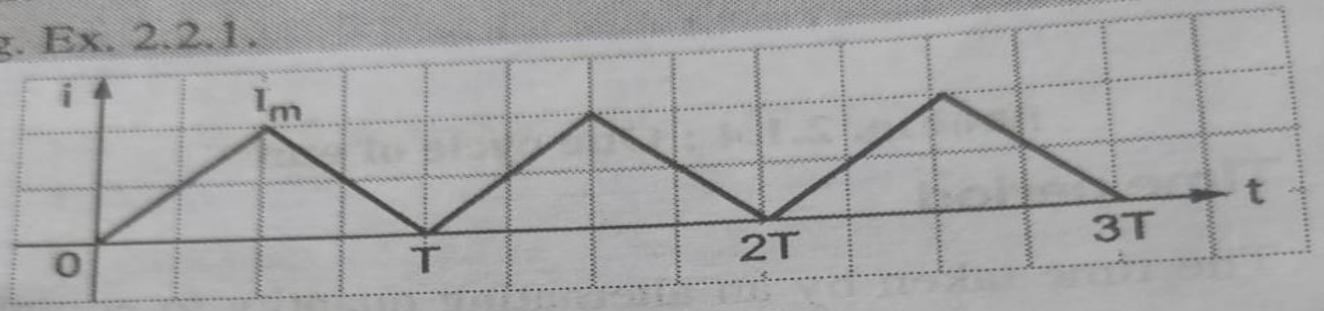


Fig. Ex. 2.2.1

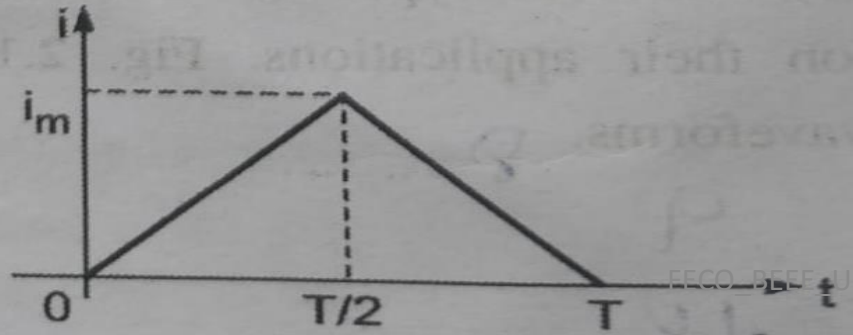
☑ **Stepwise Soln. :**

(i) for $0 \le t \le T/2$, $i = \frac{I_m}{T/2} t = \frac{2 I_m}{T} t$ $\quad \therefore \quad i^2 = \frac{4 I_m^2}{T^2} t^2$

(ii) for $T/2 \le t \le T$, $i = \frac{I_m}{T/2} t = -\frac{2 I_m}{T} t$ $\quad \therefore \quad i^2 = \frac{4 I_m^2}{T^2} t^2$

(∵ square of negative quantity is positive)

Fig. Ex. 2.2.1(a) is one cycle of the given i : t waveform



$$I_{RMS} = \sqrt{\frac{\int_0^T i^2 dt}{T}} = \sqrt{\frac{\int_0^{T/2} i^2 dt + \int_{T/2}^{T} i^2 dt}{T}}$$

$$= \sqrt{\frac{2 \times \frac{4 I_m^2}{T^2} \int_0^{T/2} t^2 dt}{T}}$$

$$= \sqrt{\frac{8 I_m^2}{T^3} \left[\frac{t^3}{3}\right]_0^{T/2}} = \sqrt{\frac{8 I_m^2}{T^3} \times \frac{T^3}{8 \times 3}} = \frac{I_m}{\sqrt{3}}$$

$$= 0.577 I_m \quad \text{...(Ar}$$

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UEx. 2.2.2 (MU - Q. 4(b), May 16, 5 Marks)

Find the average and rms value of the waveform shown in Fig. Ex. 2.2.2.

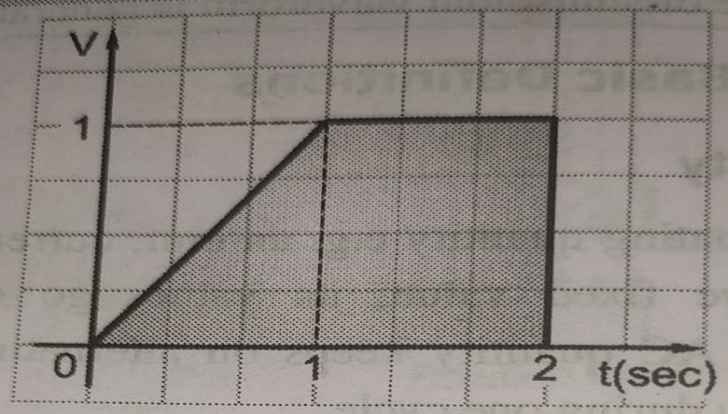


Fig. Ex. 2.2.2

☑ **Stepwise Soln. :**

(I) (1) $T = 2$ sec, (2) for $0 \le t \le 1$ sec, $v = mt = t$

(3) for $1 \le t \le 2$ sec, $v = 1$

(II) $$V_{av} = \frac{\int_0^T v\,dt}{T} = \frac{1}{T}\left(\int_0^1 v\,dt + \int_1^2 v\,dt\right)$$

$$= \frac{1}{2}\left(\int_0^1 t\,dt + \int_1^2 1\,dt\right) = \frac{1}{2}\left(\left[\frac{t^2}{2}\right]_0^1 + [t]_1^2\right)$$

$$= \frac{1}{2}\left(\left[\frac{1}{2} - 0\right] + [2-1]\right) = \frac{1}{2}\left(\frac{1}{2} + 1\right)$$

$$= \frac{1}{2}\left(\frac{3}{2}\right) = \mathbf{0.75\ V}$$

(iii) $$V_{RMS} = \sqrt{\frac{\int_0^T v^2\,dt}{T}} = \sqrt{\frac{1}{T}\left(\int_0^1 v^2\,dt + \int_1^2 v^2\,dt\right)}$$

$$= \sqrt{\frac{1}{t}\left(\int_0^1 t^2\,dt + \int_1^2 (1)^2\,dt\right)}$$

$$= \sqrt{\frac{1}{2}\left(\left[\frac{t^3}{3}\right]_0^1 + [t]_1^2\right)}$$

$$= \sqrt{\frac{1}{2}\left(\left[\frac{1}{3} - 0\right] + [2-1]\right)}$$

$$= \sqrt{\frac{1}{2}\left[\frac{1}{3} + 1\right]} = \sqrt{\frac{1}{2}\left[\frac{4}{3}\right]} = \mathbf{0.816\ V}$$

Ex. 2.2.4 (MU - Q. 1(C), May [illegible])

Determine the rms value of voltage waveform shown in Fig. Ex. 2.2.4.

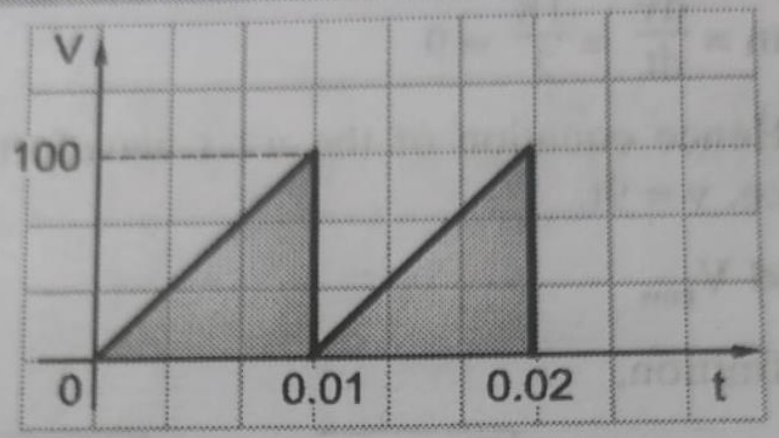


Fig. Ex. 2.2.4

☑ **Stepwise Soln. :** Fig. Ex. 2.2.4(a) is only 1 cycle.

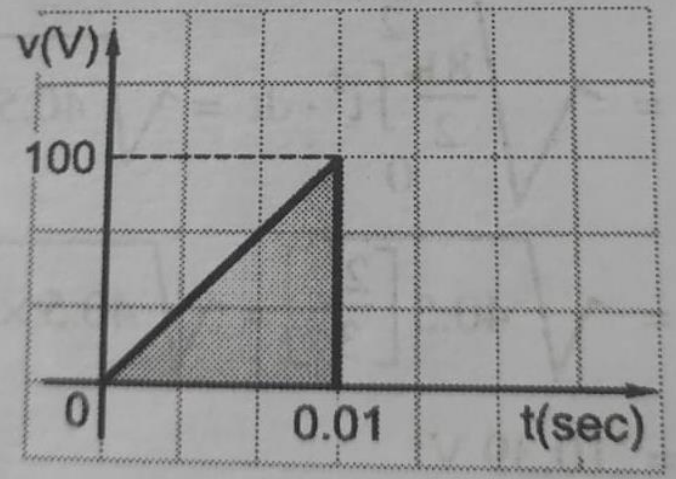


Fig. Ex. 2.2.4(a)

(I) (1) $T = 0.01$ sec.

(2) for $0 \le t \le 0.01$ sec., $v = \frac{100}{0.01} t = 10000\, t$

(II) $$V_{RMS} = \sqrt{\frac{\int_0^T v^2\, dt}{T}} = \sqrt{\frac{\int_0^{0.01} (10000\, t)^2\, dt}{0.01}}$$

$$= \sqrt{\frac{(10000)^2}{0.01} \int_0^{0.01} t^2\, dt} = \sqrt{1 \times 10^{10} \left[\frac{t^3}{3}\right]_0^{0.01}}$$

$$= \sqrt{1 \times 10^4 \left[\frac{(0.01)^3}{3}\right]} = 57.735 \text{ V}$$

UEx. 2.2.3 **(MU - Q. 4(b), May 15, 5 Marks)**

Find the rms value for the waveform shown in Fig. Ex. 2.2.3.

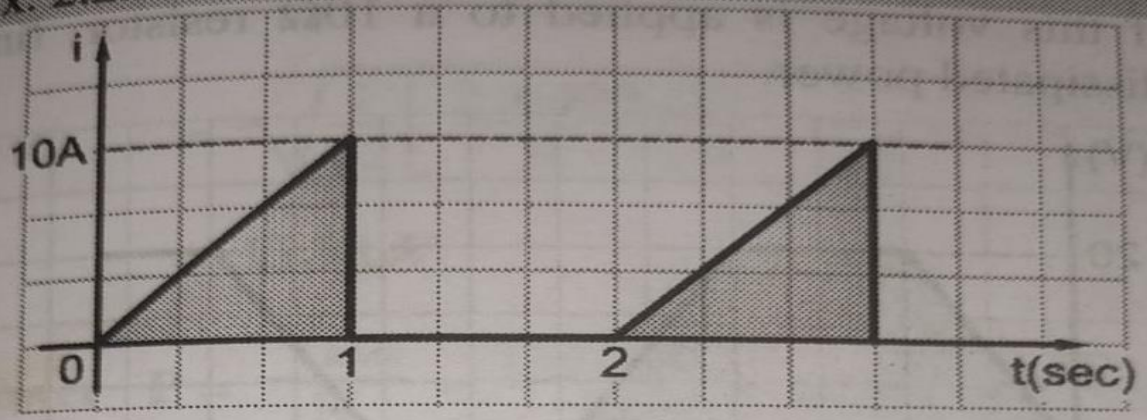


Fig. Ex. 2.2.3

☑ **Stepwise Soln. :**

Fig. Ex. 2.2.3(a) is only 1 cycle.

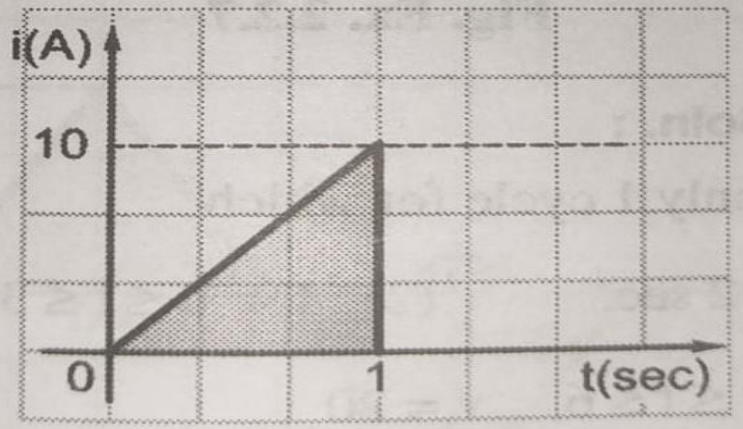


Fig. Ex. 2.2.3(a)

(I) (1) $T = 2$ sec.

(2) for $0 \le t$, $i = 10\,t$

(3) for $1 \le t \le 2$, $i = 0$.

(II) $$I_{RMS} = \sqrt{\frac{\int_0^2 i\,dt}{2}} = \sqrt{\frac{\int_0^1 (10\,t)^2\,dt}{2}}$$

$$= \sqrt{\frac{1}{2}\int_0^1 100\,t^2\,dt} = \sqrt{\frac{100}{2}\left[\frac{t^3}{3}\right]_0^1}$$

$$= \sqrt{\frac{50}{3}} = 4.082 \text{ A}$$

...Ans.

UEx. 2.2.5 **MU - Q.3(a)(i), May 2022, 10 Marks**

Find Vrms of the wave.

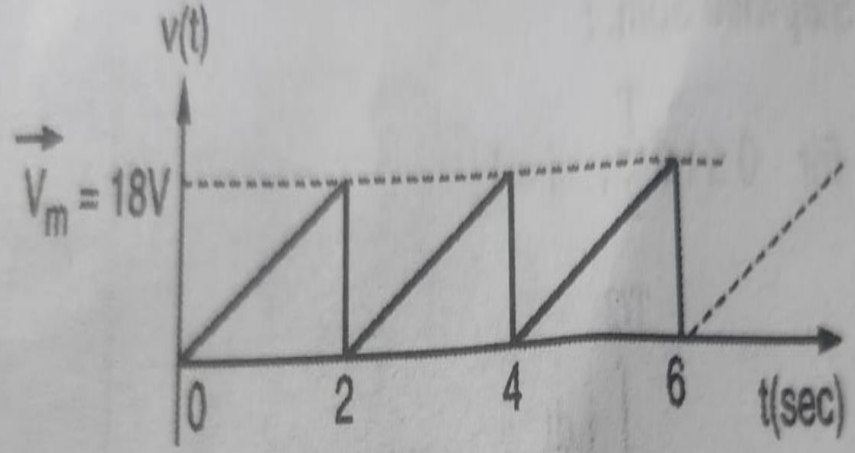


Fig. Ex. 2.2.5

☑ **Stepwise Soln. :**

Given : Triangular voltage waveform with T = 2 sec

Required : V_{rms}

(I) To find the equation of v : t waveform

(a) For one cycle of v : t,

dv = 18 V, dt = 2 Sec ∴ slope

$$m = \frac{dv}{dt} = \frac{18}{2} = 9$$

(b) Hence equation of the v : t waveform is v = mt

i.e. v = 9t.

(II) To find V_{rms}

By definition,

$$V_{RMS} = \sqrt{\int_0^T \frac{V^2 \cdot dt}{T}} = \sqrt{\int_0^2 \frac{(9t)^2}{2} dt}$$

$$V_{RMS} = \sqrt{\frac{81}{2}\int_0^2 t^2 \cdot dt} = \sqrt{40.5\left[\frac{t^3}{3}\right]_0^2}$$

$$= \sqrt{40.5\left[\frac{2^3}{3}\right]} = \sqrt{40.5 \times \frac{8}{3}}$$

$$V_{RMS} = 10.39 \text{ V}$$

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Ex. 2.2.6 : Find the average and RMS values of the current waveform cycle shown in Fig. Ex. 2.2.6.

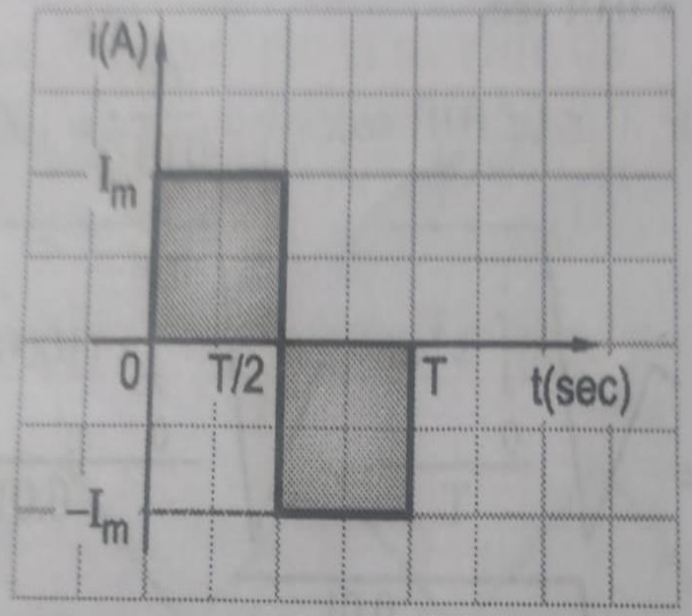


Fig. Ex. 2.2.6

Note

$\therefore \quad +\frac{1}{2}\text{cycle} = -\frac{1}{2}\text{cycle}$

$\therefore$ Consider only $\frac{1}{2}$ cycle for average and RMS values.

☑ **Stepwise Soln. :**

(I) for $0 \le t \le \frac{T}{2}$, $i = I_m$

(II) $$I_{av} = \frac{\int_0^{T/2} i\,dt}{T/2} = \frac{2}{T}\int_0^{T/2} (I_m)\,dt = \frac{2I_m}{T} \times [t]_0^{T/2}$$

$$= \frac{2I_m}{T}\left[\frac{T}{2}\right] = I_m$$

(III) $$I_{RMS} = \sqrt{\frac{\int_0^{T/2} i^2\,dt}{T/2}} = \sqrt{\frac{2}{T}\int_0^{T/2} [I_m]^2\,dt}$$

$$= \sqrt{\frac{2I_m^2}{T}\int_0^{T/2} dt} = \sqrt{\frac{2}{T} I_m^2 [t]_0^{T/2}}$$

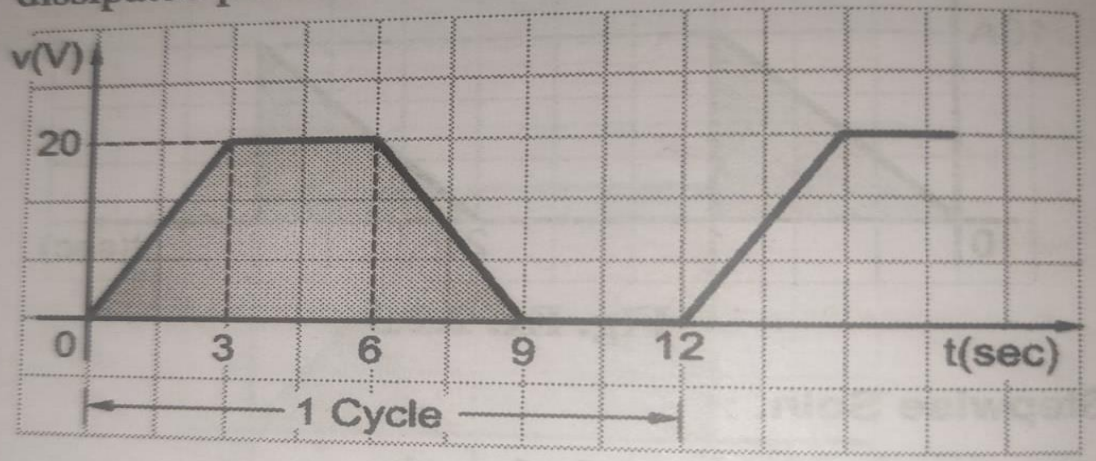
$$= \sqrt{\frac{2}{T} I_m^2 \left[\frac{T}{2}\right]} = I_m$$ ...Ans.

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Ex. 2.2.7 : A voltage wave has the variation shown in the Fig. Ex. 2.2.7.

(i) Find the average and effective values of the voltage.

(ii) If this voltage is applied to a 10Ω resistor, find the dissipated power.



☑ Stepwise Soln. :

(I) Consider only 1 cycle for which

(1) $T = 12$ sec. (2) for $0 \le t \le 3$, $v = \frac{20}{3}t$

(3) for $3 \le t \le 6$, $v = 20$

(4) for $6 \le t \le 9$, $v = -\frac{20}{3}t + 60$

(5) for $9 \le t \le 12$, $v = 0$

No need since we can merge this with (2) above

∴ No need of 'c' = 60

(II) $$V_{av} = \frac{\int_0^T v\,dt}{T} = \frac{1}{T}\int_0^T v\,dt = \frac{1}{12}\left[2\int_0^3 \frac{20}{3}t + \int_3^6 20\,dt\right]$$

$$= \frac{1}{12}\left[2 \times \left[\frac{20t^2}{3 \times 2}\right]_0^3 + [20t]_3^6\right]$$

$$= \frac{1}{12}\left[2 \times \left[\frac{20 \times (9-0)}{6}\right] + [20 \times (6-3)]\right.$$

$= 10\ V$ ...Ans.

(III) $V_{RMS} = \sqrt{\dfrac{\int_0^T v^2\,dt}{T}} = \sqrt{\dfrac{1}{12}\int_0^T v^2\,dt}$

$$= \sqrt{\frac{1}{12}\left[2 \times \int_0^3 \left(\frac{20}{3}\right) t^2\,dt + \int_3^6 (20)^2\,dt\right]}$$

($\because$ for $9 \le t \le 12$ sec. $v = 0$)

$$= \sqrt{\frac{1}{12}\left[\frac{2 \times 400}{9}\left[\frac{t^3}{3}\right]_0^3 + 400\,[t]_3^6\right]}$$

$$= \sqrt{\frac{1}{12}\left[\frac{2 \times 400}{9}\left(\frac{27}{9} - 0\right) + 400\,[6-3]\right]}$$

$= 12.91\ V$

(IV) **Power** $P = \dfrac{V_{RMS}^2}{R} = \dfrac{(12.91)^2}{10} = 16.666\ W$ ...Ans.

Find the average and RMS value of the voltage waveform-

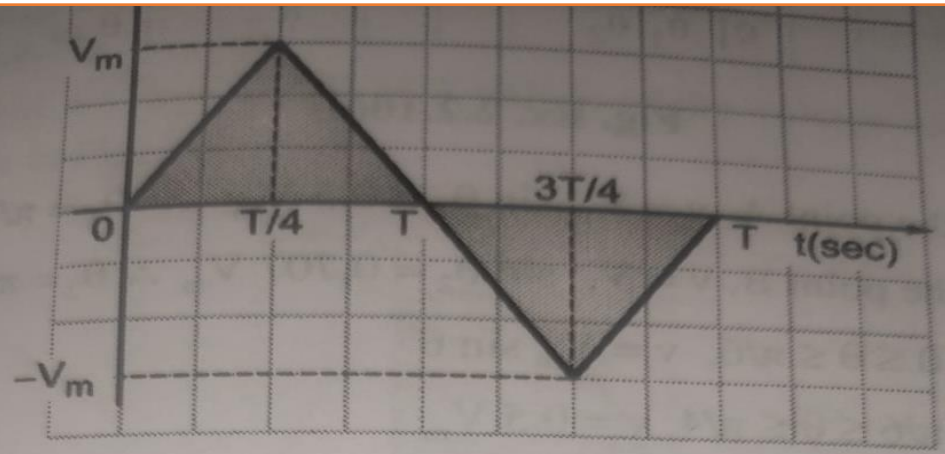


Fig. Ex. 2.2.8

Note

$\therefore \quad +\frac{1}{2} = -\frac{1}{2}$

$\therefore$ Consider only $\frac{1}{2}$ cycle for V_{av} and V_{RMS}.

☑ **Stepwise Soln. :**

(I) (1) Time period = T

(2) for $0 \le t \le \frac{T}{4}$, $V = \frac{V_m}{T/4}\, t = \frac{4\,V_m}{T}\, t$

(3) No need to consider $\frac{T}{4} \le t \le T$ as explained in the solved Ex. 2.2.8.

(II) $$V_{av} = \frac{2\int_0^{T/2} v\,dt}{T/2} = \frac{2}{T} \times 2 \times \int_0^{T/4} \left(\frac{4\,V_m}{T}\, t\right) dt$$

$$= \frac{16\,V_m}{T^2} \int_0^{T/4} t\,dt = \frac{16\,V_m}{T^2} \times \left[\frac{t^2}{2}\right]_0^{T/4}$$

$$= \frac{16\,V_m}{T^2} \times \frac{1}{2}\left[\frac{T^2}{16}\right] = \mathbf{0.5\,V_m} \quad \text{...Ans.}$$

(III) $$V_{RMS} = \sqrt{2\int_0^{T/4} \frac{v\,dt}{T/2}} = \sqrt{\frac{4}{T}\int_0^{T/4} \left(\frac{4\,V_m\,t}{T}\right)^2 dt}$$

$$= \sqrt{\frac{4}{T} \times \frac{16\,V_m^2}{T^2}\left[\frac{t^3}{3}\right]_0^{T/4}}$$

$$= \sqrt{\frac{4}{T} \times \frac{16\,V_m^2}{T^2} \times \frac{T^3}{3 \times 64}}$$

$$= \frac{V_m}{\sqrt{3}} = \mathbf{0.577\,Vm} \quad \text{...Ans}$$

2.2.2 Trigonometric Waveforms

UEx. 2.2.9 (MU - Q. 6(b), May 19, 10 Marks)

Find : (i) average value (ii) rms value

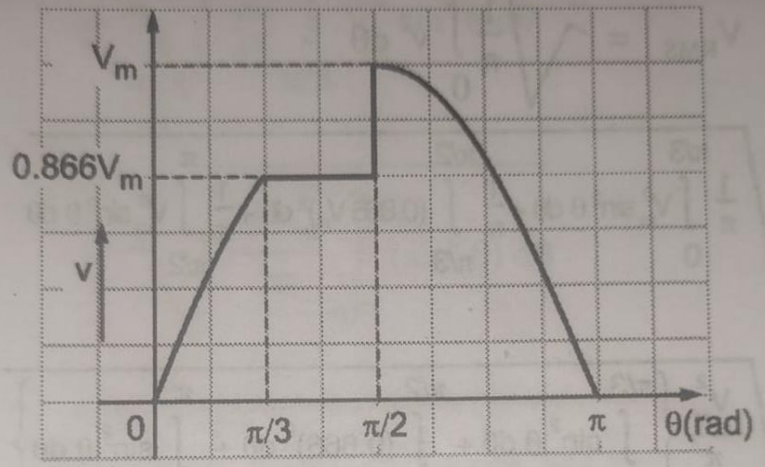
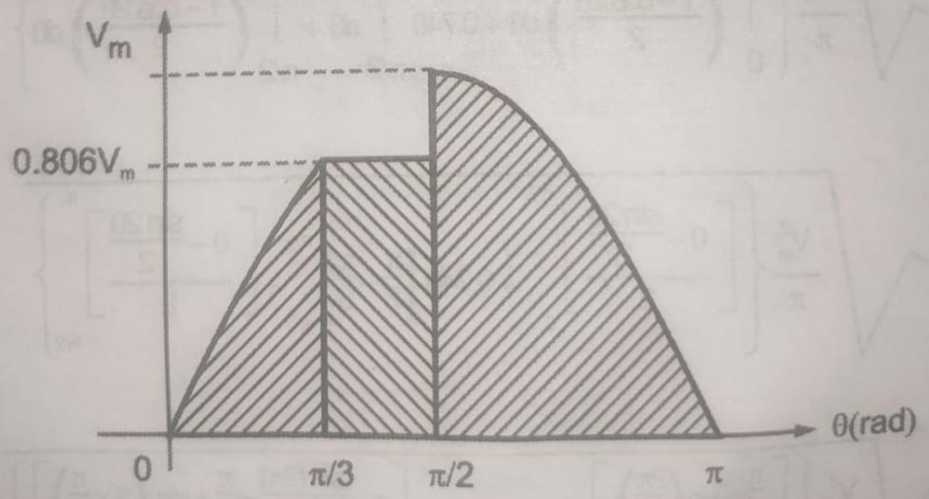


Fig. Ex. 2.2.9

☑ **Stepwise Soln. :**



(i) for $0 \le \theta \le \pi/3$, $v = V_m \sin \theta$,

(ii) for $\pi/3 \le \theta \le \pi/2$, $v = 0.866\ V_m$,

(iii) for $\pi/2 \le \theta \le \pi$, $v = V_m \sin \theta$

(I) $$V_{av} = \frac{\int_0^{\pi} v\, d\theta}{\pi}$$

$$= \frac{\int_0^{\pi/3} V_m \sin\theta\, d\theta + \int_{\pi/3}^{\pi/2} (0.866\, V_m)\, d\theta + \int_{\pi/2}^{\pi} V_m \sin\theta\, d\theta}{\pi}$$

$$= \frac{V_m}{\pi}\left\{ \int_0^{\pi/3} \sin\theta\, d\theta + 0.866 \int_{\pi/3}^{\pi/2} d\theta + \int_{\pi/2}^{\pi} \sin\theta\, d\theta \right\}$$

$$= \frac{V_m}{\pi}\left\{ [-\cos\theta]_0^{\pi/3} + 0.866\, [\theta]_{\pi/3}^{\pi/2} + [-\cos\theta]_{\pi/2}^{\pi} \right\}$$

$$= \frac{V_m}{\pi}\left\{ -\left[\cos\frac{\pi}{3} - \cos 0\right] + 0.866\left[\frac{\pi}{2} - \frac{\pi}{3}\right] - \left[\cos\pi - \cos\frac{\pi}{2}\right] \right\}$$

$$= \frac{V_m}{\pi}\{ -[0.5 - 1] + 0.866\,[0.523] - [-1 - 0] \}$$

$$= \frac{V_m}{\pi}\{1.9529\} = \mathbf{0.621\ V_m}$$

...Ans

(II) $$V_{RMS} = \sqrt{\frac{1}{\pi}\int_0^{\pi} v^2\, d\theta}$$

$$= \sqrt{\frac{1}{\pi}\int_0^{\pi/3} V_m^2 \sin^2\theta\, d\theta + \frac{1}{\pi}\int_{\pi/3}^{\pi/2} (0.866\, V_m)^2\, d\theta + \frac{1}{\pi}\int_{\pi/2}^{\pi} V_m^2 \sin^2\theta\, d\theta}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\int_0^{\pi/3} \sin^2\theta\, d\theta + \int_{\pi/3}^{\pi/2} (0.866)^2\, d\theta + \int_{\pi/2}^{\pi} \sin^2\theta\, d\theta\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\int_0^{\pi/3} \left(\frac{1-\cos 2\theta}{2}\right) d\theta + 0.749\int_{\pi/3}^{\pi/2} d\theta + \int_{\pi/2}^{\pi} \left(\frac{1-\cos 2\theta}{2}\right) d\theta\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\left[\frac{\theta - \frac{\sin 2\theta}{2}}{2}\right]_0^{\pi/3} + 0.749\left[\frac{\pi}{2} - \frac{\pi}{3}\right]_{\pi/3}^{\pi/2} + \left[\frac{\theta - \frac{\sin 2\theta}{2}}{2}\right]_{\pi/2}^{\pi}\right\}}$$

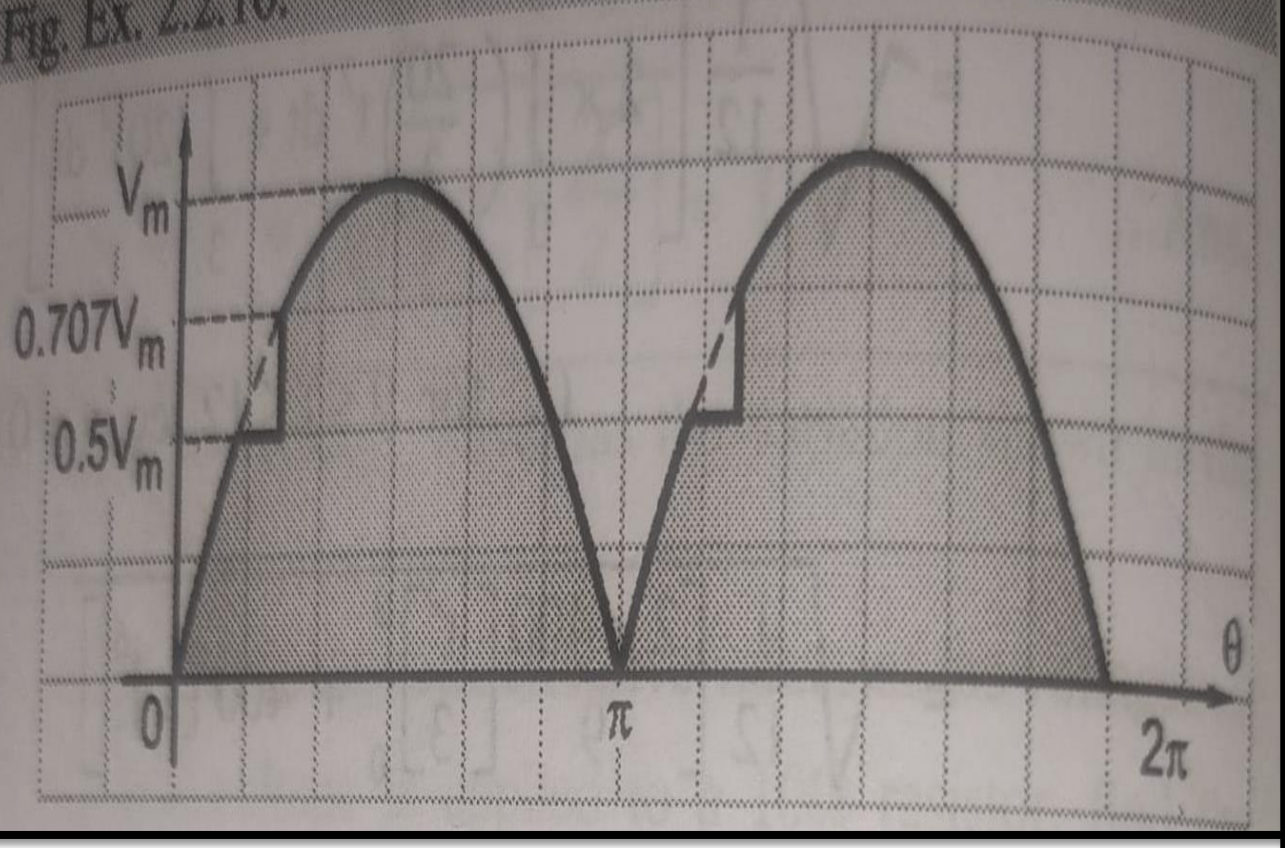
$$= \sqrt{\frac{V_m^2}{\pi}\left\{\left[\frac{\frac{\pi}{3} - \sin\left(\frac{2\pi}{3}\right)}{2}\right] + 0.749\left[\frac{\pi}{6}\right] + \left[\frac{\pi - \frac{\sin(2\pi)}{2} - \frac{\pi}{2} + \sin\left(2\times\frac{\pi}{2}\right)}{2}\right]\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\left[\frac{1.047 - 0.433}{2}\right] + [0.392] + \left[\frac{3.142 - 0 - 1.571 + 0}{2}\right]\right\}}$$

$$= 0.6352\, V_m$$ ...Ans.

UEx. 2.2.10 (MU - Q. 1(b), May 17, 4 Marks)

Find the RMS value of the voltage waveform shown in Fig. Ex. 2.2.10.



☑ **Stepwise Soln. :**

(1) Fig. Ex. 2.2.10(a) is the one cycle of v : θ waveform.

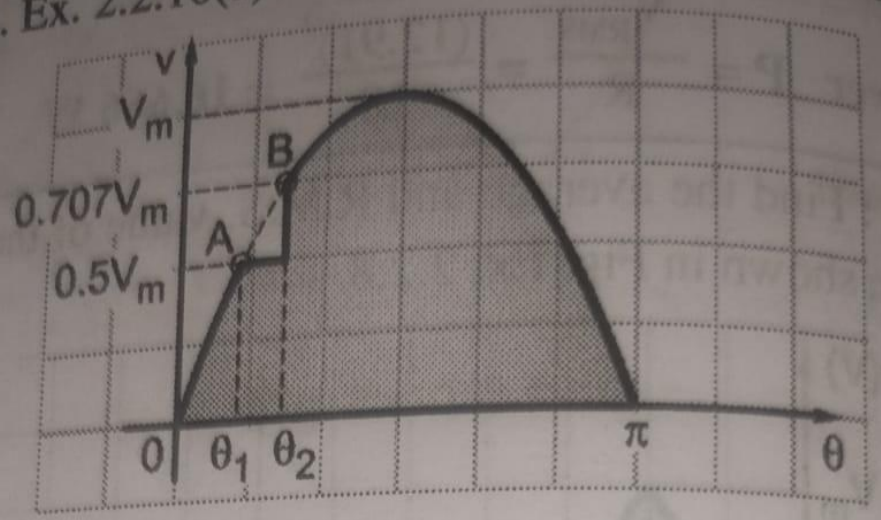


Fig. Ex. 2.2.10(a)

(1) at the point A, $v = V_m \sin \theta_1 = 0.5\ V_m$ $\therefore$ $\theta_1 = \pi/6$ rad

(2) at the point B, $v = V_m \sin \theta_2 = 0.707\ V_m$ $\therefore$ $\theta_2 = \pi/4$ rad

(3) for $0 \le \theta \le \pi/6$, $v = V_m \sin \theta$

(4) for $\pi/6 \le \theta \le \pi/4$, $v = 0.5\ V_m$

(5) for $\pi/4 \le \theta \le \pi$, $v = V_m \sin \theta$

II) $$V_{RMS} = \sqrt{\frac{\int_0^{\pi} v^2 \, d\theta}{\pi}}$$

$$= \sqrt{\frac{1}{\pi}\left\{\int_0^{\pi/6} v^2 \, d\theta + \int_{\pi/6}^{\pi/4} v^2 \, d\theta + \int_{\pi/4}^{\pi} v^2 \, d\theta\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\int_0^{\pi/6} \sin^2\theta \, d\theta + 0.25\int_{\pi/6}^{\pi/4} d\theta + \int_{\pi/4}^{\pi} \sin^2\theta \, d\theta\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\int_0^{\pi/2} \left(\frac{1-\cos 2\theta}{2}\right) d\theta + 0.25[\theta]_{\pi/6}^{\pi/4} + \int_{\pi/4}^{\pi} \left(\frac{1-\cos 2\theta}{2}\right) d\theta\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\frac{1}{2}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/2} + 0.25\left[\frac{\pi}{4} - \frac{\pi}{6}\right] + \frac{1}{2}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/2}\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\frac{1}{2}\left[\frac{\pi}{2} - \frac{\sin\pi}{2}\right] + 0.25\,[0.7853 - 0.5235] + \frac{1}{2}\left[\frac{\pi}{2} - \frac{\sin\pi}{2}\right]\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\frac{1}{2}(1.5707 - 0) + 0.25\,(0.2618) + \frac{1}{2}(1.5707 - 0)\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}[1.63615]} = \sqrt{0.5708\,V_m^2} = 0.7555\,V_m \quad \ldots A$$

Ex. 2.2.11

Find the average and R.M.S values of the voltage waveform shown in Fig. Ex. 2.2.11.

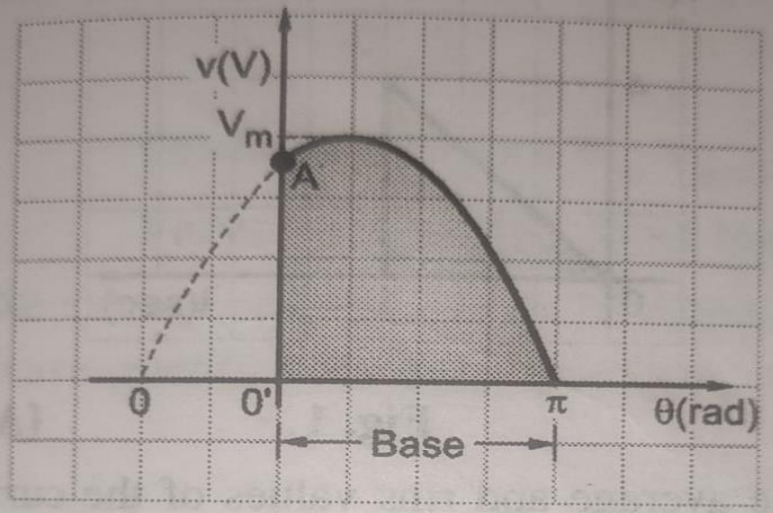


Fig. Ex. 2.2.11

☑ **Stepwise Soln. :**

Note

Actually a sine curve starts from (0, 0) but here it is given from point O′.

(I) (1) At point O′, we have to calculate θ_1, for this we proceed as follows :

(i) at this point 'A', $v = V \sin\theta$

(ii) also at point A , v is given 0.866 V_m.

(iii) $V_m \sin\theta_1 = 0.866\ V_m$

or $\theta_1 = \sin^{-1}(0.866) = 60° = \frac{\pi}{3}$

(2) $\therefore$ θ period = base = $\left(\pi - \frac{\pi}{3}\right) = 2\frac{\pi}{3}$ rad.

(II) $$V_{av} = \frac{\int_{\pi/3}^{\pi} v\, d\theta}{\frac{2\pi}{3}} = \frac{3}{2\pi}\int_{\pi/3}^{\pi} V_m \sin\theta\, d\theta$$

$$= \frac{3V_m}{2\pi}\int_{\pi/3}^{\pi} \sin\theta\, d\theta = \frac{3V_m}{2\pi}\left[-\cos\theta\right]_{\pi/3}^{\pi}$$

$$= \frac{3V_m}{2\pi}\left[-\cos(\pi) + \cos\left(\frac{\pi}{3}\right)\right]$$

$$= 3\frac{V_m}{2\pi}[-(-1) + (0.5)]$$

$$= \frac{3V_m}{2\pi} \times 1.5 = \mathbf{0.716\ V_m}$$ **...Ans.**

$$(III)V_{RMS} = \sqrt{\frac{\int_{\pi/3}^{\pi} v^2 \, d\theta}{\frac{2\pi}{3}}}$$

$$= \sqrt{\frac{3}{2\pi} \int_{\pi/3}^{\pi} V_m^2 \sin^2\theta \, d\theta}$$

$$= \sqrt{\frac{3V_m^2}{2\pi} \int_{\pi/3}^{\pi} (\sin^2\theta) \, d\theta}$$

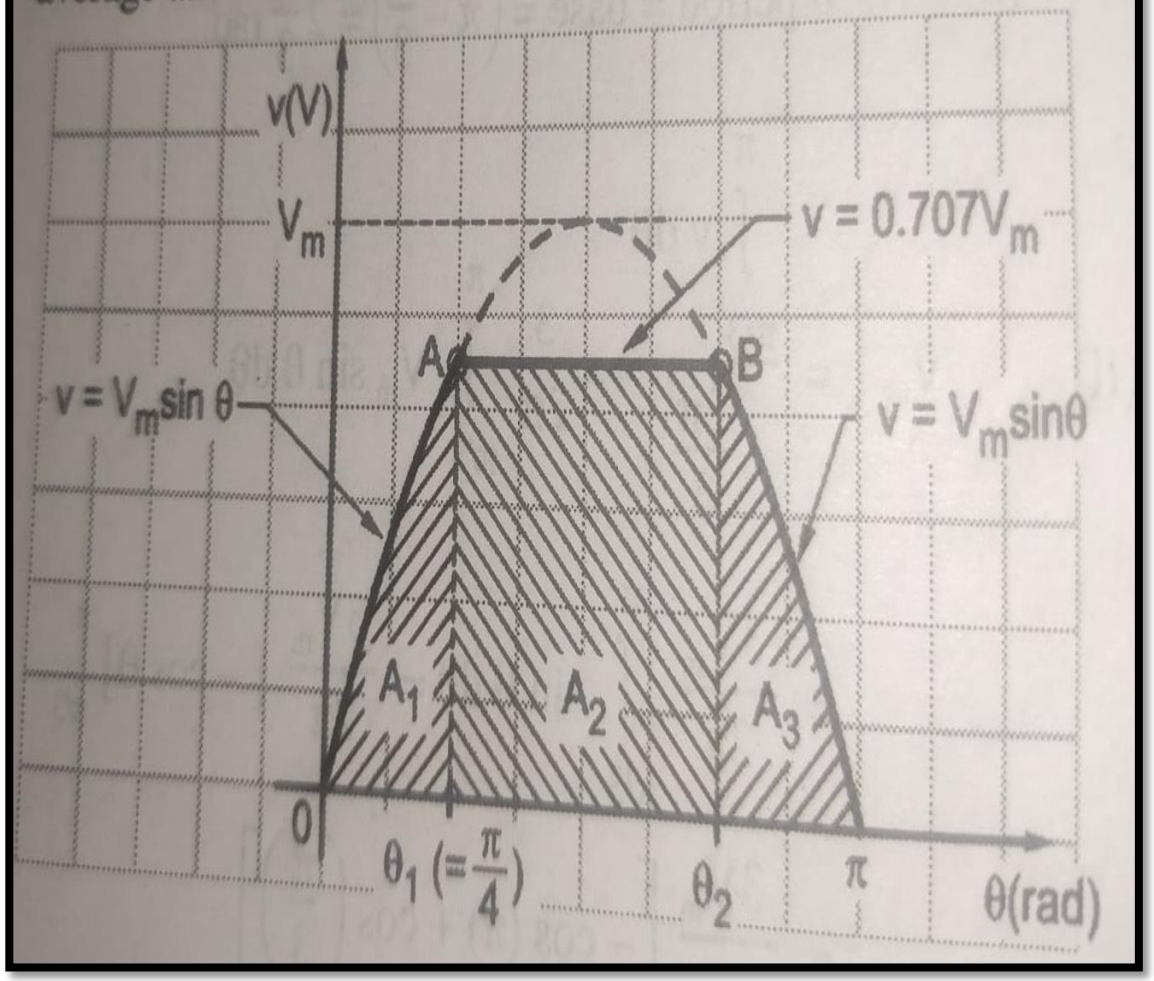
$$= \sqrt{\frac{3V_m^2}{2\pi} \int_{\pi/3}^{\pi} \left[\frac{1-\cos 2\theta}{2}\right] d\theta}$$

$$= \sqrt{\frac{3V_m^2}{2\pi} \left[\theta - \frac{\sin 2\theta}{2}\right]_{\pi/3}^{\pi}}$$

$$= \sqrt{\frac{3V_m^2}{2\pi} \left[\left(\pi - \frac{\pi}{3}\right) - (0 - 0.433)\right]}$$

$$= \sqrt{\frac{3V_m^2}{2\pi} \left[\frac{2\pi}{3} + 0.433\right]} = 0.776\, V_m$$

Ex. 2.2.12 : A full wave rectifier is clipped at 70.7% of its maximum value as shown in the Fig. Ex. 2.2.12, find its average and RMS values.



☑ **Soln. :**

(I) (i) To find value of θ_1 at point A, $v = V_m \sin \theta_1$

$= 0.707\ V_m$ $\quad \therefore\ \theta_1 = \frac{\pi}{4}$

(ii) $\because$ area $A_3 = A_1$ $\quad \therefore$ total area $= (2A_1 \cdot A_2)$

(iii) (1) θ period $= \pi$ rad

(2) for $0 \le \theta \le \frac{\pi}{4}$, $v = V_m \sin \theta$

(3) for $\frac{\pi}{4} \le \theta \le \frac{3\pi}{4}$, $v = 0.707\ V_m$

(II) $$V_{av} = \frac{\int_0^{\pi} v\, d\theta}{\frac{\pi}{4}}$$

$$= \frac{2 \times \int_0^{\pi/4} V_m \sin\theta\, d\theta + \int_{\pi/4}^{3\pi/2} 0.707\, V_m\, d\theta}{\pi}$$

$$= \frac{V_m}{\pi}\left\{2 \times \int_0^{\pi/4} \sin\theta\, d\theta + 0.707 \int_{\pi/4}^{3\pi/4} d\theta\right\}$$

$$= \frac{V_m}{\pi}\left\{2 \times [-\cos\theta]_0^{\pi/4} + 0.707\,[\theta]_{\pi/4}^{3\pi/4}\right\}$$

$$= \frac{V_m}{\pi}\left\{2 \times \left[-\cos\frac{\pi}{4} + \cos 0\right] + 0.707\left[\frac{3\pi}{4} - \frac{\pi}{4}\right]\right\}$$

$$= \frac{V_m}{\pi}\left\{2 \times [-0.707 + 1] + 0.707\left[\frac{\pi}{2}\right]\right\} = \mathbf{0.54\ V_m}$$

(III) $V_{RMS} = \sqrt{\frac{\int_0^{\pi} v^2 \, d\theta}{\pi}}$

$$= \sqrt{\frac{1}{\pi}\left\{2\int_0^{\pi/4} V_m^2 \sin^2\theta \, d\theta + \int_{\pi/4}^{3\pi/4} (0.707\, V_m)^2 \, d\theta\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{2\int_0^{\pi/4} \sin^2\theta \, d\theta + (0.707)^2 \int_{\pi/4}^{3\pi/4} d\theta\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{2\times\int_0^{\pi/4}\left[\frac{1-\cos 2\theta}{2}\right]d\theta + (0.707)^2\,[\theta]_{\pi/4}^{3\pi/4}\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/4} + (0.707)^2\left[\frac{3\pi}{4} - \frac{\pi}{4}\right]\right\}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left\{\left[\frac{\pi}{4} - 0 - \sin\left(2\times\frac{\pi}{4}\right) + \sin 0\right] + (0.707)^2 \times \frac{\pi}{2}\right\}}$$

$$= 0.584\, V_m$$ ...Ans.

Parallel circuits (RL,RC,RLC)

2.3 AC Parallel Circuits

An ac circuit that has number of branches connected in parallel such that the voltage across them is same, is called an **ac parallel circuit**. The current in any one branch depends upon the impedance of that branch. The total line current supplied to the circuit is the phasor sum of branch currents. Parallel circuits are used more often in practice as all lighting and power circuits are constant voltage circuits and the loads/equipments are connected in parallel.

While analysing an ac parallel circuit, two important points must be kept in mind. First, a parallel circuit, in fact, consists of two or more series circuits connected in parallel. Therefore, each branch of the circuit can be analysed separately as a series circuit and then the effect of the separate branches can be combined. Secondly, alternating voltages and currents are phasor quantities. This implies that both magnitudes and phase angles must be taken account while carrying out circuit calculations. There are three methods of solving ac parallel circuits, namely;

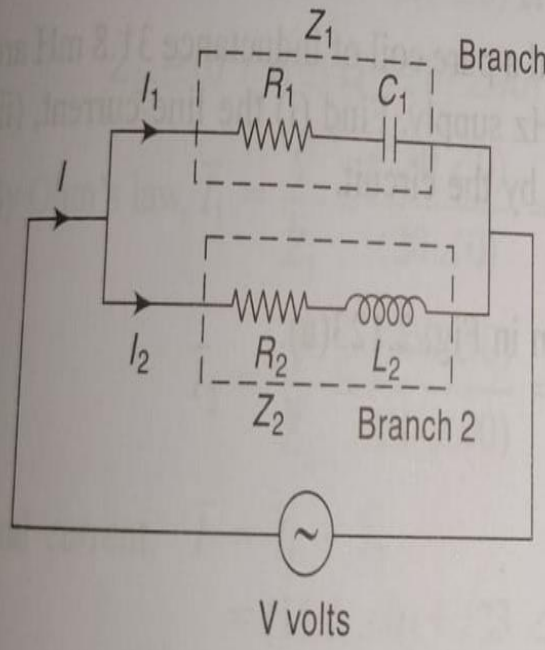
1. By phasor diagram
2. By phasor algebra
3. Admittance method

The use of a particular method will depend upon the conditions of the problem. However, in general, that method should be used which yields quick results.

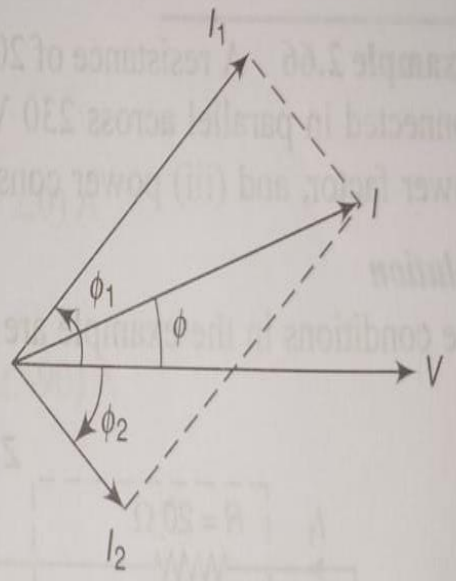
EEC 302 BDEU/NTE Part 2 Nimala

2.3.1 Phasor diagram

In this method, we find the magnitude and phase angle of each branch current. We then draw the phasor diagram taking voltage as reference phasor. The circuit or line current is the phasor sum of the branch currents and can be determined by parallelogram law. Consider a parallel circuit as shown in Fig. 2.122(a), consisting of two branches and connected to an alternating voltage of V volts.



(a) Circuit diagram



(b) Phasor diagram

Fig. 2.122 Analysis of parallel circuit by phasor diagram

Branch 1 $Z_1 = \sqrt{R_1^2 + X_{C1}^2}\ ;\ I_1 = \dfrac{V}{Z_1};\ \phi_1 = \tan^{-1}\dfrac{X_{C1}}{R_1}$

The current I_1 in branch 1 leads the applied voltage V by ϕ_1°.

Branch 2 $Z_2 = \sqrt{R_2^2 + X_{L2}^2}\ ;\ I_2 = \dfrac{V}{Z_2};\ \phi_2 = \tan^{-1}\dfrac{X_{L2}}{R_2}$

The current I_2 in branch 2 lags behind the applied voltage V by ϕ_2°.

The phasor diagram is shown in Fig. 2.122(b). The line current I is the phasor sum of I_1 and I_2.

The phasor diagram method is suitable only when the parallel circuit is simple and contains two branches. However, if the parallel circuit is complex having more than two branches, this method becomes very inconvenient. In such cases, use of phasor algebra is recommended to solve parallel-circuit problems.

2.3.2 Phasor Algebra

In this method, voltages, currents and impedances are expressed in the complex form, i.e., either in the rectangular or polar form. Since complex form includes both magnitude and phase angle, the solution of parallel circuit problems can be obtained mathematically by using the rules of phasor algebra. This eliminates the need of phasor diagram. Referring back to the parallel circuit shown in Fig. 2.122(a), we have

$$\bar{V} = (V - j\,0) = (V \angle 0)$$

$$\bar{Z}_1 = (R_1 - j\,X_{C1}) = (Z_1 \angle -\phi_1)$$

$$\bar{Z}_2 = (R_2 + j\,X_{L2}) = (Z_2 \angle \phi_2)$$

Branch current, $\bar{I}_1 = \dfrac{\bar{V}}{\bar{Z}_1}$

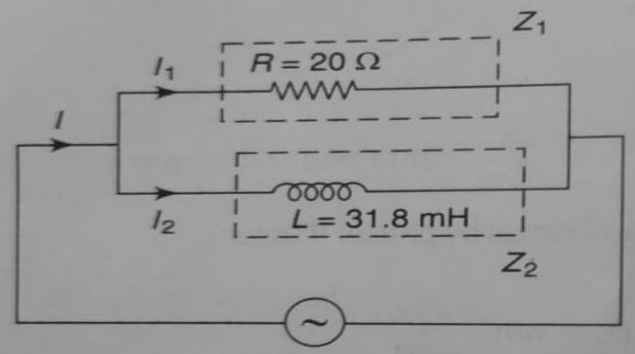
Branch current, $\bar{I}_2 = \dfrac{\bar{V}}{\bar{Z}_2}$

Total circuit current, $\bar{I} = \bar{I}_1 + \bar{I}_2$

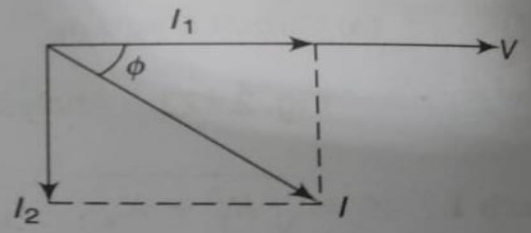
Example 2.66 A resistance of 20 Ω and a pure coil of inductance 31.8 mH are connected in parallel across 230 V, 50 Hz supply. Find (i) the line current, (ii) power factor, and (iii) power consumed by the circuit.

Solution

The conditions in the example are shown in Fig. 2.123(a).



(a) Circuit diagram



(b) Phasor diagram

Given: $V = 230$ V

$f = 50$ Hz

$L = 31.8 \times 10^{-3}$ H $\Rightarrow$ $X_L = 2\pi fL = 2\pi \times 50 \times 31.8 \times 10^{-3} = 10\ \Omega$

Branch current, $I_1 = \dfrac{V}{R} = \dfrac{230}{20} = 11.5$ A

The current I_1 is in phase with the applied voltage.

Branch current, $I_2 = \dfrac{V}{X_L} = \dfrac{230}{10} = 23$ A

The current I_2 lags behind the applied voltage by 90°.

So, line current, $I = \sqrt{I_1^2 + I_2^2}$

$= \sqrt{(11.5)^2 + (23)^2}$

$= 25.71$ A

From phasor diagram,

$$\text{pf} = \cos\phi = \frac{I_1}{I} = \frac{11.5}{25.71} = 0.447 \text{ lag}$$

Power consumed, $P = VI \cos\phi$

$= 230 \times 25.71 \times 0.447$

$= 2643$ W

Alternative method:

Taking V as reference phasor,

$\bar{V} = (230 \angle 0)$ V

$\bar{Z}_1 = (20 + j0)\ \Omega = (20 \angle 0)\ \Omega$

$\bar{Z}_2 = (0 + j10)\ \Omega = (10 \angle 90)\ \Omega$

By Ohm's law, $\bar{I}_1 = \dfrac{\bar{V}}{\bar{Z}_1} = \dfrac{(230 \angle 0)}{(20 \angle 0)} = (11.5 \angle 0)$ A

$$\bar{I}_2 = \frac{\bar{V}}{\bar{Z}_2} = \frac{(230 \angle 0)}{(10 \angle 90)} = (23 \angle -90) \text{ A}$$

Total current, $\bar{I} = \bar{I}_1 + \bar{I}_2$

$= (11.5 \angle 0) + (23 \angle -90)$

$= (11.5 + j0) + (0 - j23)$

$= (11.5 - j23)$ A

$= (25.71 \angle -63.43)$ A

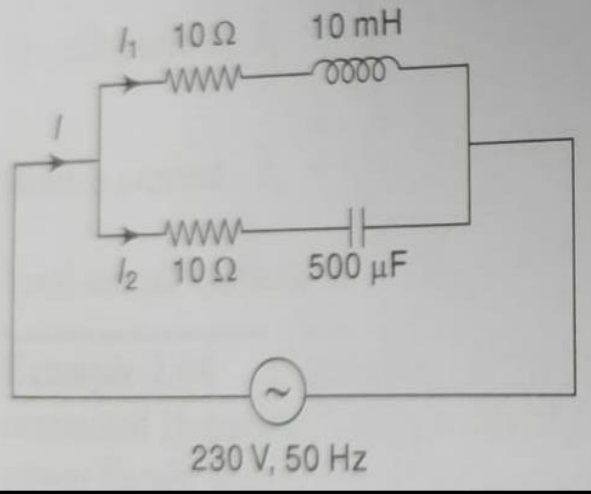
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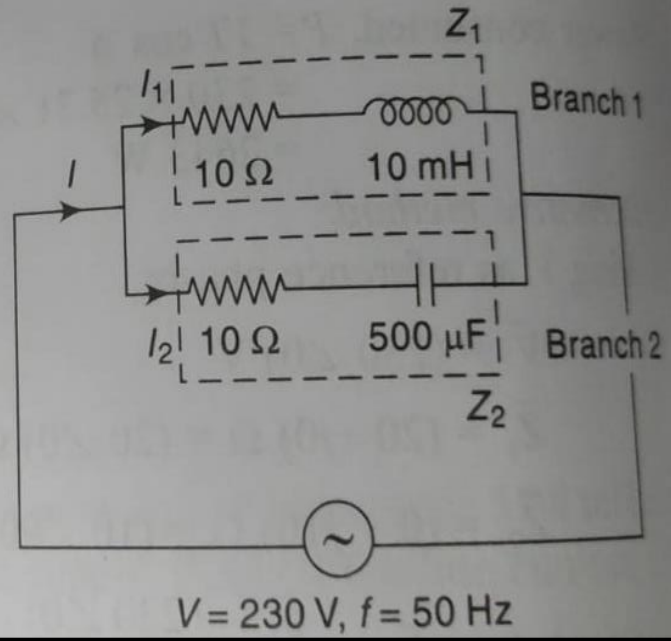
So, $\text{pf} = \cos\phi$
$= \cos 63.43$
$= 0.447$ lagging

Power, $P = VI\cos\phi$
$= 230 \times 25.71 \times 0.447$
$= 2643$ W

Example 2.67 For a circuit shown in Fig. 2.124, determine:

(i) Total impedance of the circuit and total current
(ii) Branch current I_1 and I_2
(iii) Power factor of each branch and total power factor
(iv) Power consumed by each branch





Solution

Assuming the different impedances, we get Fig. 2.125.

Taking applied voltage as reference phasor,

$$\bar{V} = (230\ \angle 0)$$

$$L = 10 \times 10^{-3}\ \text{H};\ X_L = 2\pi fL = 2\pi \times 50 \times 10 \times 10^{-3} = 3.14\ \Omega$$

$$C = 50 \times 10^{-6}\ \text{F};\ X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 500 \times 10^{-6}} = 6.37\ \Omega$$

$$\bar{Z}_1 = (10 + j3.14)\ \Omega = (10.48\ \angle 17.43)\ \Omega$$

$$\bar{Z}_2 = (10 - j6.37)\ \Omega = (11.86\ \angle -32.5)\ \Omega$$

(i) Total impedance of the circuit,

$$\bar{Z} = \frac{\bar{Z}_1 \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2}$$

$$= \frac{(10.48\ \angle 17.43)\ (11.86\ \angle -32.5)}{(10 + j3.14) + (10 - j6.37)}$$

$$= \frac{(124.29\ \angle -15.07)}{(20 - j3.23)}$$

$$= \frac{(124.29\ \angle -15.07)}{(20.26\ \angle -9.17)}$$

$$= (6.13\ \angle -5.9)\ \Omega$$

Total current, $I = \dfrac{\bar{V}}{\bar{Z}} = \dfrac{(230\ \angle 0)}{(6.13\ \angle -5.9)} = (37.52\ \angle 5.9)$ A

(ii) Branch current, $\bar{I}_1 = \frac{\bar{V}}{\bar{Z}_1} = \frac{(230\angle 0)}{(10.48\angle 17.43)} = (21.95\angle -17.43)\,A$

$$\bar{I}_2 = \frac{\bar{V}}{\bar{Z}_2} = \frac{(230\angle 0)}{(11.86\angle -32.5)} = (19.39\angle 32.5)\,A$$

(iii) Power factor of branch 1,

$$(pf)_1 = \cos\phi_1$$

where ϕ_1 is the phase angle of branch 1. From polar form of impedance Z_1, $\phi_1 = 17.43$. Branch 1 is inductive.

So, $(pf)_1 = \cos 17.43$

$= 0.954$ lagging

Power factor of branch 2,

$$(pf)_2 = \cos\phi_2$$

where ϕ_2 is the phase angle of branch 2. From polar form of impedance Z_2, $\phi_2 = -32.5$. Branch 2 is capacitive.

So, $(pf)_2 = \cos(-32.5)$

$= 0.843$ leading

Total power factor.

$$pf = \cos\phi$$

where ϕ is the phase angle total circuit. From polar form of impedance Z, $\phi = -5.9$. As phase angle is negative total circuit is capacitive.

So, $pf = \cos(-5.9)$

$= 0.995$ leading

(iv) Power consumed by branch 1,

$$P_1 = I_1^2 R = (21.95)^2 \times 10 = 4.818\ kW$$

Power consumed by branch 2,

$$P_2 = I_2^2 R = (19.39)^2 \times 10 = 3.76\ kW$$

Example 2.68 In the circuit shown in Fig. 2.126, a voltage $v = 100 \sin 3t$ is applied. Determine (i) branch current I_1 and I_2 with their phase angles and (ii) total current supplied by the source and its phase angle.

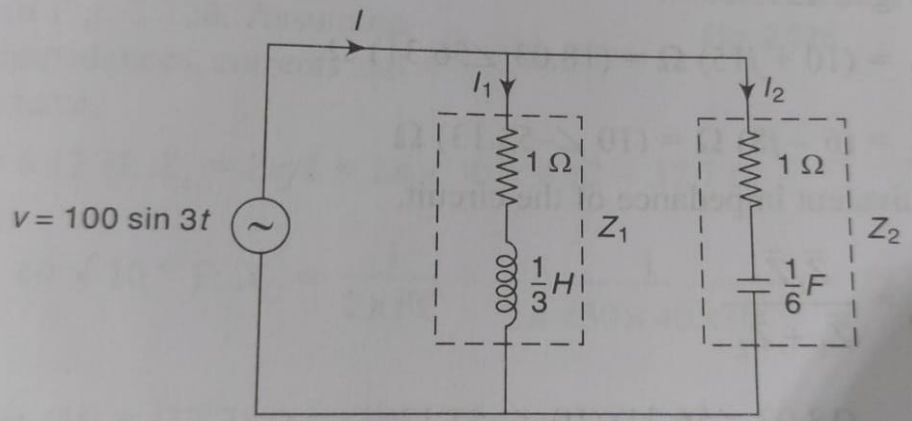


Fig. 2.126

Solution

We have $v = 100 \sin 3t$. Converting the standard sinusoidal form into polar form,

$$\bar{V} = (70.71 \angle 0)\ \text{V}$$

From equation of applied voltage, $\omega = 3$ rad/sec

$$L = \frac{1}{3}\ \text{H}; X_L = \omega L = 3 \times \frac{1}{3} = 1\ \Omega$$

$$C = \frac{1}{6}\ \text{F}; X_C = \frac{1}{\omega C} = \frac{1}{3 \times \frac{1}{6}} = 2\ \Omega$$

$$\bar{Z}_1 = (1 + j1)\ \Omega = (1.414 \angle 45)\ \Omega$$

$$\bar{Z}_2 = (1 - j2)\ \Omega = (2.236 \angle -63.43)\ \Omega$$

(i) Branch current, $\bar{I}_1 = \dfrac{\bar{V}}{\bar{Z}_1} = \dfrac{(70.71 \angle 0)}{(1.414 \angle 45)} = (50 \angle -45)\ \text{A}$

Branch current, $\bar{I}_2 = \dfrac{\bar{V}}{\bar{Z}_2} = \dfrac{(70.71 \angle 0)}{(2.236 \angle -63.43)} = (31.62 \angle 63.43)\ \text{A}$

(ii) Total current supplied by the source,

$$\bar{I} = \bar{I}_1 + \bar{I}_2$$
$$= (50 \angle -45) + (31.62 \angle 63.43)$$
$$= (35.36 - j35.36) + (14.14 + j28.28)$$
$$= (49.5 - j7.08)\ \text{A}$$
$$= (50 \angle -8.14)\ \text{A}$$

Example 2.69 Two impedances, $\bar{Z}_1 = (10 + j15)\ \Omega$ and $\bar{Z}_2 = (6 - j8)\ \Omega$ are connected in parallel. The total current supplied is 15 A. What is the power taken by each branch?

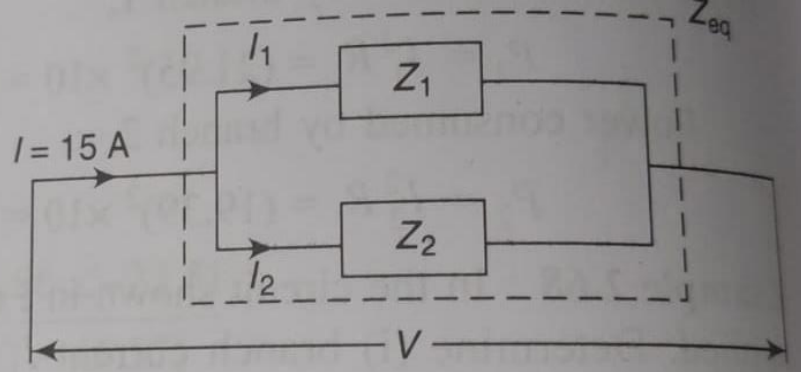


Fig. 2.127

Solution

The conditions in the example are shown in Fig. 2.127.

$$\bar{Z}_1 = (10 + j15)\ \Omega = (18.03\ \angle 56.31)\ \Omega$$

$$\bar{Z}_2 = (6 - j8)\ \Omega = (10\ \angle -53.13)\ \Omega$$

Total or equivalent impedance of the circuit,

$$\bar{Z}_{eq} = \frac{\bar{Z}_1 \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2}$$

$$= \frac{(18.03\ \angle 56.31)\ (10\ \angle -53.13)}{(10 + j15) + (6 - j8)}$$

$$= \frac{(180.3\ \angle 3.18)}{(16 + j7)}$$

$$= \frac{(180.3\ \angle 3.18)}{(17.46\ \angle 23.63)}$$

$$= (10.33\ \angle -20.45)\ \Omega$$

By Ohm's law,

Voltage across the circuit, $V = IZ_{eq}$

$= 15 \times 10.33$

$= 154.95$ V

Branch current, $I_1 = \frac{V}{Z_1} = \frac{154.95}{18.03} = 8.59$ A

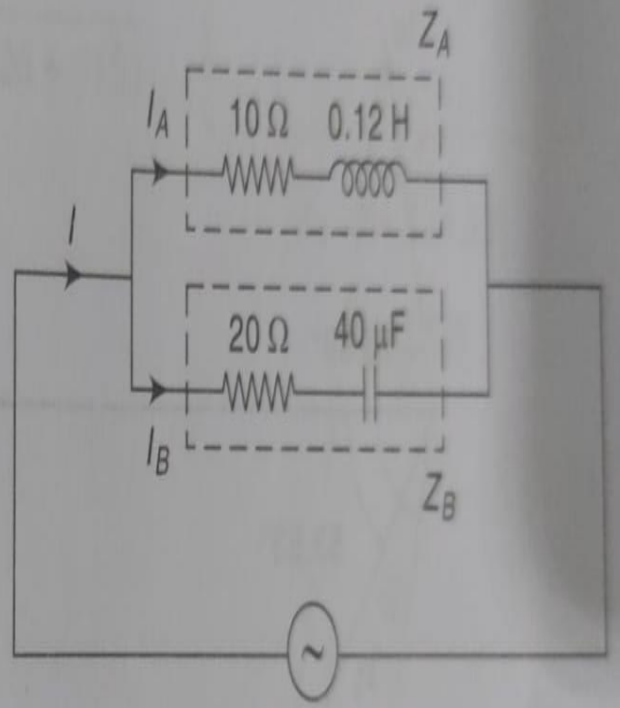
Branch current, $I_2 = \frac{V}{Z_2} = \frac{154.95}{10} = 15.495$ A

Hence,

Power taken by the first branch, $P_1 = I_1^2 R_1 = (8.59)^2 \times 10 = 737.88$ W

Power taken by the second branch, $P_2 = I_2^2 R_2 = (15.495)^2 \times 6 = 1440.57$ W

Example 2.70 Two circuits *A* and *B* are connected in parallel across a 200 V, 50 Hz mains. Circuit *A* consists of a resistance of 10 Ω and an inductance of 0.12 H connected in series. Circuit *B* consists of a resistance of 20 Ω in series with a capacitor of 40 μF. Calculate (i) current in each branch and (ii) power factor. Draw phasor diagram.



$V = 200$ V, 50 Hz

Fig. 2.128

Solution

The conditions in the example are shown in Fig. 2.128. Assuming the various impedances, currents and voltage, we have,

$$L = 0.12 \text{ H}; X_L = 2\pi fL = 2\pi \times 50 \times 0.12 = 37.7\ \Omega$$

$$C = 40 \times 10^{-6} \text{ F}; X_C = \frac{1}{2\pi FC} = \frac{1}{2\pi \times 50 \times 40 \times 10^{-6}} = 79.58\ \Omega$$

$$\bar{Z}_A = (10 + j37.7)\ \Omega = (39\ \angle 75.14)\ \Omega$$

$$\bar{Z}_B = (20 - j79.58)\ \Omega = (82.05\ \angle -75.89)\ \Omega$$

Total or equivalent impedance of the circuit,

$$Z_{eq} = \frac{\bar{Z}_A \bar{Z}_B}{\bar{Z}_A + \bar{Z}_B} = \frac{(39\angle 75.14)(82.05\angle -75.89)}{(10 + j37.7)(20 - j79.58)}$$

$$= \frac{(3200\angle -0.75)}{(30 - j41.88)} = \frac{(3200\angle -0.75)}{(51.52\angle -54.38)}$$

$$= (62.11\ \angle 53.63)\ \Omega$$

Taking applied voltage as reference, we have

$$\bar{V} = (200\ \angle 0)\ \text{V}$$

(i) Branch current, $\bar{I}_A = \dfrac{\bar{V}}{\bar{Z}_A} = \dfrac{(200 \angle 0)}{(39 \angle 75.14)} = (5.13 \angle -75.14)\ \text{A}$

Branch current, $\bar{I}_B = \dfrac{\bar{V}}{\bar{Z}_B} = \dfrac{(200 \angle 0)}{(82.05 \angle -75.89)} = (2.44 \angle 75.89)\ \text{A}$

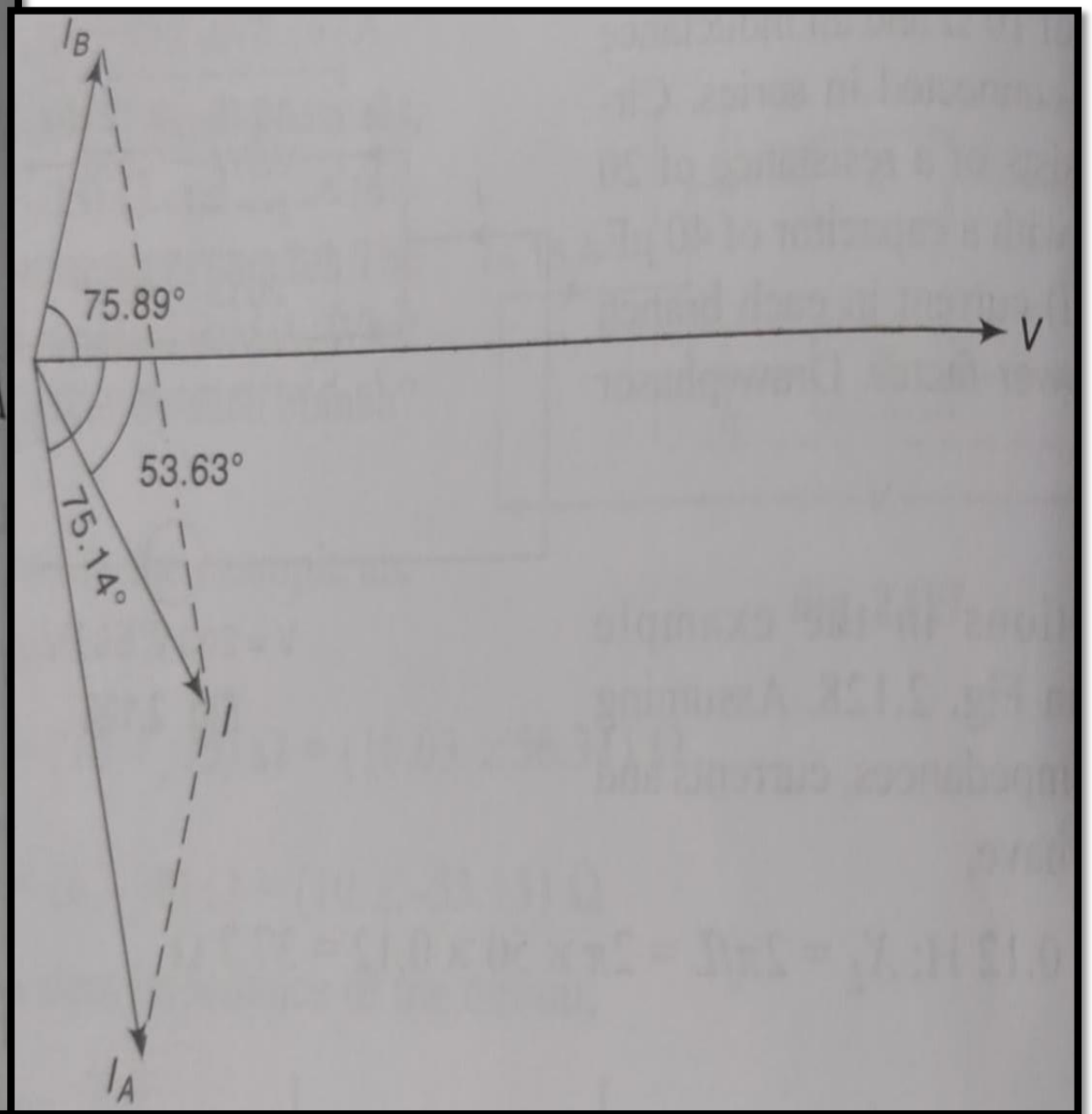
(ii) pf = cos 53.63

= 0.59 lagging

Phasor diagram (Fig. 2.129):

Taking applied voltage as reference phasor.

From circuit diagram, $\bar{I} = \bar{I}_A + \bar{I}_B$



Example 2.71 Two impedances $(20 \angle -45)\ \Omega$ and $(30 \angle 30)\ \Omega$ are connected in series across a certain ac supply and the resulting current is drawn to be 10 A. If the supply voltage remains unchanged, calculate the supply current, when the two impedances are connected in parallel.

Solution

Let $\bar{Z}_1 = (20 \angle -45)\ \Omega = (14.14 - j14.14)\ \Omega$

$\bar{Z}_2 = (30 \angle 30)\ \Omega = (26 + j15)\ \Omega$

When impedances are in series, current is 10 A (Fig. 2.130):

Total impedances,

$$\bar{Z}_{eq} = \bar{Z}_1 + \bar{Z}_2$$

$$= (14.14 - j14.14) + (26 + j15)$$

$$= (40.14 + j0.86)\ \Omega$$

$$= (40.15 \angle 1.23)\ \Omega$$

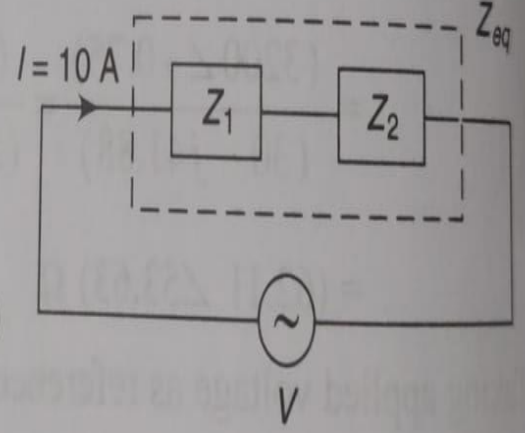


Fig. 2.130

Now, taking 'I' as reference, $\bar{I} = (10 \angle 0)$

$\therefore$ Supplied voltage $= \bar{I}\bar{Z}_{eq}$

$$= (10 \angle 0)\,(40.15 \angle 1.23)$$

$$= (401.5 \angle 1.23)\ \text{V}$$

When impedances are in parallel across the same supply (Fig. 2.131):

Total impedances,

$$\bar{Z}_{eq} = \frac{\bar{Z}_1 \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2}$$

$$= \frac{(20 \angle -45)\,(30 \angle 30)}{(14.14 - j14.14) + (26 + j15)}$$

$$= \frac{(600 \angle -15)}{(40.14 + j0.86)}$$

$$= \frac{(600 \angle -15)}{(40.15 \angle 1.23)}$$

$$= (14.94 \angle -16.23)\ \Omega$$

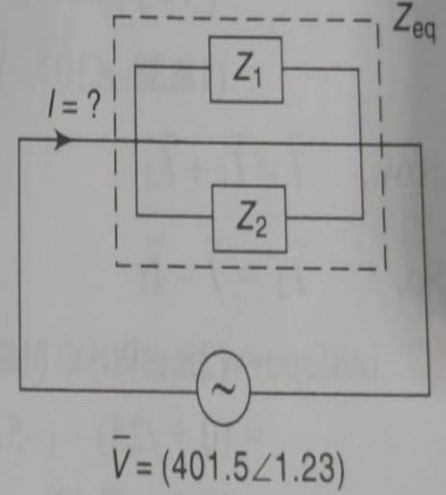


Fig. 2.131

Hence, Supply current, $I = \dfrac{\bar{V}}{\bar{Z}_{eq}}$

$$= \frac{(401.5\angle 1.23)}{(14.94\angle -16.23)}$$

$$= (26.87 \angle 17.46)\ \text{A}$$

Example 2.72 Find I_1 and I_2 in the circuit shown in Fig. 2.132.

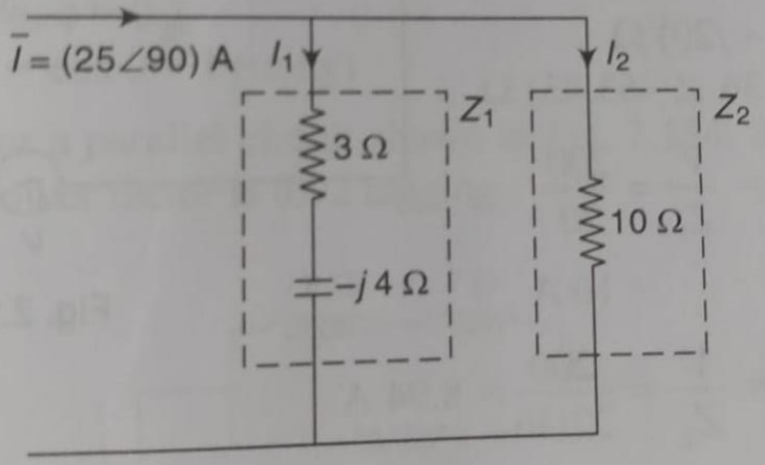


Fig. 2.132

Solution

$$\bar{I} = (25 \angle 90)\ \text{A}$$

$$\bar{Z}_1 = (3 - j4)\ \Omega = (5 \angle -53.13)\ \Omega$$

$$\bar{Z}_2 = (10 + j0)\ \Omega = (10 \angle 0)\ \Omega$$

By current division rule,

$$\bar{I}_1 = \frac{\bar{I} \times \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2}$$

$$= \frac{(25\angle 90)(10\angle 0)}{(3 - j4) + (10 + j0)} = \frac{(250\angle 90)}{(13 - j4)} = \frac{(250\angle 90)}{(13.6\angle -17.1)}$$

$$= (18.38 \angle 107.1)\ \text{A}$$

Now, $\bar{I} = \bar{I}_1 + \bar{I}_2$

So, $\bar{I}_2 = \bar{I} - \bar{I}_1$

$$= (25 \angle 90) - (18.38 \angle 107.1)$$

$$= (0 + j25) - (-5.4 + j17.57)$$

$$= (5.4 + j7.43)$$

$$= (9.19 \angle 53.99)\ \text{A}$$

FEEE_BEEE_UNIT 2-Part 2_Nirmala k

Example 2.73 A voltage of (200 ∠53.13) V is applied across two impedances in parallel. The values of the impedances are (12 + *j*16) Ω and (10 – *j*20) Ω. Determine kVA, kVAR, and kW in each branch and power factor of the whole circuit.

Solution

The conditions in the example are shown in Fig. 2.133.

Given:

$$\overline{V} = (200 \angle 53.13)\ \text{V}$$

Let $\overline{Z}_1 = (12 + j16)\ \Omega$

$= (20 \angle 53.13)\ \Omega$

$\overline{Z}_2 = (10 - j20)\ \Omega$

$= (22.36 \angle -63.43)\ \Omega$

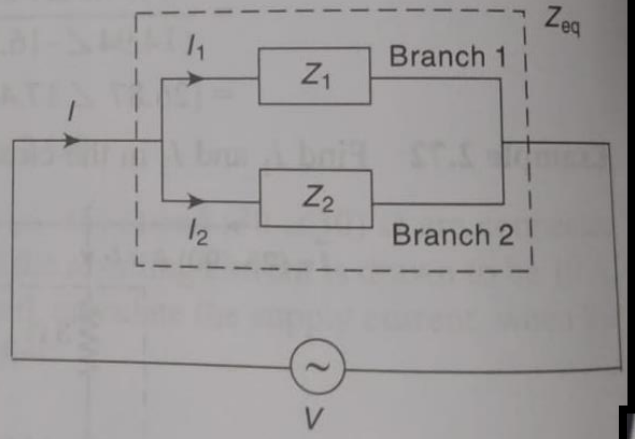


Fig. 2.133

Branch current, $I_1 = \dfrac{V}{Z_1} = \dfrac{200}{20}$

$= 10$ A

Branch current, $I_2 = \dfrac{V}{Z_2} = \dfrac{200}{22.36} = 8.94$ A

For branch 1 (i.e., impedance Z_1):

$$\text{kW} = \frac{VI_1 \cos\phi}{1000} = \frac{200 \times 10 \times \cos 53.13^\circ}{1000} = 1.2$$

$$\text{kVAR} = \frac{VI_1 \sin\phi}{1000} = \frac{200 \times 10 \times \sin 53.13^\circ}{1000} = 1.6$$

$$\text{kVA} = \frac{VI_1}{1000} = \frac{200 \times 10}{1000} = 2$$

For branch 2 (i.e., impedance Z_2):

$$\text{kW} = \frac{VI_2 \cos\phi}{1000} = \frac{200 \times 8.94 \times \cos(-63.43)}{1000} = 0.8$$

$$\text{kVAR} = \frac{VI_2 \sin\phi}{1000} = \frac{200 \times 8.94 \times \sin(-63.43)}{1000} = -1.6$$

$$\text{kVA} = \frac{VI_2}{1000} = \frac{200 \times 8.94}{1000} = 1.788$$

For power factor of the circuit, the phase angle (ϕ) of the circuit is required. The phase angle (ϕ) of the circuit can be calculated as:

Total impedance, $\overline{Z}_{eq} = \dfrac{\overline{Z}_1\,\overline{Z}_2}{\overline{Z}_1 + \overline{Z}_2}$

$$= \frac{(20 \angle 53.13)(22.36 \angle -63.43)}{(12 + j16) + (10 - j20)}$$

$$= \frac{(447.2 \angle -10.3)}{(22 - j4)}$$

$$= \frac{(447.2 \angle -10.3)}{(22.36 \angle -10.3)}$$

$= (20 \angle 0)\ \Omega$

Thus, from polar form of impedance,

phase angle, $\phi = 0$

So, pf $= \cos\phi = \cos 0 = 1$ (unity)

FECO_UNIT_UNIT 2-Part 2_Nirmala k

Example 2.74 For a parallel circuit shown in Fig. 2.134, find the value of R_2 when the overall power factor is 0.92 lagging.

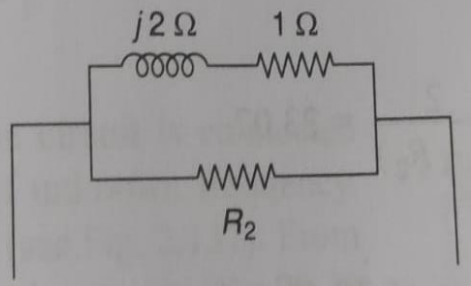


Fig. 2.134

Solution

Assuming the various impedances (Fig. 2.135), we get Fig. 2.135.

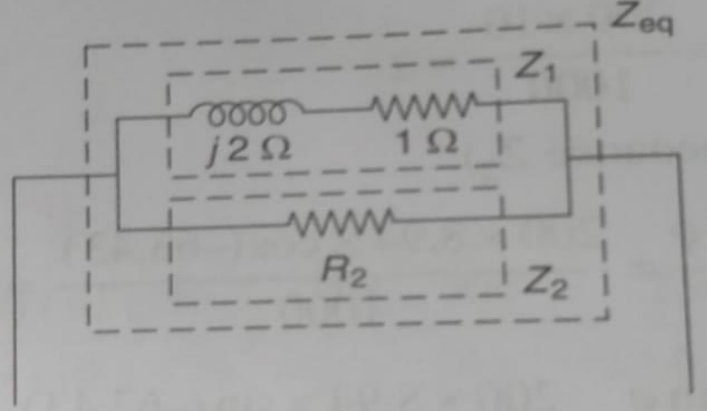


Fig. 2.135

Given: $\bar{Z}_1 = (1 + j2)\ \Omega = (2.24\ \angle 63.43)\ \Omega$

$\bar{Z}_2 = (R_2 + j0)\ \Omega = (R_2\ \angle 0)\ \Omega$

Total impedance, $\bar{Z}_{eq} = \dfrac{\bar{Z}_1\ \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2}$

$$= \frac{(2.24\ \angle 63.43)\ (R_2\ \angle 0)}{(1 + j2) + (R_2 + j0)}$$

$$= \frac{(2.24 R_2\ \angle 63.43)}{[(1 + R_2) + j2]}$$

$$= \frac{(2.24 R_2\ \angle 63.43)}{\left[\sqrt{(1+R_2)^2 + (2)^2}\ \angle \tan^{-1}\dfrac{2}{(1+R_2)}\right]}$$

$$= \left(\underbrace{\frac{2.24 R_2}{\sqrt{(1+R_2)^2 + (2)^2}}}_{\text{magnitude } (Z)} \angle \underbrace{63.43 - \tan^{-1}\frac{2}{(1+R_2)}}_{\text{phase angle } (\phi)} \right) \Omega$$

But the power factor of overall circuit is given as 0.92 lagging, i.e.,

$\cos \phi = 0.92$

So, $\phi = 23.07°$

or $63.43 - \tan^{-1}\dfrac{2}{(1+R_2)} = 23.07$

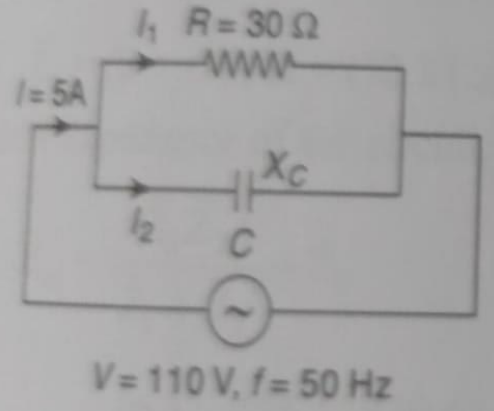
or $R_2 = 1.353\ \Omega$

FECO_BEEE_UNIT 2-Part 2_Nirmala k

Example 2.75 A resistor of 30 Ω and capacitor of unknown value are connected in parallel across a 110 V, 50 Hz, 1 – ϕ supply. The combination draws a current of 5 A from the supply. Find the value of unknown capacitance of the capacitor. This combination is again connected across a 110 V supply of unknown frequency. It is observed that the total current drawn from the mains falls to 4 A. Determine the frequency of the supply. Draw the relevant diagrams.

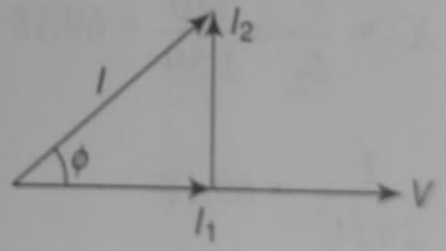
Solution

The conditions in the example are shown in Fig. 2.136(a). The phasor diagram is shown in Fig. 2.136(b).



V = 110 V, f = 50 Hz

(a) Circuit diagram



(b) Phasor diagram

From circuit diagram, by Ohm's law

Branch current, $I_1 = \frac{V}{R} = \frac{110}{30} = 3.67$ A

From phasor diagram,

$$I = \sqrt{I_1^2 + I_2^2}$$

or $$5 = \sqrt{(3.67)^2 + I_2^2}$$

or $$I_2 = 3.4 \text{ A}$$

Now, $$V = I_2 X_C$$

or $$110 = 3.4 \times X_C$$

or $$X_C = 32.35\ \Omega$$

Now, $$\frac{1}{2\pi f C} = 32.35$$

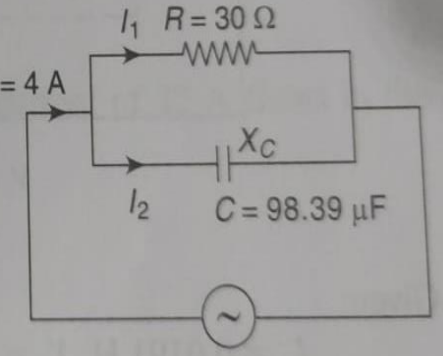
or $$C = \frac{1}{2\pi \times 50 \times 32.35}$$

or $$C = 98.39 \times 10^{-6} \text{ F}$$
$$= 98.39\ \mu\text{F}$$

Case (ii) Now the same circuit is connected across a 110 V supply of unknown frequency. The circuit current is 4 A (see Fig. 2.137). From case (i), we know the capacitance, i.e., $C = 98.39$ μF.

From circuit diagram, by Ohm's law,

Branch current, $I_1 = \frac{V}{R} = \frac{110}{30} = 3.67$ A



V = 110 V, f = ?

Fig. 2.137

FECO_BEEE_UNIT 2-Part B_Nirmala

or $$C = \frac{1}{2\pi \times 50 \times 32.35}$$

or $$C = 98.39 \times 10^{-6}\ \text{F}$$
$$= 98.39\ \mu\text{F}$$

Case (ii) Now the same circuit is connected across a 110 V supply of unknown frequency. The circuit current is 4 A (see Fig. 2.137). From case (i), we know the capacitance, i.e., $C = 98.39\ \mu$F.

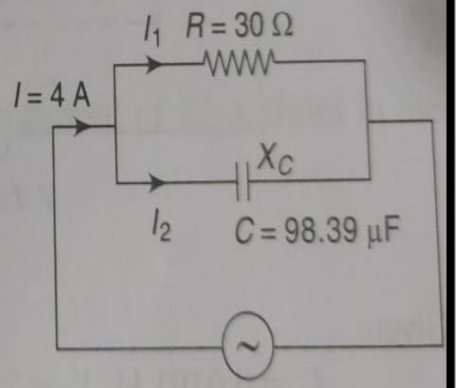


Fig. 2.137

From circuit diagram, by Ohm's law,

Branch current, $I_1 = \frac{V}{R} = \frac{110}{30} = 3.67\ \text{A}$

As $$I = \sqrt{I_1^2 + I_2^2},$$

$$4 = \sqrt{(3.67)^2 + I_2^2}$$

So, $$I_2 = 1.59\ \text{A}$$

Hence, $$X_C = \frac{V}{I_2} = \frac{110}{1.59} = 69.18\ \Omega$$

So, $$\frac{1}{2\pi fC} = 69.18$$

or $$f = \frac{1}{2\pi C \times 69.18} = \frac{1}{2\pi \times 98.39 \times 10^{-6} \times 69.18} = 23.38\ \text{Hz}$$

Example 2.76 Determine the equivalent impedance of the circuit shown in Fig. 2.138.

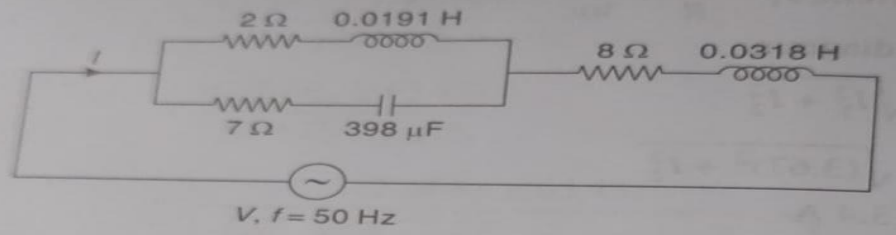
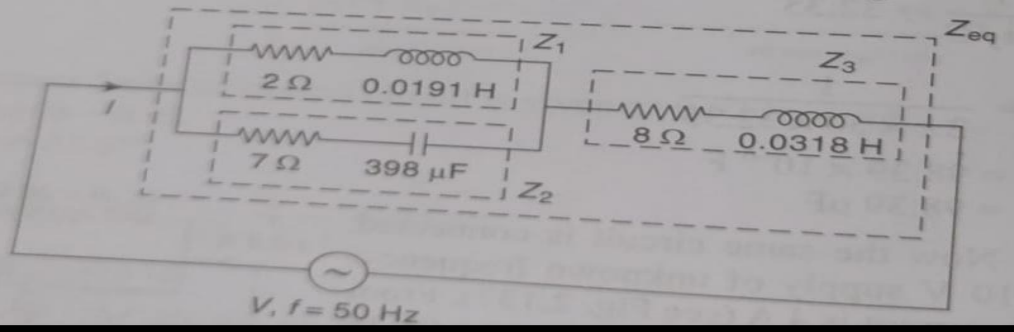


Fig. 2.138

Solution

By marking the various impedances in Fig. 2.138, we get Fig. 2.139.



Given:

$$L_1 = 0.0191\ \text{H}, X_L = 2\pi f L_1 = 2\pi \times 50 \times 0.0191 = 6\ \Omega$$

$$C = 398 \times 10^{-6}\ \text{F}, X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 398 \times 10^{-6}} = 8\ \Omega$$

$$L_3 = 0.0318\ \text{H}; X_L = 2\pi f L_3 = 2\pi \times 50 \times 0.0318 = 10\ \Omega$$

$$\bar{Z}_1 = (2 + j6)\ \Omega = (6.32\ \angle 71.57)\ \Omega$$

$$\bar{Z}_2 = (7 - j8)\ \Omega = (10.63\ \angle -48.81)\ \Omega$$

$$\bar{Z}_3 = (8 + j10)\ \Omega = (12.81\ \angle 51.34)\ \Omega$$

Equivalent impedance of the circuit,

$$\bar{Z}_{eq} = \frac{\bar{Z}_1 \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2} + \bar{Z}_3$$

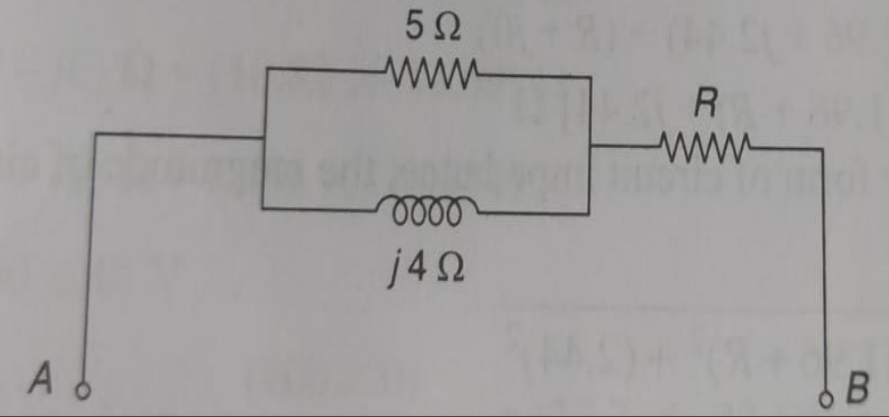
$$= \frac{(6.32 \angle 71.57)(10.63 \angle -48.81)}{(2 + j6) + (7 - j8)} + (8 + j10)$$

$$= \frac{(67.18 \angle 22.76)}{(9 - j2)} + (8 + j10)$$

$$= \frac{(67.18 \angle 22.76)}{(9.22 \angle -12.53)} + (8 + j10)$$

$$= (7.29 \angle 35.29) + (8 + j10)$$
$$= (5.95 + j4.21) + (8 + j10)$$
$$= (13.95 + j14.21)\ \Omega$$
$$= (19.91 \angle 45.53)\ \Omega$$

Example 2.77 The voltage of 150 V applied between terminals *AB*, produces a current of 32 A for the circuit shown in Fig. 2.140. Find the value of *R* and the power factor of the circuit.



Solution

When 150 V is applied between terminals *AB*, a current of 32 A flows in the circuit (see Fig. 2.141).

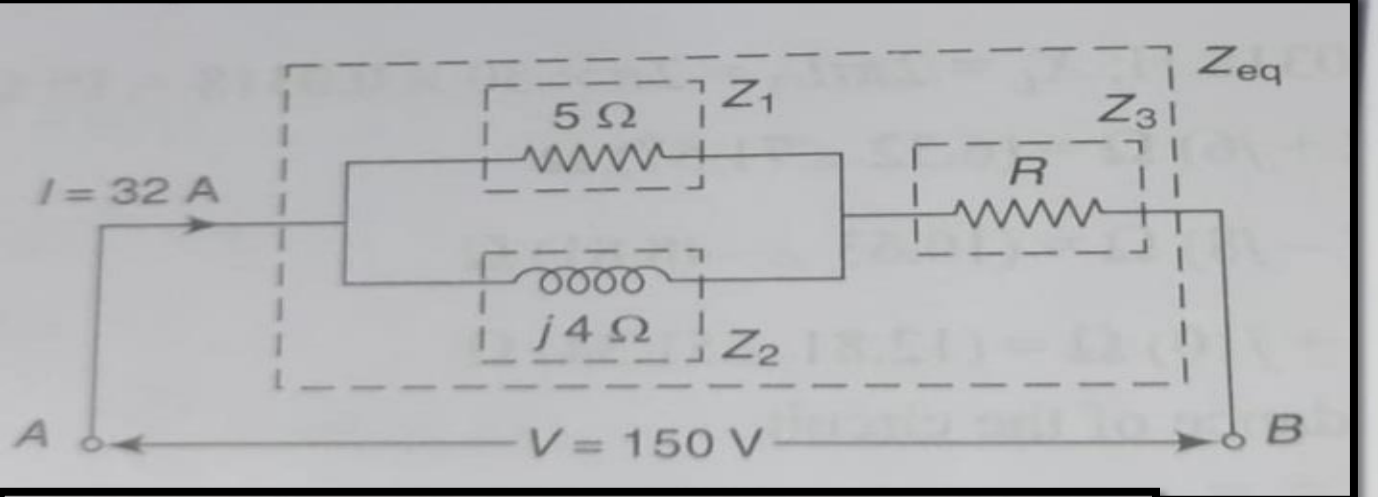
Given:

$$\bar{Z}_1 = (5 + j0)\ \Omega = (5 \angle 0)\ \Omega$$

$$\bar{Z}_2 = (0 + j4)\ \Omega = (4 \angle 90)\ \Omega$$

$$\bar{Z}_3 = (R + j0)\ \Omega = (R \angle 0)\ \Omega$$

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From polar form of equivalent impedance,

phase angle, $\phi = 31.32$

(as ϕ is positive, circuit is inductive)

So, $pf = \cos 31.32$

$= 0.854$ lagging

Magnitude of the circuit impedance,

$$Z_{eq} = \frac{V}{I} = \frac{150}{32} = 4.69\ \Omega$$

Now, the circuit impedance can be expressed as

$$\bar{Z}_{eq} = \frac{\bar{Z}_1 \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2} + \bar{Z}_3$$

$$= \frac{(5\angle 0)(4\angle 90)}{(5 + j0) + (0 + j4)} + (R + j0)$$

$$= \frac{(20\angle 90)}{(5 + j4)} + (R + j0)$$

$$= \frac{(20\angle 90)}{(6.4\angle 38.66)} + (R + j0)$$

$$= (3.13\angle 51.34) + (R + j0)$$

$$= (1.96 + j2.44) + (R + j0)$$

$$= [(1.96 + R) + j2.44]\ \Omega$$

From rectangular form of circuit impedance, the magnitude of circuit impedance can be written as

$$Z_{eq} = \sqrt{(1.96 + R)^2 + (2.44)^2}$$

So, $4.69 = \sqrt{(1.96 + R)^2 + (2.44)^2}$ $(\because Z_{eq} = 4.69\ \Omega)$

or $R = 2.05\ \Omega$

Now, by substituting the value of R in the rectangular form of circuit impedance, we get

$$\bar{Z}_{eq} = [(1.96 + R) + j2.44]\ \Omega$$

$$= [(1.96 + 2.05) + j2.44]\ \Omega$$

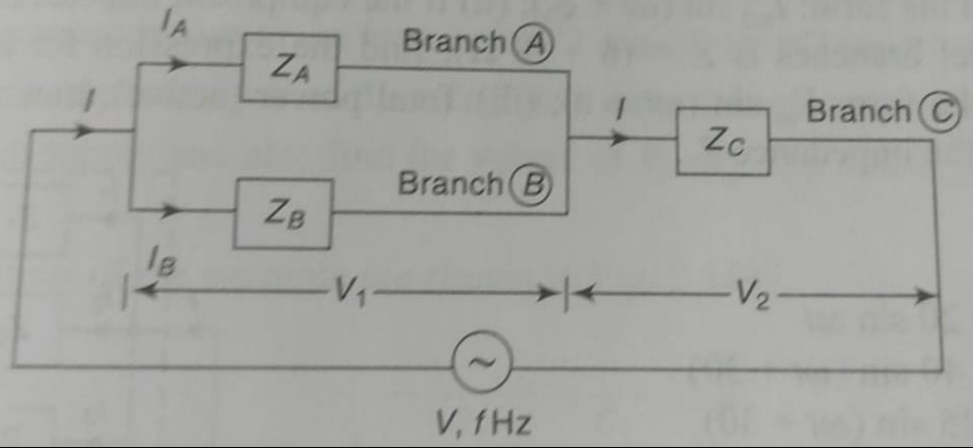
$$= (4.01 + j2.44)\ \Omega$$

$$= (4.69\angle 31.32)\ \Omega$$

Example 2.78 In a series-parallel circuit, two parallel branches A and B are in series with C. The impedances are $\bar{Z}_A = (10 + j8)\ \Omega$, $\bar{Z}_B = (9 - j6)\ \Omega$, and $\bar{Z}_C = (3 + j2)\ \Omega$. If the voltage across Z_C is $(100\ \angle 0)$ V, determine the values of I_A and I_B.

Solution

Various voltages, impedances and current given in the example are marked in the circuit of Fig. 2.142.



Given:

$$\bar{Z}_A = (10 + j8)\ \Omega = (12.81\ \angle 38.66)\ \Omega$$

$$\bar{Z}_B = (9 - j6)\ \Omega = (10.82\ \angle -33.69)\ \Omega$$

$$\bar{Z}_C = (3 + j2)\ \Omega = (3.61\ \angle 33.69)\ \Omega$$

$$\bar{V}_2 = (100\ \angle 0)\ \text{V}$$

Circuit current, $\bar{I} = \dfrac{\bar{V}_2}{\bar{Z}_C} = \dfrac{(100\ \angle 0)}{(3.61\ \angle 33.69)} = (27.7\ \angle -33.69)\ \text{A}$

By current division rule,

Branch current, $\bar{I}_A = \dfrac{\bar{I}\,\bar{Z}_B}{\bar{Z}_A + \bar{Z}_B}$

$$= \frac{(27.7\ \angle -33.69)(10.82\ \angle -33.69)}{(10 + j8) + (9 - j6)}$$

$$= \frac{(299.71\ \angle -67.38)}{(19 + j2)}$$

$$= \frac{(299.71\ \angle -67.38)}{(19.1\ \angle 6)}$$

$$= (15.69\ \angle -73.38)\ \text{A}$$

Branch current, $\bar{I}_B = \bar{I} - \bar{I}_A$

$$= (27.7\ \angle -33.69) - (15.69\ \angle -73.38)$$

$$= (23.05 - j15.37) - (4.49 - j15.03)$$

$$= (18.56 - j0.34)\ \text{A}$$

$$= (18.56\ \angle -1.05)\ \text{A}$$

Example 2.79 Three impedances Z_1, Z_2 and Z_3 are connected in parallel. A sinusoidal voltage $V_m \sin(\omega t \pm \phi)$ is applied to it. The current drawn by these impedances are $i_1 = 20 \sin \omega t$, $i_2 = 40 \sin(\omega t + 30)$, $i_3 = I_{m3} \sin(\omega t \pm \phi_3)$ and the total current drawn from the supply is $i = 25\sin(\omega t + 30)$ amp. Calculate (i) the current i_3 in the form: $I_{m3} \sin(\omega t \pm \phi_3)$; (ii) if the equipment impedance of these three parallel branches is $\bar{Z} = (6 + j8\ \Omega)$, find the expression for the supply voltage in the form $V_m \sin(\omega t \pm \phi)$; (iii) Total power (active) drawn from the mains; (iv) the impedance Z_1.

Solution

Given:

$i_1 = 20 \sin \omega t$

$i_2 = 40 \sin(\omega t + 30)$

$i = 25 \sin(\omega t + 30)$

$\bar{Z} = (6 + j8)\ \Omega = (10 \angle 53.13)\ \Omega$

Converting the standard sinusoidal form into polar forms,

$\bar{I}_1 = (14.14 \angle 0)$ A

$\bar{I}_2 = (28.28 \angle 30)$ A

$I = (17.68 \angle 30)$ A

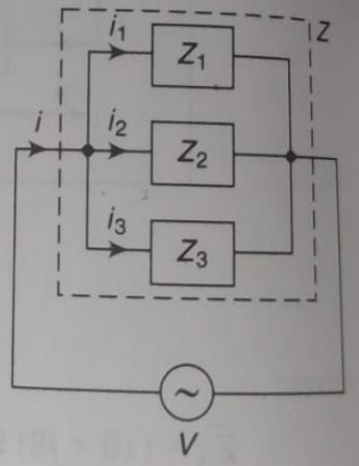


Fig. 2.143

(i) $\bar{I} = \bar{I}_1 + \bar{I}_2 + \bar{I}_3$

So, $\bar{I}_3 = \bar{I} - (\bar{I}_1 + \bar{I}_2)$

$= (17.68 \angle 30) - [(14.14 \angle 0) + (28.28 \angle 30)]$

$= (15.31 + j8.84) - [(14.14 + j0) + (24.49 + j14.14)]$

$= (15.31 + j8.84) - (38.63 + j14.14)$

$= (-23.32 - j5.3)$ A

$= (23.91 \angle -167.2)$ A

Converting the polar form into standard sinusoidal form,

$i_3 = 33.81 \sin(\omega t - 167.2)$

Supply voltage, $\bar{V} = \bar{I}\bar{Z}$

$= (17.68 \angle 30)\ (10 \angle 53.13)$

$= (176.8 \angle 83.13)$ V

Converting the polar form into standard sinusoidal form

$v = 250 \sin(\omega t + 83.13)$

(iii) Active power, $P = VI \cos \phi$

$= 176.8 \times 17.68 \times \cos 53.13$

$= 1875$ W

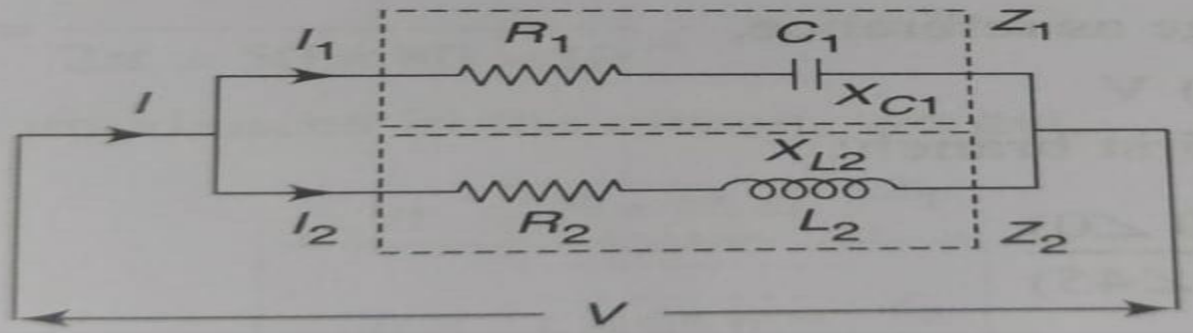
(iv) Impedance, $Z_1 = \dfrac{\bar{V}}{\bar{I}_1} = \dfrac{(176.8 \angle 83.13)}{(14.14 \angle 0)}$

$= (12.5 \angle 83.13)\ \Omega$

Example 2.80 Two impedances $(R_1 - jX_{C1})$ and $(R_2 + jX_{L2})$ are connected in parallel across supply voltage, $v = 100\sqrt{2}\sin(314t)$. The current flowing through the two impedances are given by $i_1 = 10\sqrt{2}\sin(314t + \pi/4)$ and $i_2 = 10\sqrt{2}\sin(314t + \pi/4)$, respectively. Find the equation of instantaneous value of total current drawn from supply and also find the values of R_1, R_2, X_{c1}, and X_{L2}.

Solution

The conditions of the example are shown in Fig. 2.144.



Equation of branch current i_1 is given as:

$$i_1 = 10\sqrt{2}\sin(314t + \pi/4)$$

So, $$I_1 = \frac{I_m}{\sqrt{2}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10 \text{ A}$$

So, $$\bar{I}_1 = (10 \angle 45°) \text{ A} \quad \text{(Polar form)}$$

$$\bar{I}_1 = (7.071 + j7.071) \text{ A} \quad \text{(Rectangular form)}$$

Equation of branch current i_2 is given as:

$$i_2 = 10\sqrt{2}\sin(314t + \pi/4)$$

So, $I_2 = \frac{I_m}{\sqrt{2}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10\ \text{A}$

So, $\bar{I}_2 = (10 \angle -45°)\ \text{A}$ (Polar form)

$\bar{I}_2 = (7.071 - j7.071)\ \text{A}$ (Rectangular form)

Now, total current of the circuit:

$$\bar{I} = \bar{I}_1 + \bar{I}_2$$

$\therefore \quad \bar{I} = (7.071 + j7.071) + (7.071 - j7.071)$

$\therefore \quad \bar{I} = (14.142 + j0)$

$\therefore \quad \bar{I} = (14.142 \angle 0)$

Now, equation of instantaneous value of total current:

$$I_m = \sqrt{2}I = \sqrt{2} \times 14.142 = 20\ \text{A}$$

$\therefore \quad i = 20 \sin (314t)$

From equation of voltage,

$$V = \frac{V_m}{\sqrt{2}} = \frac{100\sqrt{2}}{\sqrt{2}} = 100\ \text{V}$$

Taking applied voltage as reference,

$$\bar{V} = (100 \angle 0)\ \text{V}$$

Now, impedance of first branch:

$$\bar{Z}_1 = \frac{\bar{V}}{\bar{I}_1} = \frac{(100 \angle 0)}{(10 \angle 45)}$$

$\therefore \quad \bar{Z}_1 = (10 \angle -45)\ \Omega$ (Polar form)

$\therefore \quad \bar{Z}_1 = (7.071 - j7.071)\ \Omega$ (Rectangular form)

So, $R_1 = 7.071\ \Omega$ and $X_{C1} = 7.071\ \Omega$

Now, impedance of second branch:

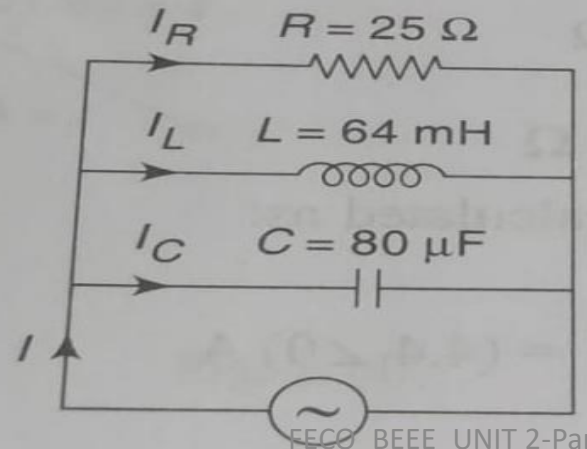
$$\bar{Z}_2 = \frac{\bar{V}}{\bar{I}_2} = \frac{(100\angle 0)}{(10\angle -45)}$$

$\therefore \quad \bar{Z}_2 = (10\angle 45)\ \Omega$ (Polar form)

$\therefore \quad \bar{Z}_2 = (7.071 + j7.071)$ (Rectangular form)

So, $R_2 = 7.071\ \Omega$ and $X_{L2} = 7.071\ \Omega$

Example 2.81 A parallel circuit of 25 Ω resistor, 64 mH inductor, and 80 μF capacitor is connected across: 110 V, 50 Hz, single phase supply, as shown in Fig. 2.145. Calculate the current in individual element, the total current drawn from the supply, and the overall power factor of the circuit. Draw a neat phasor diagram.



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Solution

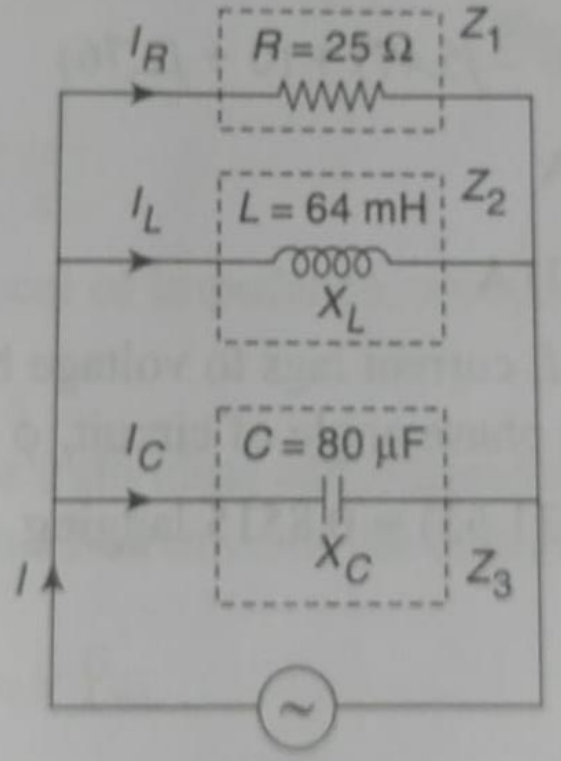
Inductive reactance,

$$X_L = 2\pi fL = 2\pi \times 50 \times 64 \times 10^{-3} = 20.106\ \Omega$$

Capacitive reactance,

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 80 \times 10^{-6}} = 39.79\ \Omega$$

Marking the different impdances in the circuit, we get,



$V = 110$ V, 50 Hz

Taking applied voltage as reference,

$$\bar{V} = (110\ \angle 0)\ \text{V}$$

Now, $\bar{Z}_1 = (25\ \angle 0)\ \Omega$

Now, total current drawn from supply,

$$\bar{I} = \bar{I}_R + \bar{I}_L + \bar{I}_C$$

$$= (4.4\ \angle 0) + (5.47\ \angle -90) + (2.76\ \angle 90)$$

$$= (4.4 + j0) + (0 - j5.47) + (0 + j2.76)$$

$$= (4.4 - j2.71)\ \text{A} \quad \text{(Rectangular form}$$

$$= (5.17\ \angle -31.63)\ \text{A} \quad \text{(Polar form)}$$

From polar form of V and I, current lags to voltage by 31.63°. Thus, the over circuit is inductive and the phase angle of circuit, $\phi = 31.63$.

$\therefore$ $pf = \cos\phi = \cos(31.63) = 0.8515$ lagging

Phasor diagram:

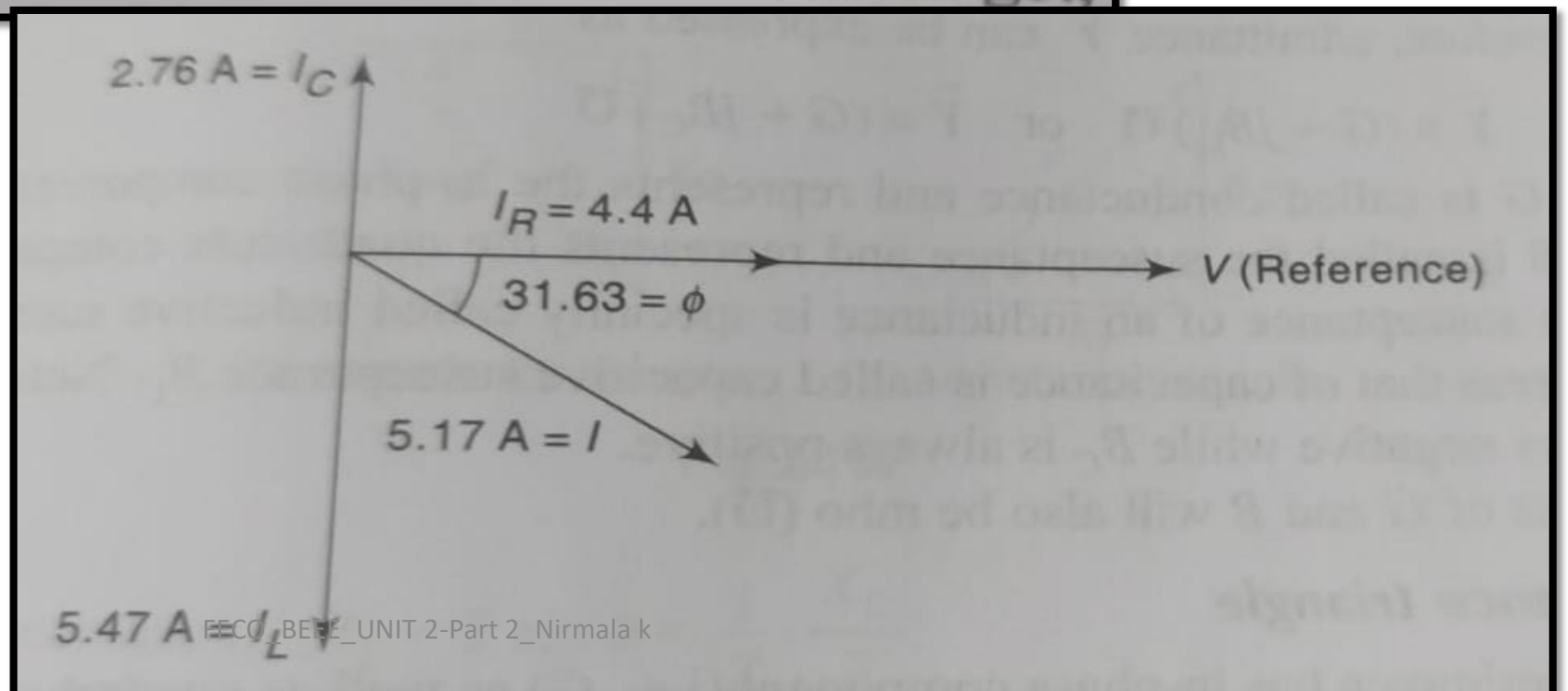
We have, $\bar{I}_R = (4.4 \angle 0)\ A$

$\bar{I}_L = (5.47 \angle -90)\ A$

$\bar{I}_C = (2.76 \angle 90)\ A$

$I = (5.17 \angle -31.63)\ A$

Taking scale, 1 cm = 1 A; and taking V as reference, we get,



Admittance Y

2.3.3 Admittance (Y)

The admittance is defined as the reciprocal of impedance, i.e.,

$$\text{Admittance, } Y = \frac{1}{Z} = \frac{I}{V} \quad \text{or} \quad \bar{Y} = \frac{1}{\bar{Z}} = \frac{\bar{I}}{\bar{V}}$$

The unit of admittance is mho (ohm spelt backward) and its symbol is ℧. The admittance of the circuit may be considered as a measure of the ease with which a circuit can conduct alternating current. Thus, a circuit with higher value of admittance will have a higher value of current.

Consider a parallel circuit as shown in Fig. 2.148.

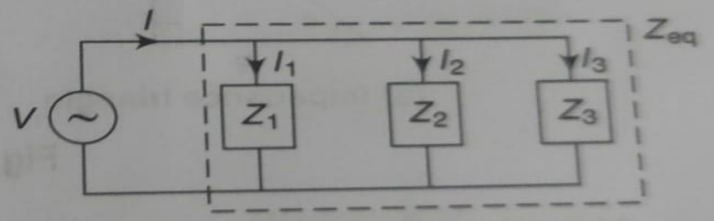


Fig. 2.148 Parallel circuit

By law of parallel circuit,

$$\frac{1}{\bar{Z}_{eq}} = \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \frac{1}{\bar{Z}_3}$$

Since admittance is reciprocal of impedance, we have

$$\bar{Y}_{eq} = \bar{Y}_1 + \bar{Y}_2 + \bar{Y}_3$$

where Y_1, Y_2, and Y_3 are the individual admittances of the parallel branches and Y_{eq} is the total or equivalent admittance of the circuit.

So, Line current, $\bar{I} = \dfrac{\bar{V}}{\bar{Z}_{eq}} = \bar{V}\,\bar{Y}_{eq}$

Components of admittance

The impedance of the circuit can be expressed in the complex form as $\bar{Z} = (R + jX_L)\ \Omega$ or $\bar{Z} = (R - jX_C)\ \Omega$ depending upon the nature of reactance, where R is the resistance or in phase component of Z while X_L or X_C is the reactive or quadrature component of Z. The reciprocal of impedance (i.e., admittance) will also have a complex form.

Therefore, admittance $\bar{Y}$ can be expressed as

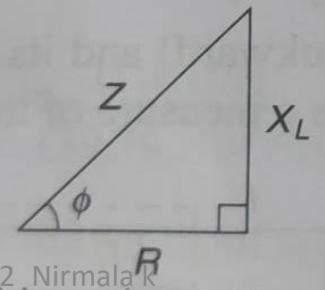
$$\bar{Y} = (G - jB_L)\,\mho \quad \text{or} \quad \bar{Y} = (G + jB_C)\,\mho$$

where G is called conductance and represents the in-phase component of $\bar{Y}$, while B is called the susceptance and represents the quadrature component of $\bar{Y}$. The susceptance of an inductance is specially called inductive susceptance B_L whereas that of capacitance is called capacitive susceptance B_C. Note that B_L is always negative while B_C is always positive. The units of G and B will also be mho ($\mho$).

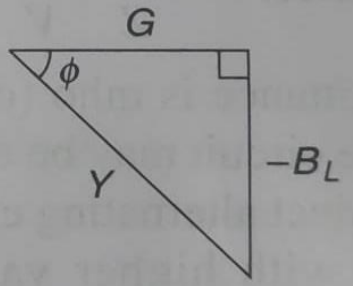
Admittance triangle

Since admittance has in-phase component (i.e., G) as well as quadrature component (i.e., B_L or B_C), it can be represented by a right-angled triangle, called admittance triangle.

(i) For an inductive circuit, the impedance and admittance triangle are shown in Fig. 2.149.



(a) Impedance triangle



(b) Admittance triangle

Note that admittance angle is equal to the impedance angle but is negative.

Conductance, $G = Y\cos\phi = \frac{1}{Z} \times \frac{R}{Z}$

So, $G = \frac{R}{Z^2} = \frac{R}{R^2 + X_L^2}$ ℧

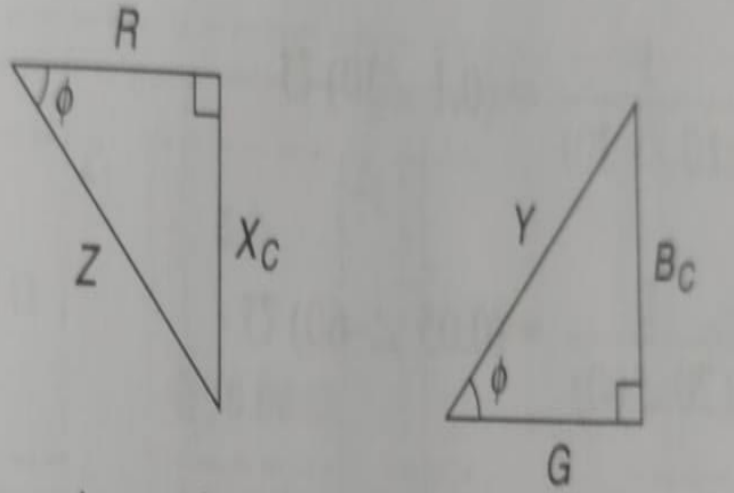
Susceptance, $B_L = Y\sin\phi = \frac{1}{Z} \times \frac{X_L}{Z}$

So, $B_L = \frac{X_L}{Z^2} = \frac{X_L}{R^2 + X_L^2}$ ℧

(ii) For a capacitive circuit, the impedance and admittance triangle are shown in Fig. 2.150.

Conductance, $G = Y\cos\phi = \frac{1}{Z} \times \frac{R}{Z}$

So, $G = \frac{R}{Z^2} = \frac{R}{R^2 + X_C^2}$ ℧



(a) Impedance triangle (b) Admittance triangle

Fig. 2.150

Susceptance, $B_C = Y\sin\phi = \frac{1}{Z} \cdot \frac{X_C}{Z}$

So, $B_C = \frac{X_C}{Z^2} = \frac{X_C}{R^2 + X_C^2}$ ℧

Example 2.82 Calculate the admittance ($\bar{Y}$) and draw the admittance triangle of the circuit shown in Fig. 2.151.

Solution

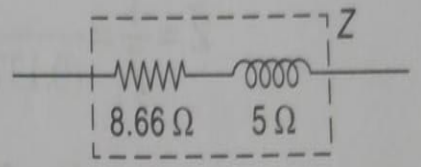


Fig. 2.151

Circuit impedance, $\bar{Z} = (8.66 + j5)\ \Omega$

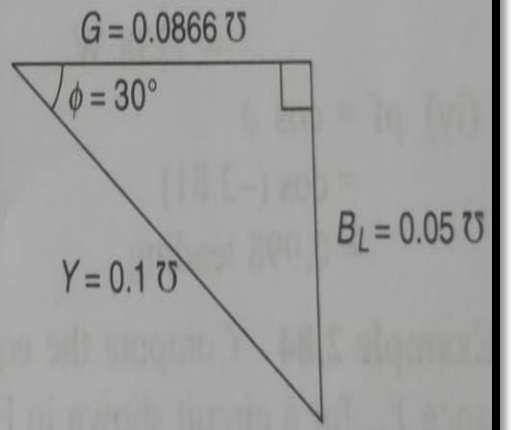
$= (10 \angle 30)\ \Omega$

So, Circuit admittance, $\bar{Y} = \dfrac{1}{\bar{Z}}$

$= \dfrac{1}{(10 \angle 30)}$

$= (0.1 \angle -30)\ \mho$

$= (0.0866 - j0.05)\ \mho$



From polar form of admittance, $Y = 0.1\ \mho$ and $\phi = -30°$.

From rectangular form of admittance,

$G = 0.0866\ \mho$ and $B_L = 0.05\ \mho$

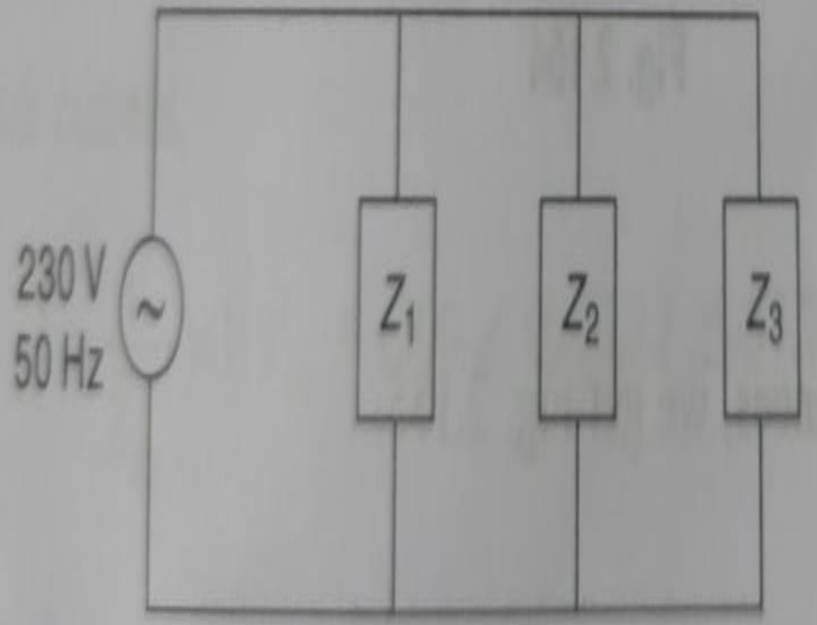
Fig. 2.152

The admittance triangle is shown in Fig. 2.152.

Example 2.83 Three loads are placed across 230 V, 50 Hz supply. The are $(10 \angle -30)\ \Omega$; $(20 \angle 60)\ \Omega$ and $(40 \angle 0)\ \Omega$. Determine (i) admittance, (ii) lent impedance, (iii) power consumed, and (iv) power factor.

Solution

The conditions in the example are shown in Fig. 2.153.



Let

$\bar{Z}_1 = (10 \angle -30)\Omega$

$\bar{Z}_2 = (20 \angle 60)\Omega$

$\bar{Z}_3 = (40 \angle 0)\Omega$

(i) $\bar{Y}_1 = \frac{1}{\bar{Z}_1} = \frac{1}{(10 \angle -30)} = (0.1 \angle 30)$ ℧

$\bar{Y}_2 = \frac{1}{\bar{Z}_2} = \frac{1}{(20 \angle 60)} = (0.05 \angle -60)$ ℧

$\bar{Y}_3 = \frac{1}{\bar{Z}_3} = \frac{1}{(40 \angle 0)} = (0.025 \angle 0)$ ℧

The total admittance of the circuit,

$$\bar{Y} = \bar{Y}_1 + \bar{Y}_2 + \bar{Y}_3$$
$$= (0.1 \angle 30) + (0.05 \angle -60) + (0.025 \angle 0)$$
$$= (0.0866 + j0.05) + (0.025 - j0.0433) + (0.025 + j0)$$
$$= (0.1366 + j0.0067)$$ ℧
$$= (0.137 \angle 2.81)$$ ℧

(ii) Equivalent impedance of the circuit,

$$\bar{Z} = \frac{1}{\bar{Y}} = \frac{1}{(0.137 \angle 2.81)} = (7.3 \angle -2.81)\ \Omega$$

(iii) Total current, $I = \frac{V}{Z} = \frac{230}{7.3} = 31.51$ A

Power consumed by the circuit,

$$P = VI \cos \phi$$
$$= 230 \times 31.51 \times \cos(-2.81)$$
$$= 7238 \text{ W}$$

(iv) $pf = \cos \phi$
$= \cos(-2.81)$
$= 0.998$ leading

Example 2.84 Compute the equivalent impedance Z_{eq} and equivalent admittance Y_{eq} for a circuit shown in Fig. 2.154. Also calculate the total current.

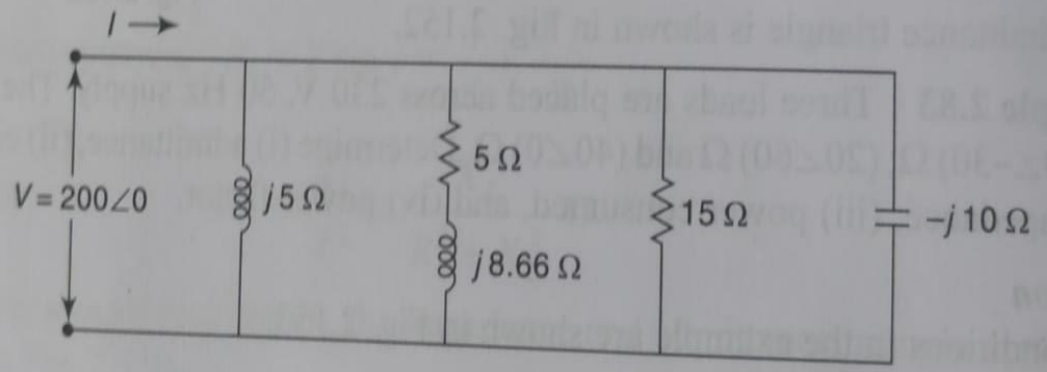


Fig. 2.154

Solution

Marking the various impedances, we get Fig. 2.155.

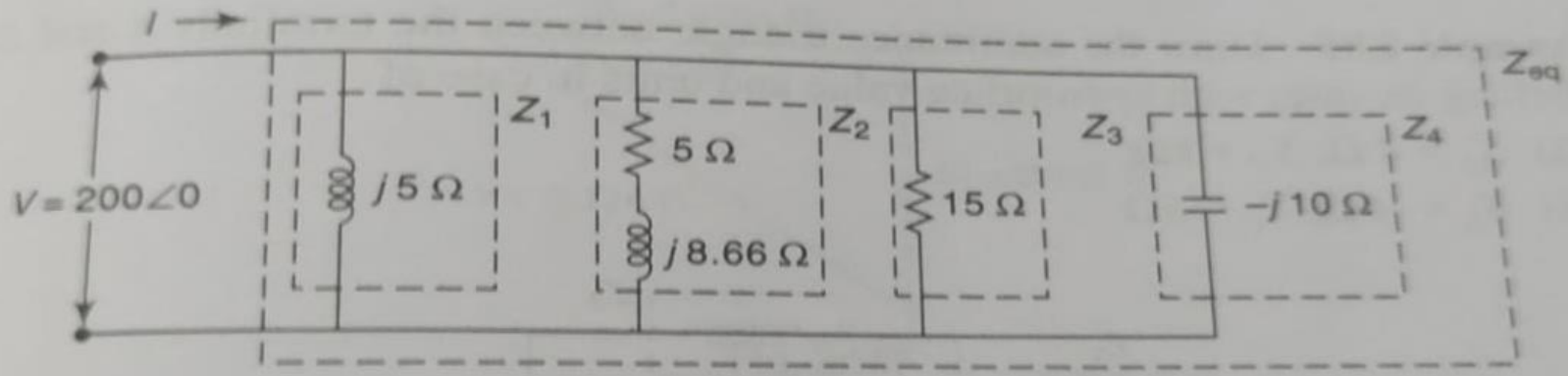


Fig. 2.155

Given: $\bar{Z}_1 = (0 + j5)\ \Omega = (5\ \angle 90)\ \Omega$

$\bar{Z}_2 = (5 + j8.66)\ \Omega = (10\ \angle 60)\ \Omega$

$\bar{Z}_3 = (15 + j0)\ \Omega = (15\ \angle 0)\ \Omega$

$\bar{Z}_4 = (0 - j10)\ \Omega = (10\ \angle -90)\ \Omega$

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The individual admittances of the parallel branches are:

$$\bar{Y}_1 = \frac{1}{\bar{Z}_1} = \frac{1}{(5\angle 90)} = (0.2\angle -90)\ \text{℧}$$

$$\bar{Y}_2 = \frac{1}{\bar{Z}_2} = \frac{1}{(10\angle 60)} = (0.1\angle -60)\ \text{℧}$$

$$\bar{Y}_3 = \frac{1}{\bar{Z}_3} = \frac{1}{(15\angle 0)} = (0.0667\angle 0)\ \text{℧}$$

$$\bar{Y}_4 = \frac{1}{\bar{Z}_4} = \frac{1}{(10\angle -90)} = (0.1\angle 90)\ \text{℧}$$

The equivalent admittance of the circuit,

$$\bar{Y}_{eq} = \bar{Y}_1 + \bar{Y}_2 + \bar{Y}_3 + \bar{Y}_4$$

$$= (0.2\angle -90) + (0.1\angle -60) + (0.0677\angle 0) + (0.1\angle 90)$$

$$= (0 - j0.2) + (0.05 - j0.0866) + (0.0677 + j0) + (0 + j0.1)$$

$$= (0.1177 - j0.1866)\ \text{℧}$$

$$= (0.22\angle -57.76)\ \text{℧}$$

The equivalent impedance of the circuit,

$$\bar{Z}_{eq} = \frac{1}{\bar{Y}_{eq}} = \frac{1}{(0.22\angle -57.76)} = (4.545\angle 57.76)\ \Omega$$

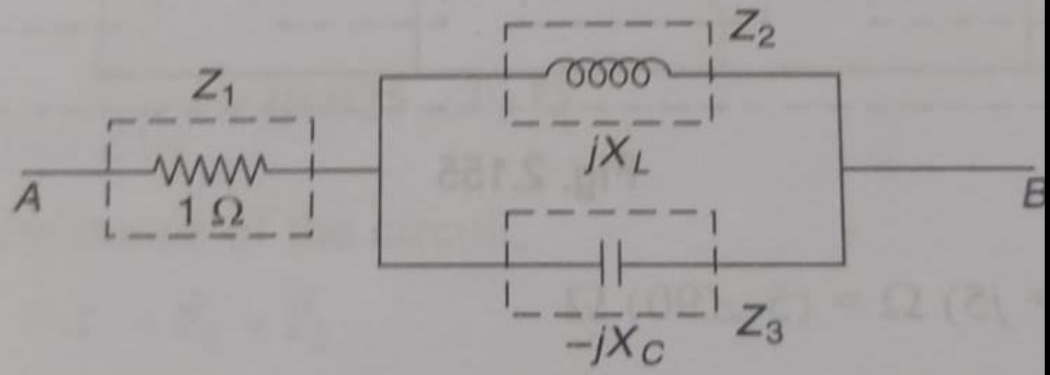
The total current,

$$\bar{I} = \frac{\bar{V}}{\bar{Z}_{eq}} = \bar{V}\bar{Y}_{eq} = (200\angle 0)(0.22\angle -57.76) = (44\angle -57.76)\ \text{A}$$

Example 2.85 Draw the admittance triangle between the terminals A and B, labelling its sides with appropriate value and units in case of

(i) $X_L = 4\ \Omega, X_C = 8\Omega$

(ii) $X_L = 10\Omega, X_C = 5\Omega$



Solution

Case (i) When $X_L = 4\ \Omega$ and $X_C = 8\ \Omega$

$$\bar{Z}_1 = (1 + j0)\ \Omega = (1\ \angle 0)\ \Omega$$

$$\bar{Z}_2 = (0 + j4)\ \Omega = (4\ \angle 90)\ \Omega$$

$$\bar{Z}_3 = (0 - j8)\ \Omega = (8\ \angle -90)\ \Omega$$

$$\bar{Z}_{eq} = \bar{Z}_1 + \frac{\bar{Z}_2 \bar{Z}_3}{\bar{Z}_2 + \bar{Z}_3}$$

$$= (1 + j0) + \frac{(4\angle 90)(8\angle -90)}{(0 + j4) + (0 - j8}$$

$$= (1 + j0) + \frac{(32\angle 0)}{(0 - j4)}$$

$$= (1 + j0) + \frac{(32\angle 0)}{(4\angle -90)}$$

$$= (1 + j0) + (8\angle 90)$$

$$= (1 + j0) + (0 + j8)$$

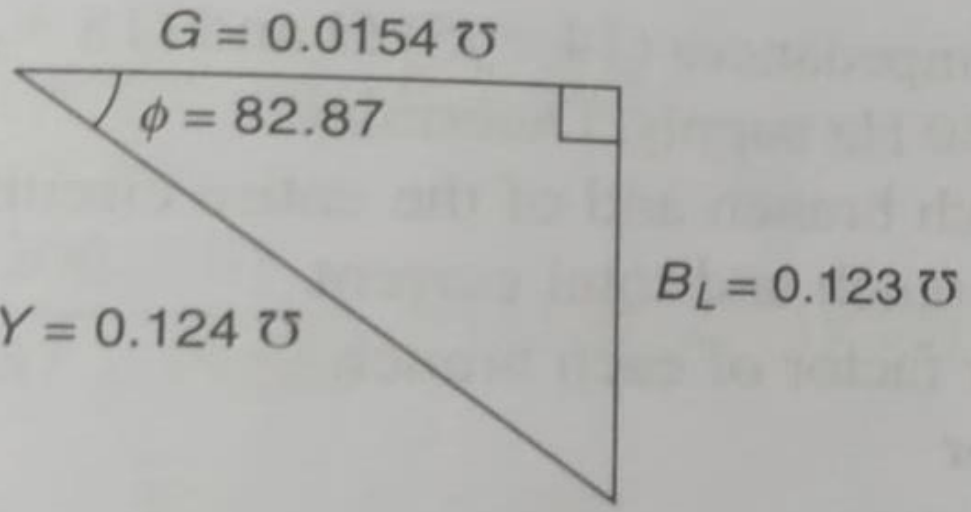
$$= (1 + j8)\ \Omega$$

$$= (8.062\ \angle 82.87)\ \Omega$$

Equivalent admittance, $\bar{Y} = \frac{1}{\bar{Z}_{eq}} = \frac{1}{(8.062\ \angle 82.87)} = (0.124\ \angle -82.87)\ \mho$

$$= (0.0154 - j0.123)\ \mho$$

The admittance triangle is shown in Fig. 2.157.



Equivalent admittance, $\bar{Y}_{eq} = \frac{1}{\bar{Z}_{eq}} = \frac{1}{(10.05 \angle -84.29)} = (0.0995 \angle 84.29)\ ℧$

$= (0.009899 + j0.099)\ ℧$

The admittance triangle is shown in Fig. 2.158.

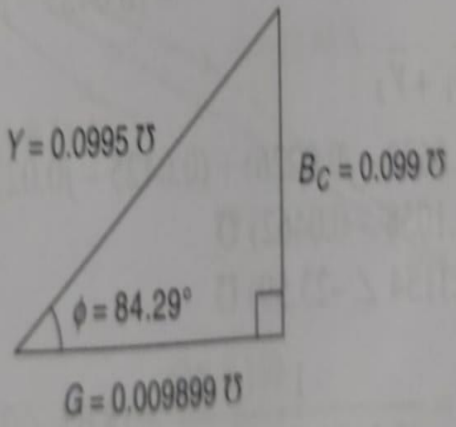


Fig. 2.158

Case (ii) When $X_L = 10\ \Omega$ and $X_C = 5\ \Omega$

We have $\bar{Z}_1 = (1 + j0)\ \Omega = (1 \angle 0)\ \Omega$

$\bar{Z}_2 = (0 + j10)\ \Omega = (10 \angle 90)\ \Omega$

$\bar{Z}_3 = (0 - j5)\ \Omega = (5 \angle -90)\ \Omega$

$$\bar{Z}_{eq} = \bar{Z}_1 + \frac{\bar{Z}_2 \bar{Z}_3}{\bar{Z}_2 + \bar{Z}_3}$$

$$= (1 + j0) + \frac{(10 \angle 90)(5 \angle -90)}{(0 + j10) + (0 - j5)}$$

$$= (1 + j0) + \frac{(50 \angle 0)}{(0 + j5)}$$

$$= (1 + j0) + \frac{(50 \angle 0)}{(5 \angle 90)}$$

$$= (1 + j0) + (10 \angle -90)$$

$$= (1 + j0) + (0 - j10)$$

$$= (1 - j10)\ \Omega$$

$$= (10.05 \angle -84.29)\ \Omega$$

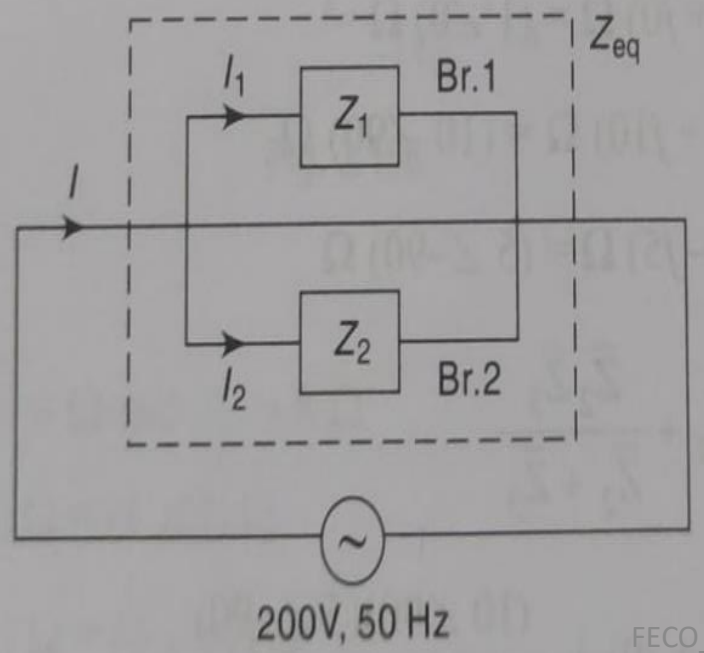
Example 2.86 Two impedances $(14 + j5)\ \Omega$ and $(18 + j10)\ \Omega$ are connected in parallel across 200 V, 50 Hz supply. Determine:

(i) admittance of each branch and of the entire circuit
(ii) current in each branch and total current
(iii) power and power factor of each branch
(iv) total power factor

Also draw phasor diagram.

Solution

The conditions in the example are shown in Fig. 2.159.



Let $\bar{Z}_1 = (14 + j5)\ \Omega = (14.87 \angle 19.65)\ \Omega$

$\bar{Z}_2 = (18 + j10)\ \Omega = (20.59 \angle 29.05)\ \Omega$

Taking V as reference,

$\bar{V} = (200 \angle 0)\ \text{V}$

(i) Admittance of branch 1,

$$\bar{Y}_1 = \frac{1}{\bar{Z}_1} = \frac{1}{(14.87 \angle 19.65)} = (0.0672 \angle -19.65)\ \text{℧}$$

$$= (0.0633 - j0.0226)\ \text{℧}$$

Admittance of branch 2,

$$\bar{Y}_2 = \frac{1}{\bar{Z}_2} = \frac{1}{(20.59 \angle 29.05)} = (0.0486 \angle -29.05)\ \text{℧}$$

$$= (0.0425 - j0.0236)\ \text{℧}$$

Now, $\bar{Y}_{eq} = \bar{Y}_1 + \bar{Y}_2$

$= (0.0633 - j0.0226) + (0.0425 - j0.0236)$
$= (0.1058 - 0.0462)\ \text{℧}$
$= (0.1154 \angle -23.59)\ \text{℧}$

Also $\bar{Z}_{eq} = \frac{1}{\bar{Y}_{eq}} = \frac{1}{(0.1154 \angle -23.59)} = (8.67 \angle 23.59)\ \Omega$

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(ii) Branch current, $\bar{I}_1 = \dfrac{\bar{V}}{\bar{Z}_1}$ or $\bar{I}_1 = \bar{V}\,\bar{Y}_1$

So, $\bar{I}_1 = \dfrac{(200 \angle 0)}{(14.87 \angle 19.65)} = (13.45 \angle -19.65)\ \text{A}$

Branch current, $\bar{I}_2 = \dfrac{\bar{V}}{\bar{Z}_2}$ or $\bar{I}_2 = \bar{V}\,\bar{Y}_2$

So, $\bar{I}_2 = \dfrac{(200 \angle 0)}{(20.59 \angle 29.05)} = (9.71 \angle -29.05)\ \text{A}$

Total current, $\bar{I} = \dfrac{\bar{V}}{\bar{Z}_{eq}}$ or $\bar{I} = \bar{Y}\,\bar{V}$ or $\bar{I} = \bar{I}_1 + \bar{I}_2$

So, $\bar{I} = \dfrac{(200 \angle 0)}{(8.67 \angle 23.59)} = (23.07 \angle -23.59)\ \text{A}$

(iii) Power consumed by branch 1,

$P_1 = I_1^2 R_1 = (13.45)^2 \times 14 = 2.533\ \text{kW}$

Power consumed by branch 2,

$P_2 = I_2^2 R_2 = (9.71)^2 \times 18 = 1.697\ \text{kW}$

Power factor of branch 1, $(\text{pf})_1 = \cos \phi_1 = \cos (19.65) = 0.942$ lagging

Power factor of branch 2, $(\text{pf})_2 = \cos \phi_2 = \cos (29.05) = 0.874$ lagging

(iv) Total power factor, $(\text{pf}) = \cos \phi = \cos (23.59) = 0.9164$ lagging

(v) Phasor diagram:

We know that, $\bar{I}_1 = (13.45 \angle -19.65)\ \text{A}$

$\bar{I}_2 = (9.71 \angle -29.05)\ \text{A}$

Scale: 1 cm = 2 A

Taking V as reference phasor,

$\bar{I} = \bar{I}_1 + \bar{I}_2$

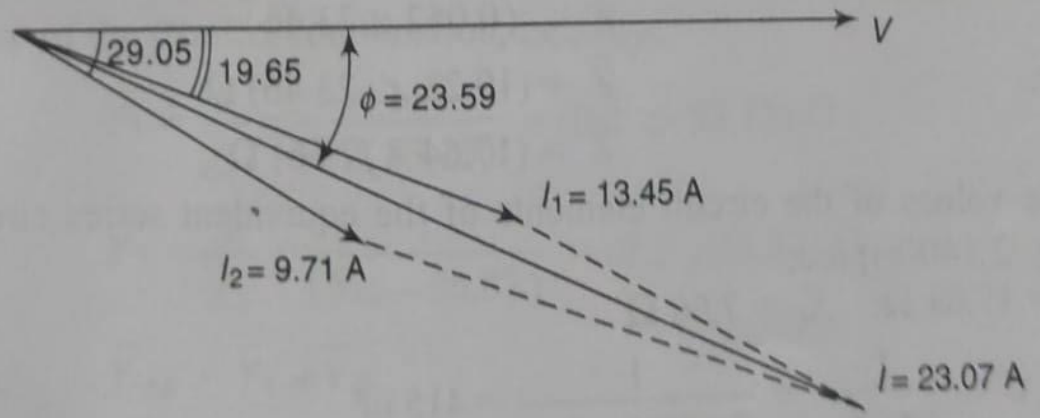


Fig. 2.160

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Example 2.87 Two parallel circuits comprise respectively (a) a coil of resistance 20 Ω and inductance 0.08 H (b) a capacitor of capacitance 200 μF in series with e resistance of 10 Ω. If the circuit is connected to 230 V, 50 Hz supply, determine

(i) supply current and power factor
(ii) equivalent series circuit
(iii) the value and nature of reactance to be connected in series with the circuit to bring the overall power factor to unity.

Solution

The conditions in the example are shown in Fig. 2.140(a).

$$L = 0.08 \text{ H}; X_L = 2\pi fL = 2\pi \times 50 \times 0.08 = 25.13 \ \Omega$$

$$C = 200 \times 10^{-6} \text{ F}; X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 200 \times 10^{-6}} = 15.92 \ \Omega$$

$$\bar{Z}_1 = (20 + j25.13) \ \Omega = (32.12 \angle 51.49) \ \Omega$$

$$\bar{Z}_2 = (10 - j15.92) \ \Omega = (18.8 \angle -57.87) \ \Omega$$

$$\bar{Y}_1 = \frac{1}{\bar{Z}_1} = \frac{1}{(32.12 \angle 51.49)} = (0.0311 \angle -51.49) \ \text{℧}$$

$$\bar{Y}_2 = \frac{1}{\bar{Z}_2} = \frac{1}{(18.8 \angle -57.87)} = (0.0532 \angle 57.87) \ \text{℧}$$

So, Total admittance,

$$\bar{Y} = \bar{Y}_1 + \bar{Y}_2$$
$$= (0.0311 \angle -51.49) + (0.0532 \angle 57.87)$$
$$= (0.0194 - j0.0243) + (0.0283 + j0.045)$$
$$= (0.0477 + j0.0207) \ \text{℧}$$
$$= (0.052 + \angle 23.46) \ \text{℧}$$

(i) Now, supply current, $\bar{I} = \bar{V}\bar{Y} = (230\angle 0)(0.052\angle 23.46)$

$= (11.96\angle 23.46)$ A

Power factor, pf $= \cos\phi$

$= \cos 23.46$

$= 0.917$ lead

(ii) Total impedance, $\bar{Z} = \dfrac{1}{\bar{Y}} = \dfrac{1}{(0.052\angle 23.46)}$

So, $\bar{Z} = (19.23\angle -23.46)\ \Omega$

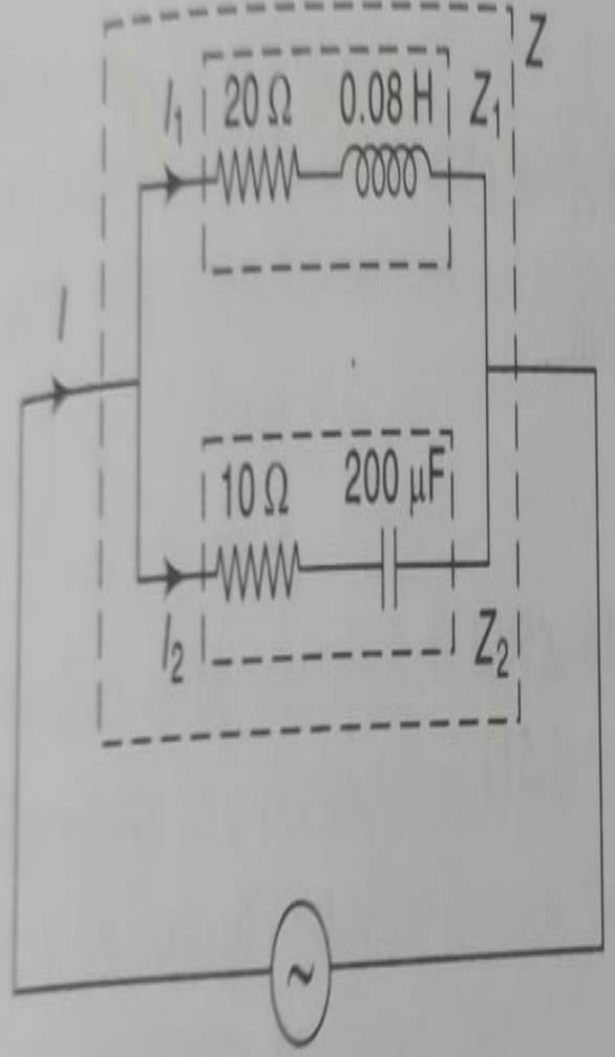
or $\bar{Z} = (17.64 - j7.66)\ \Omega$

The values of the circuit elements of the equivalent series circuit [see Fig. 2.140(b)] are:

$R = 17.64\ \Omega$; $X_C = 7.66\ \Omega$

So, $C = \dfrac{1}{2\pi f X_C} = \dfrac{1}{2\pi \times 50 \times 7.66} = 415\ \mu$F

(iii) In order that the circuit power factor is unity, the reactance of the circuit should be zero. To do so, we would connect an inductive reactance $X_L = 7.66\ \Omega$ in series with the circuit.



$V = 230$ V, 50 Hz

(a)

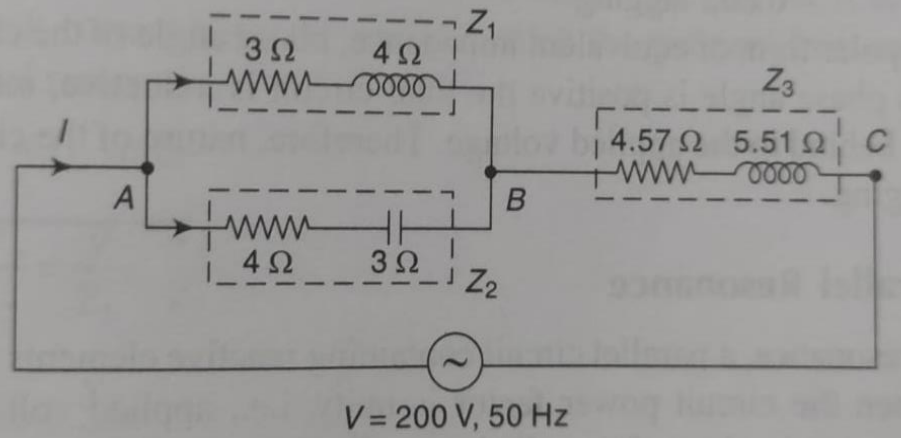


$R = 17.64\ \Omega$ $X_C = 7.66\ \Omega$

Equivalent circuit

(b)

Example 2.88 Applying admittance method to series-parallel circuit shown in Fig. 2.162. Find: (i) total impedance, (ii) supply current, and (iii) pf of entire circuit.



Solution

Given: $\bar{Z}_1 = (3 + j4)\ \Omega = (5\ \angle 53.13)\ \Omega$

$\bar{Z}_2 = (4 - j3)\ \Omega = (5\ \angle -36.87)\ \Omega$

$\bar{Z}_3 = (4.57 + j5.51)\ \Omega = (7.16\ \angle 50.33)\ \Omega$

(i) It is a series–parallel circuit. We will first calculate the equivalent impedance of the parallel combination, i.e., Z_{AB}.

$$\bar{Y}_1 = \frac{1}{\bar{Z}_1} = \frac{1}{(5\ \angle 53.13)} = (0.2\ \angle -53.13)\ \text{℧}$$

$$\bar{Y}_2 = \frac{1}{\bar{Z}_2} = \frac{1}{(5\ \angle -36.87)} = (0.2\ \angle 36.87)\ \text{℧}$$

So, $\bar{Y}_{AB} = \bar{Y}_1 + \bar{Y}_2$

$= (0.2\ \angle -53.13) + (0.2\ \angle 36.87)$

$= (0.12 - j0.16) + (0.16 + j0.12)$

$= (0.28 - j0.04)\ \text{℧}$

$= (0.2828\ \angle -8.13)\ \text{℧}$

Now, $\bar{Z}_{AB} = \frac{1}{\bar{Y}_{AB}} = \frac{1}{(0.2828\ \angle -8.13)} = (3.54\ \angle 8.13)\ \Omega$

Equivalent impedance of the circuit,

$\bar{Z} = \bar{Z}_{AB} + \bar{Z}_{BC}$

$= (3.54\ \angle 8.13) + (7.16\ \angle 50.33)$

$= (3.5 + j0.5) + (4.57 + j5.51)$

$= (8.07 + j6.01)\ \Omega$

$= (10.06\ \angle 36.68)\ \Omega$

(ii) Supply current, $\bar{I} = \frac{\bar{V}}{\bar{Z}} = \frac{(200\ \angle 0)}{(10.06\ \angle 36.68)} = (19.88\ \angle -36.68)\ \text{A}$

(iii) Power factor of the entire circuit,

$\text{pf} = \cos\phi$

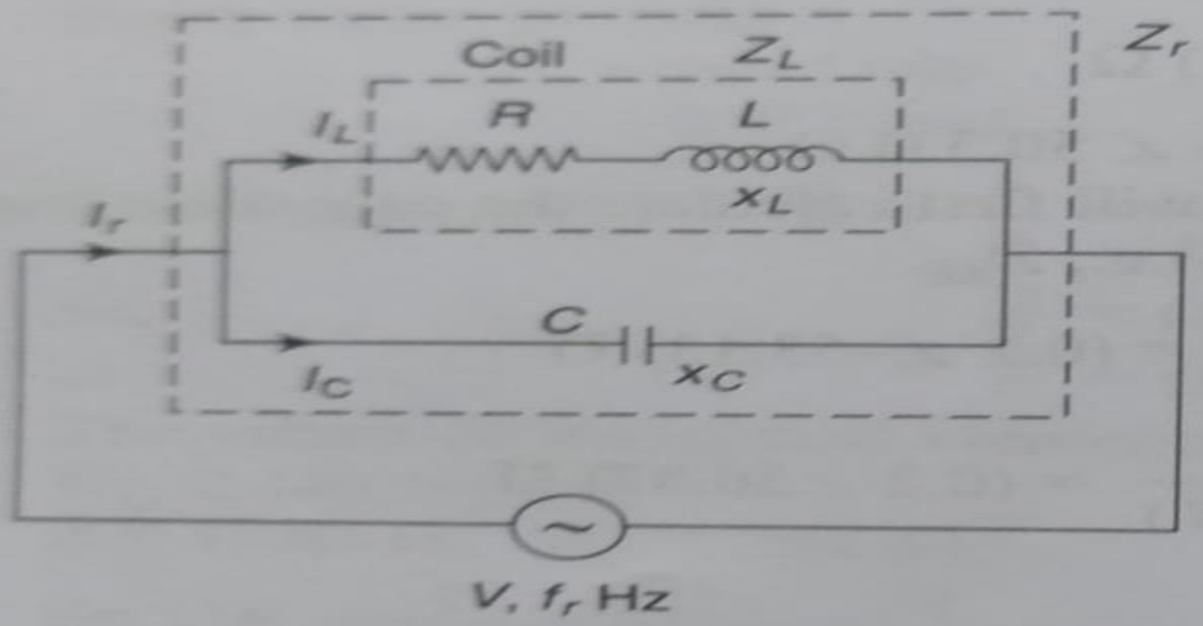
$= \cos 36.68$

$= 0.802$ lagging

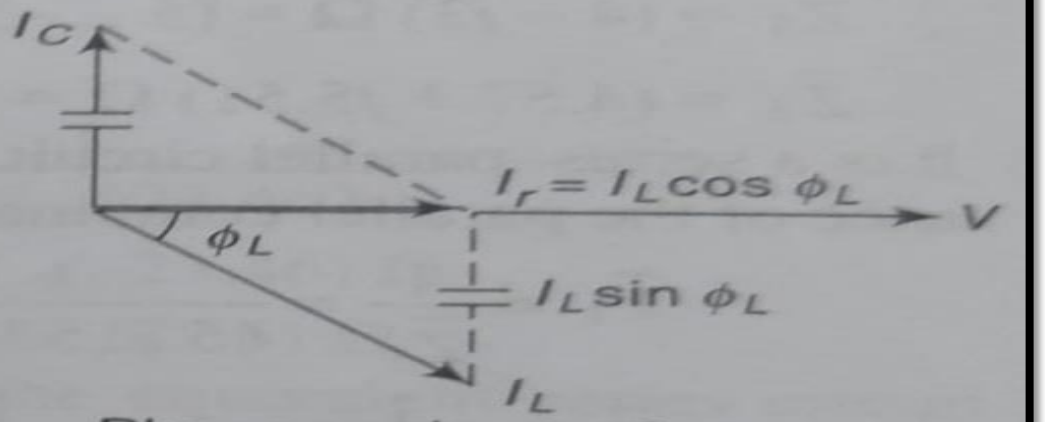
Note: From polar form of equivalent impedance, phase angle of the circuit, i.e., $\phi = 36.68$. As phase angle is positive the total circuit is inductive, means supply current lags behind to the applied voltage. Therefore, nature of the circuit power factor is lagging.

2.3.4 Parallel Resonance

Like series resonance, a parallel circuit containing reactive elements (L and C) is resonant when the circuit power factor is unity, i.e., applied voltage and the supply current are in phase. It is called a parallel resonance because it concerns a parallel circuit. The most common resonant parallel circuit is an inductor (coil) in parallel with a pure capacitor C as shown in Fig. 2.163(a). The phasor diagram of this parallel circuit is shown in Fig. 2.163(b). The coil current I_L has two



(a) Circuit diagram



Phase angle, $\phi = 0$
$pf = 1$ (unity)
$I_C = I_L \sin \phi_L$
and $I = I_L \cos \phi_L$

(b) Phasor diagram (when $I_C = I_L \sin \phi_L$)

components, viz active component $I_L \cos \phi_L$ and reactive component $I_L \sin \phi_L$. This parallel circuit will resonant when the circuit power factor is unity. This is possible only when the net reactive component of the circuit is zero,

i.e., $I_C - I_L \sin \phi_L = 0$

or $I_C = I_L \sin \phi_L$

or Reactive component of I_C = Reactive component of I_L

Resonance in a parallel circuit can be achieved by changing the supply frequency. Increase of frequency increase the reactance of the coil and consequently the coil impedance increases. The coil current I_L, therefore, decreases and lags behind the applied voltage V by a progressively greater angle. The capacitive branch current on the other hand, increases although it will always lead V by 90°. At some frequency f_r (called resonant frequency), I_C becomes equal to $I_L \sin \phi_L$ and resonance occurs.

$$(2\pi f_r L)^2 = \frac{L}{C} - R^2$$

$$f_r = \frac{1}{2\pi L}\sqrt{\frac{L}{C} - R^2}$$

$$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

(i) Resonance/resonant frequency (f_r)

The supply frequency at which parallel resonance occurs (i.e., reactive component of circuit current becomes zero) is called the resonant frequency.

At parallel resonance,

$$I_C = I_L \sin \phi_L$$

$$\frac{V}{X_C} = \frac{V}{Z_L} \cdot \frac{X_L}{Z_L}$$

$$\frac{1}{X_C} = \frac{X_L}{Z_L^2}$$

$$Z_L^2 = \frac{L}{C}$$

$$R^2 + X_L^2 = \frac{L}{C}$$

$$R^2 + (2\pi f_r L)^2 = \frac{L}{C}$$

(2.2

If R is very very small,

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

(ii) Impedance at resonance (Z_r)

At parallel resonance,

$$I = I_L \cos \phi_L$$

or $$\frac{V}{Z_r} = \frac{V}{Z_L} \cdot \frac{R}{Z_L}$$

or $$\frac{1}{Z_r} = \frac{R}{Z_L^2}$$

or $$Z_r = \frac{Z_L^2}{R}$$

From Eq. (2.25), $Z_L^2 = \frac{L}{C}$

So, Circuit impedance, $Z_r = \frac{L}{CR}\Omega$ (2.28)

Thus, at parallel resonance, the circuit impedance is equal to L/CR. This is known as equivalent or dynamic impedance of the parallel circuit at resonance.

Z_r is the pure resistance because in the expression of Z_r, frequency term is not present. Also Z_r is very high because the ratio L/C is very large at parallel resonance.

As supply frequency changes, value of circuit impedance also varies. If we plot impedance–frequency graph for a parallel circuit shown in Fig. 2.163(a), the shape of the curve will be as shown in Fig. 2.164. Note that impedance of the circuit is maximum at resonance. As the frequency changes from resonance, the circuit impedance decreases very rapidly.

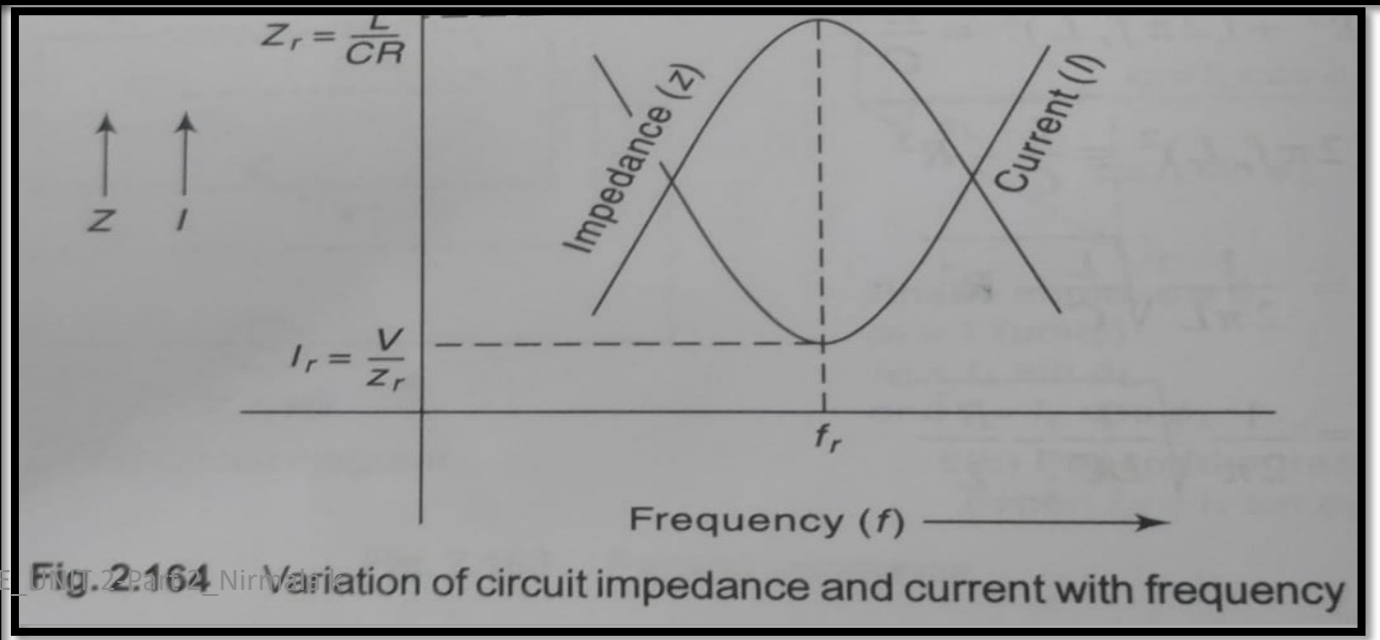


Fig. 2.164 Variation of circuit impedance and current with frequency

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(iii) Circuit current at resonance (I_r)

At parallel resonance, the circuit impedance (Z_r) becomes maximum and the circuit current (I_r) is at minimum value. The current–frequency curve of the parallel circuit is shown in Fig. 2.143. Note that the value of line current is minimum at resonance.

At parallel resonance, the impedance of each branch (i.e., X_C and Z_L) is relatively small compared with circuit impedance (Z_r). Therefore, current flowing through the capacitor and coil are much greater than the line current I_r. But these currents are approximately 180° out of phase, therefore resultant current I_r is extremely small.

(iv) Q-factor of parallel resonance circuit

At parallel resonance, the current circulating between the two branches is many times greater than the line current. The current amplification produced by the resonance is termed as Q-factor of the parallel resonant circuit, i.e.,

$$Q\text{-factor} = \frac{I_L \text{ or } I_C}{I_r} \quad (\because I_L \cong I_C) \tag{2.29}$$

Let $$Q\text{-factor} = \frac{I_C}{I_r}$$

Now, $$I_C = \frac{V}{X_C} = 2\pi f_r CV \quad \text{and} \quad I_r = \frac{V}{(L/CR)}$$

So, $$Q\text{-factor} = \frac{2\pi f_r CV}{\frac{V}{\left(\frac{L}{CR}\right)}}$$

$$= \frac{2\pi f_r L}{R}$$

Now, $$f_r = \frac{1}{2\pi\sqrt{LC}}$$

By substituting the value of f_r in Eq. (2.30), we get

$$Q\text{-factor} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

2.3.5 Comparison of Series and Parallel Resonant Circuits

Particular	Series circuit	Parallel circuit
1. Impedance at resonance	Minimum ($Z_r = R$)	Maximum ($Z_r = L/CR$)
2. Current at resonance	Maximum $\left(I_r = \frac{V}{R}\right)$	Minimum $\left(I_r = \frac{V}{Z_r}\right)$
3. Resonant frequency	$f_r = \frac{1}{2\pi\sqrt{LC}}$ Hz	$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$ Hz
4. Q-Factor	$= \frac{V_L \text{ or } V_C}{V}$	$= \frac{I_L \text{ or } I_C}{I}$
5. It magnifies	Voltage	Current
6. When $f < f_r$	Circuit is capacitive.	Circuit is inductive.
7. When $f > f_r$	Circuit is inductive.	Circuit is capacitive.

FECO_BEEE_UNIT 2-Part 2_Nirmala R

Example 2.89 A parallel circuit consists of a 2.5 μF capacitor and a coil whose resistance and inductance are 15 Ω and 260 mH, respectively. Determine (i) the resonant frequency, (ii) Q-factor of the circuit at resonance, and (iii) dynamic impedance of the circuit.

Solution

(i) Resonant frequency (f_r),

$$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{1}{260\times10^{-3}\times2.5\times10^{-6}} - \frac{(15)^2}{(260\times10^{-3})^2}}$$

$$= 197.19 \text{ Hz}$$

(ii) Q-factor $= \frac{2\pi f_r L}{R} = \frac{2\pi\times197.19\times260\times10^{-3}}{15} = 21.48$

(iii) Dynamic impedance (Z_r),

$$Z_r = \frac{L}{CR} = \frac{260\times10^{-3}}{2.5\times10^{-6}\times15} = 6933.33\ \Omega$$

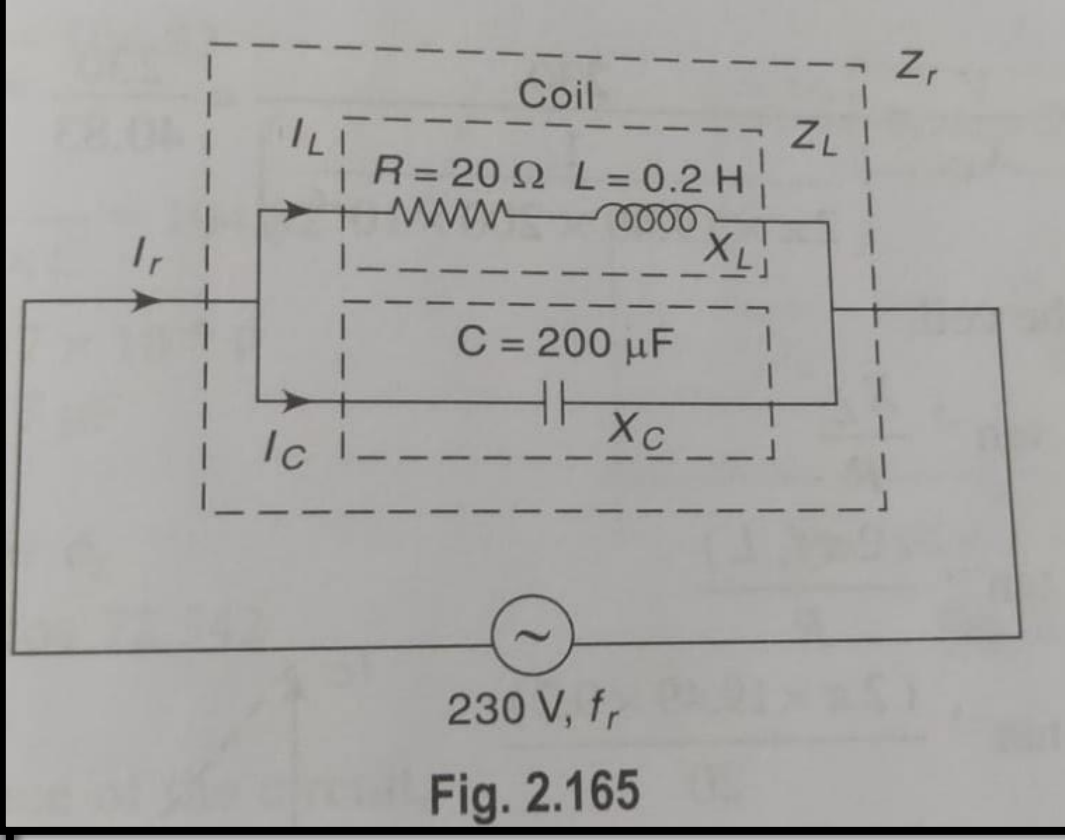


Fig. 2.165

Example 2.90 An inductive coil of resistance 20 Ω and inductance 0.2 H is connected in parallel with 200 μF capacitor with variable frequency, 230 V supply. Find the resonant frequency at which the total current taken from the supply is in phase with supply voltage. Also find the value of this current. Draw the phasor diagram.

Solution

The conditions in the example are shown in Fig. 2.165

FECO_BEEE_UNIT 2-Part 2_Nirmala K

Resonant frequency,

$$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \text{ Hz}$$

$$= \frac{1}{2\pi}\sqrt{\frac{1}{0.2 \times 200 \times 10^{-6}} - \frac{20^2}{(0.2)^2}}$$

$$= \frac{1}{2\pi}\sqrt{25000 - 10000}$$

$$= \frac{1}{2\pi}\sqrt{15000}$$

$$= 19.49 \text{ Hz}$$

Dynamic impedance of the circuit,

$$Z_r = \frac{L}{CR}$$

$$= \frac{0.2}{200 \times 10^{-6} \times 20}$$

$$= 50\ \Omega$$

Circuit current at resonance,

$$I_r = \frac{V}{Z_r} = \frac{230}{50} = 4.6 \text{ A}$$

Phasor diagram (see Fig. 2.145):

For phasor diagram, we need to calculate the values of branch currents (I_L and I_C) and phase angle of the coil (ϕ_L).

Now, $$I_L = \frac{V}{Z_L} = \frac{230}{\sqrt{(20)^2 + (2\pi \times 19.49 \times 0.2)^2}} = \frac{230}{31.62} = 7.274 \text{ A}$$

$$I_C = \frac{V}{X_C} = \frac{230}{\left(\frac{1}{2\pi \times 19.49 \times 200 \times 10^{-6}}\right)} = \frac{230}{40.83} = 5.63 \text{ A}$$

Phase angle of the coil,

$$\phi_L = \tan^{-1}\frac{X_L}{R}$$

$$= \tan^{-1}\frac{(2\pi f_r L)}{R}$$

$$I_C = \frac{V}{X_C} = \frac{230}{\left(\frac{1}{2\pi \times 19.49 \times 200 \times 10^{-6}}\right)} = \frac{230}{40.83} = 5.63 \text{ A}$$

Phase angle of the coil,

$$\phi_L = \tan^{-1}\frac{X_L}{R}$$

$$= \tan^{-1}\frac{(2\pi f_r L)}{R}$$

$$= \tan^{-1}\frac{(2\pi \times 19.49 \times 0.2)}{20}$$

$$= \tan^{-1}\frac{24.49}{20}$$

So, $\phi_L = 50.76$

Take V as reference phasor.

We know that $\bar{I}_r = \bar{I}_L + \bar{I}_C$.

Now, $I_L = 7.274$ A

$I_C = 5.63$ A

$I_r = 4.6$ A

$\phi_L = 50.76°$

Taking scale 1 cm = 1 A, the phasor diagram can be drawn as shown in Fig. 2.166.

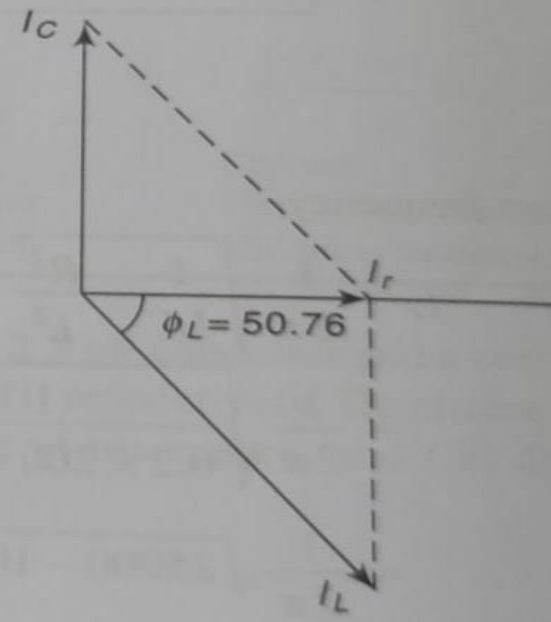


Fig. 2.166

Example 2.91 A coil takes a current of 1 A at 0.3 pf when connected to a 100 V, 50 Hz supply. Determine the value of the capacitance, which when connected in parallel with the coil, will reduce the line current to a minimum. Calculate the impedance of the parallel circuit at 50 Hz.

Solution

When a coil is connected to 100 V, 50 Hz supply, it takes a current of 1 A at 0.3 pf (see Fig. 2.167).

$$(\text{pf})_{\text{coil}} = 0.3$$

or $\cos \phi_L = 0.3$

or $\phi_L = 72.542$ (ϕ_L is the phase angle of the coil)

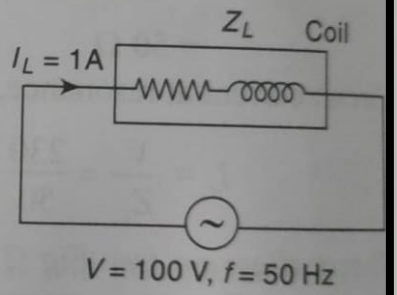


Fig. 2.167

Now, capacitor is connected across the coil (see Fig. 2.168). We need a value of C, so that the line current reduces to a minimum value, i.e., parallel resonance occurs at $f = 50$ Hz.

At resonance, $I_C = I_L \sin \phi_L$

So, $I_C = 1 \times \sin (72.542)$

or $I_C = 0.954$ A

Now, $X_C = \dfrac{V}{I_C} = \dfrac{100}{0.954} = 104.82\ \Omega$

So, $\dfrac{1}{2\pi f_r C} = 104.82$

or $\dfrac{1}{2\pi \times 50 \times C} = 104.82$

or $C = 30.37 \times 10^{-6}$ F

$= 30.37\ \mu$F

At resonance,

$I_r = I_L \cos \phi_L$

or $I_r = 1 \times \cos 72.542$

or $I_r = 0.3$ A

Dynamic impedance of the circuit,

$$Z_r = \frac{V}{I_r} = \frac{100}{0.3} = 333.33\ \Omega$$

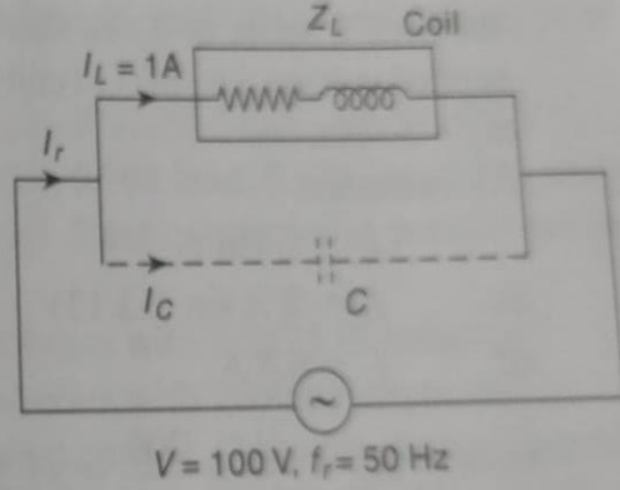


Fig. 2.168

Example 2.92 A circuit has $X_L = 20\ \Omega$ at 50 Hz, its resistance being 15 Ω. For an applied voltage of 200 V at 50 Hz, calculate (i) the pf, (ii) the current, (iii) the value of shunting capacitance to bring the resultant current into phase with the applied voltage, and (iv) the resultant current in case (iii).

Solution

Impedance of the circuit, $Z_L = \sqrt{(15)^2 + (20)^2}$

$= 25\ \Omega$

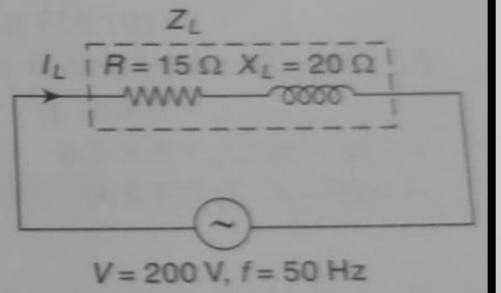


Fig. 2.169

(i) $(\text{pf})_{\text{coil}} = \cos \phi_L = \dfrac{R}{Z_L} = \dfrac{15}{25}$

$= 0.6$ lagging

So, $\phi_L = \cos^{-1} 0.6 = 53.15$

(ii) Current, $I_L = \dfrac{V}{Z_L} = \dfrac{200}{25} = 8$ A

(iii) Now, capacitance is connected in parallel with the above circuit (see Fig. 2.170).

We need a value of C, so that resulting current comes in phase with the applied voltage, i.e., pf = 1 (unity). In other words, parallel resonance occurs at $f = 50$ Hz.

At resonance,

$I_C = I_L \sin \phi_L$

So, $I_C = 8 \times \sin (53.13)$

or $I_C = 6.4$ A

Now, $X_C = \dfrac{V}{I_C} = \dfrac{200}{6.4} = 31.25\ \Omega$

So, $\dfrac{1}{2\pi f_r C} = 31.25$

or $\dfrac{1}{2\pi \times 50 \times C} = 31.25$

So, $C = \dfrac{1}{2\pi \times 50 \times 31.25}$

or $C = 101.859 \times 10^{-6}$ F

$= 101.859\ \mu$F

(ii) Now, capacitance is connected in parallel with the above circuit (see Fig. 2.170).

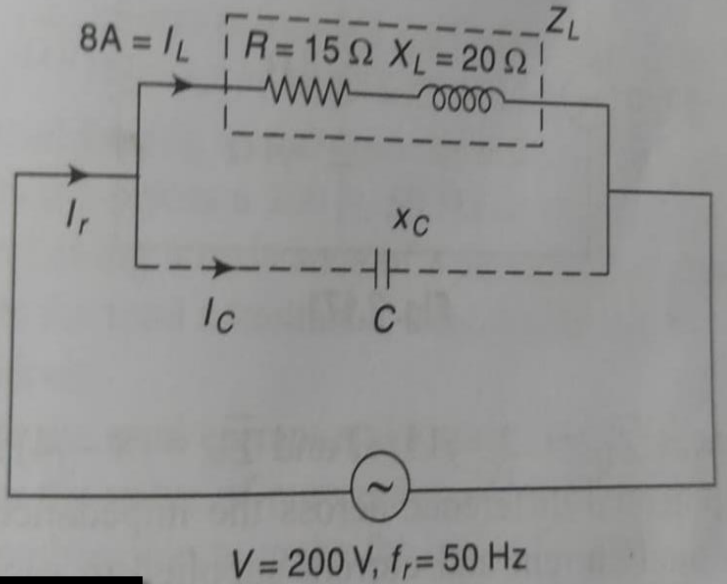


Fig. 2.170

(iv) The circuit current at resonance,

$I_r = I_L \cos \phi_L$

or $I_r = 8 \times 0.6$

or $I_r = 4.8$ A

Concept of Active , reactive and apparent power.

- Power in AC circuits is defined as the amount of energy delivered per unit of time by the circuit. It is used to calculate how much total power is required to run a load.
- In an AC circuit, the power factor is defined as the ratio of real power(P) and apparent power(S).
- The Electrical power is the rate at which energy is consumed inside a circuit. Therefore, all electrical and electronic devices and gadgets have a maximum amount of electrical power they can safely handle.

Types of Power- In a DC circuit, electrical power is described as the multiplication of voltage and current. Mathematically: $P=VI$ (W)

There are three categories - 1.Active power

2.Reactive power

3. Apparent Power

1. Active Power P - Active or genuine or real power refers to the quantity of power that is dissipated or that does productive work in the circuit. In power systems, it is calculated in watts but is more commonly expressed in (kilowatts) and (megawatts).It is equivalent to an average amount of $P=VI\cos\varphi$. (W/KW)

2.Reactive Power Q -Reactive power is the power that comes back and forth between the source and the load. Reactive power, indicated by the letter Q, is the component that is proportionate to

$$Q=VI\sin\varphi \qquad \text{(VA/VAR)}$$

This is a power, but it is not calculated in watts because it is a non-active power, so it is calculated in Volt-Amperes-Reactive (VAR). The load power factor determines whether this reactive power is negative or positive.

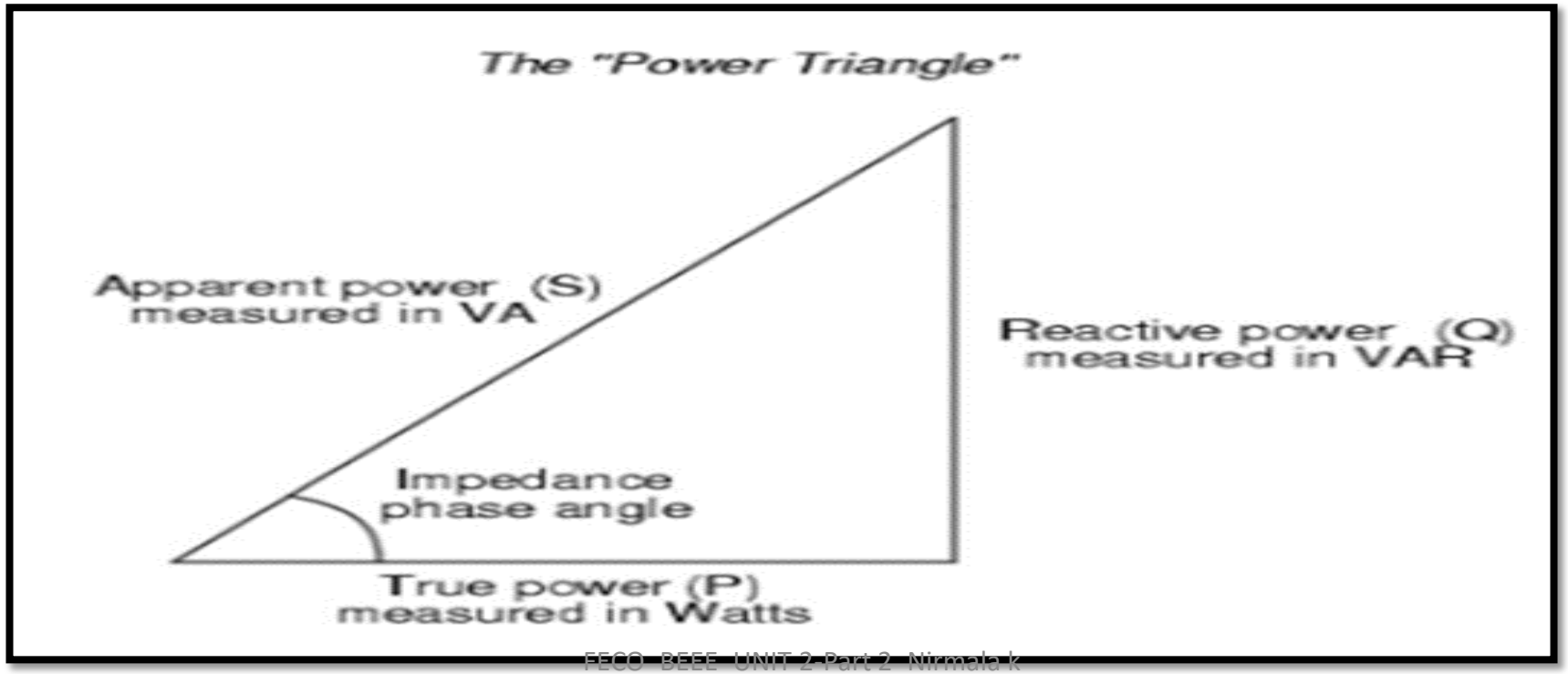
3.Apparent Power S -The term apparent power refers to the complicated combination of true power and reactive power, equal to the multiplication of voltage and current, regardless of phase angle.

The perceived power is helpful in determining the rating of power equipment. It can alternatively be written as the square of current times the impedance /resistance of the circuit.

$$S=P+jQ = I^2Z \qquad \text{(VA/KVA/MVA)}$$

Power Triangle -A power triangle is the relationship between active power, reactive power and apparent power that can be explained by describing numbers like the vector in the geometrical form. Power factor can be expressed as a decimal or as a percentage.

- The total of the squares of two sides (active power and reactive power) equals the squares of the diagonal, according to the Pythagoras theorem (apparent power). i.e., $S^2=P^2+Q^2$, $S = \sqrt{((P^2+Q^2))}$



Power Factor - The ratio of the genuine power running across the circuit to an apparent power existing in the circuit is known as the power factor of the alternating current.

Power Factor = True power/Apparent power=cosΦ = R/Z

Here, R represents resistance and Z represents impedance.

Importance of Power Factor -Power factor is critical because it dictates the quantity of currents flowing through the circuit and, as a result, the wire size that must be used. This factor has an impact on the voltage and current waveforms.

A low Power factor indicates that a great deal of energy has been wasted as heat as a result of inefficient electricity use, which can lead to overheating and equipment failure.

Comparison of power

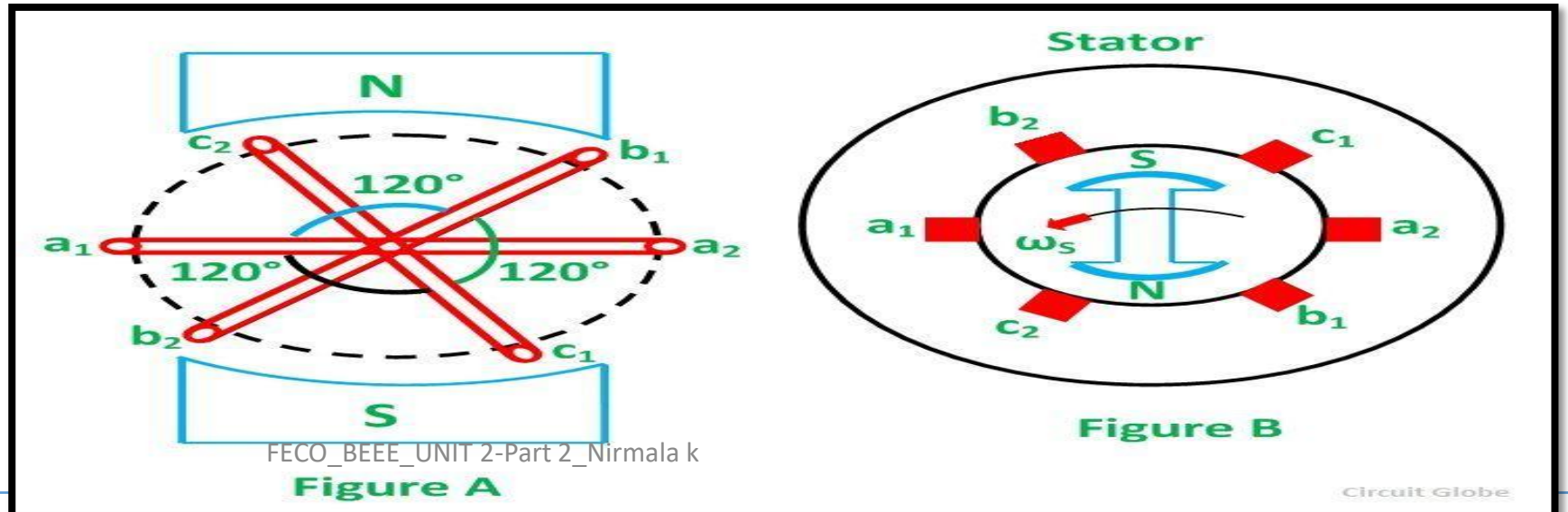
Active Power	Reactive Power	Apparent Power
Active or genuine or real power refers to the quantity of power that is dissipated or that does productive work in the circuit.	Reactive power is the power that comes back and forth between the source and the load.	Apparent power refers to the complicated combination of true power and reactive power.
Indicated by the letter P	Indicated by the letter Q	Indicated by the letter S
Formula is $P = VI \cos \varphi$	Formula is $Q = VI \sin \varphi$	Formula is $S=I^2Z$
Calculated in watts	Calculated in volt-ampere reactive	Calculated in volt-ampere

Generation of 3-phase AC Voltage

Three-phase power is a three-wire ac power circuit with each phase ac signal 120 electrical degrees apart. Residential homes are usually served by a single-phase power supply, while commercial and industrial facilities usually use a three-phase supply for heavy loads (industrial motors, electric grids, aircraft, data centres, shipboards, and system which have to load more than 1000 watts).

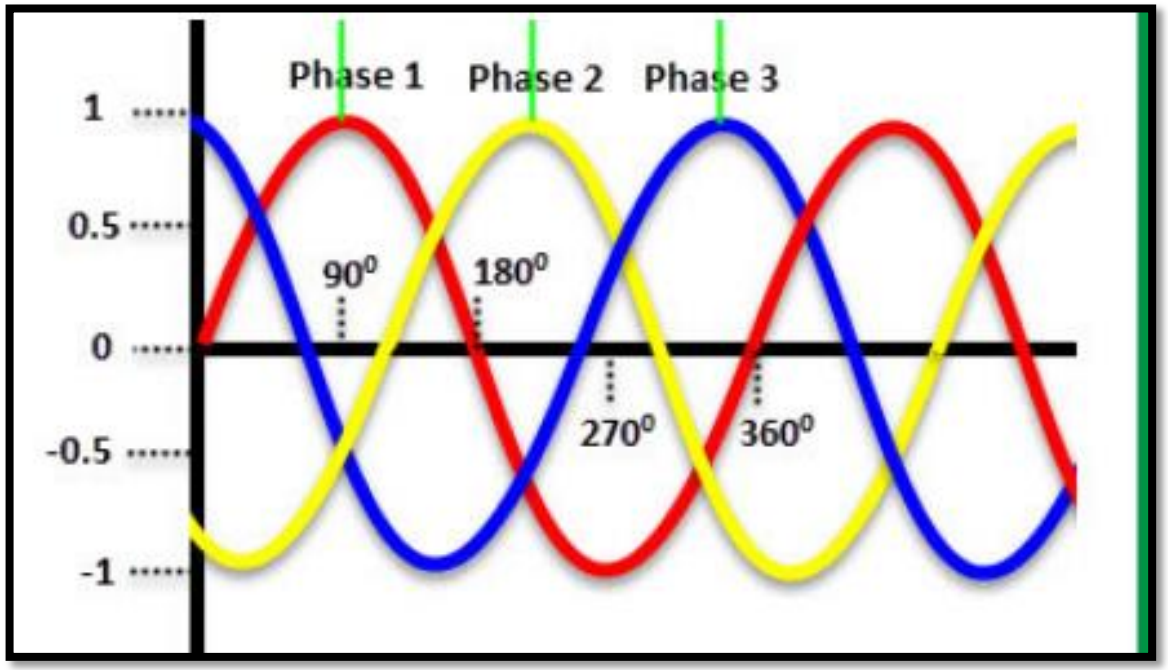
In a 3 phase system, there are three equal voltages or EMFs of the same frequency having a phase difference of 120 degrees. These voltages can be produced by a three-phase AC generator having three identical windings displaced apart from each other by 120 degrees electrical.

When these windings are kept stationary, and the magnetic field is rotated as shown in the figure A below or when the windings are kept stationary, and the magnetic field is rotated as shown below in figure B, an emf is induced in each winding. The magnitude and frequency of these EMFs are the same but are displaced apart from one another by an angle of 120 degrees.

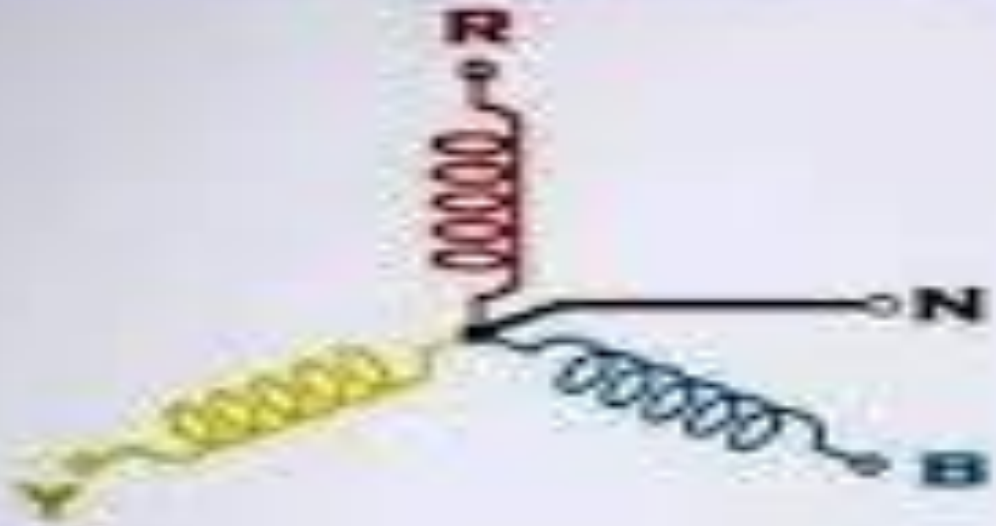


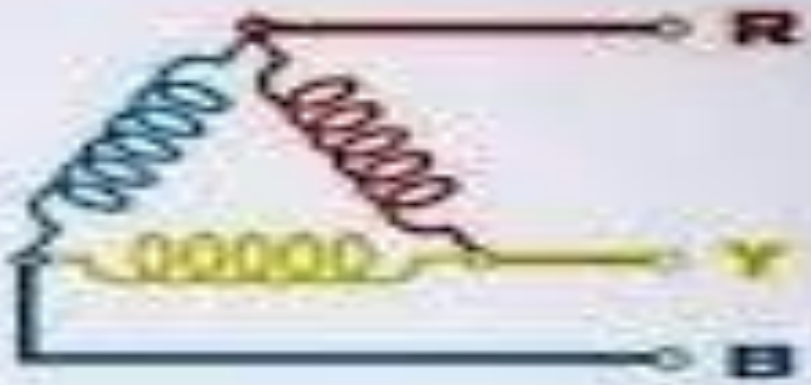
Circuit Globe

- Consider three identical coils a1a2, b1b2 and c1c2 as shown in the above figure. In this figure a1, b1 and c1 are the starting terminals, whereas a2, b2 and c2 are the finish terminals of the three coils. The phase difference of 120 degrees has to be maintained between the start terminals a1, b1 and c1.
- Now, let the three coils mounted on the same axis, and they are rotated by either keeping coil stationary and moving the magnetic field or vice versa in an anticlockwise direction at (ω) radians per seconds. Three EMFs are induced in the three coils respectively



3-phase star delta connection





Star V/S Delta

Line current = Phase current

$I_L = I_P$

Line voltage = √3 * Phase voltage

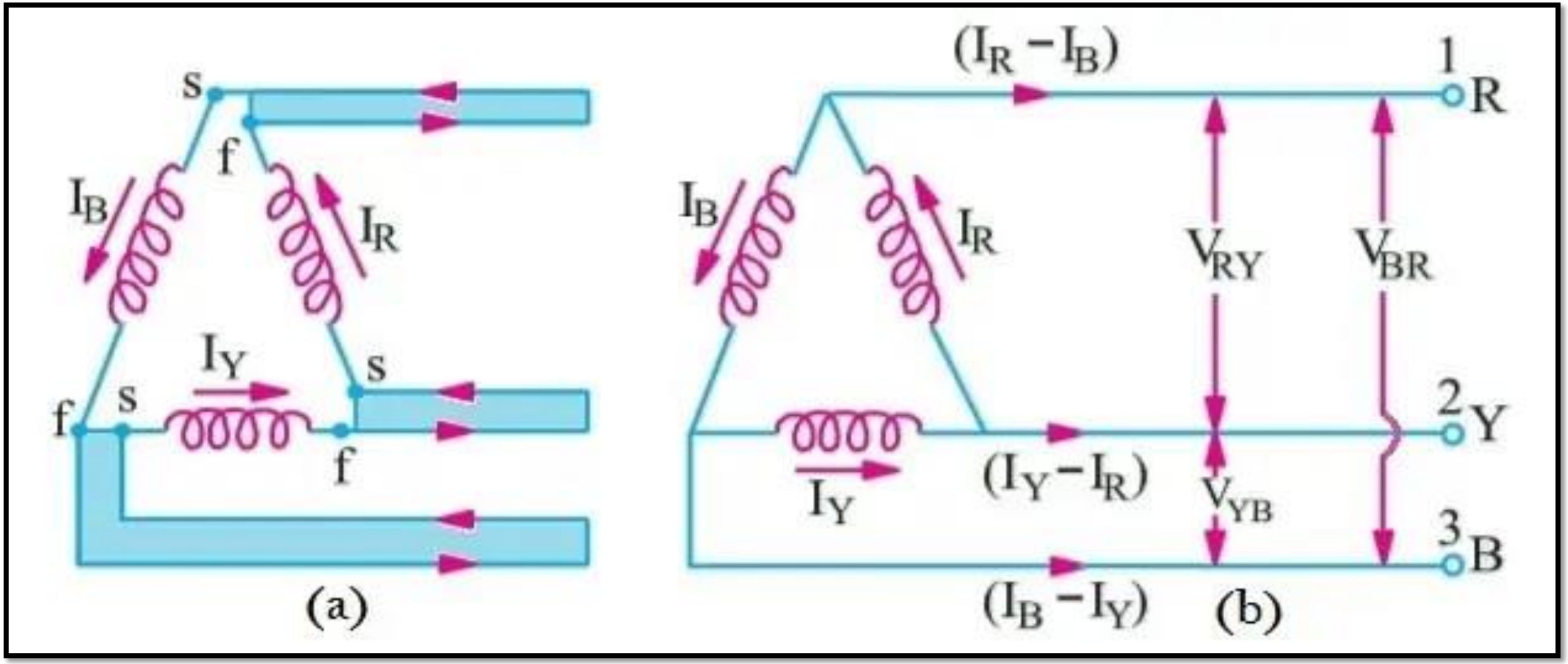
$V_L = \sqrt{3} * V_P$

Line voltage = Phase voltage

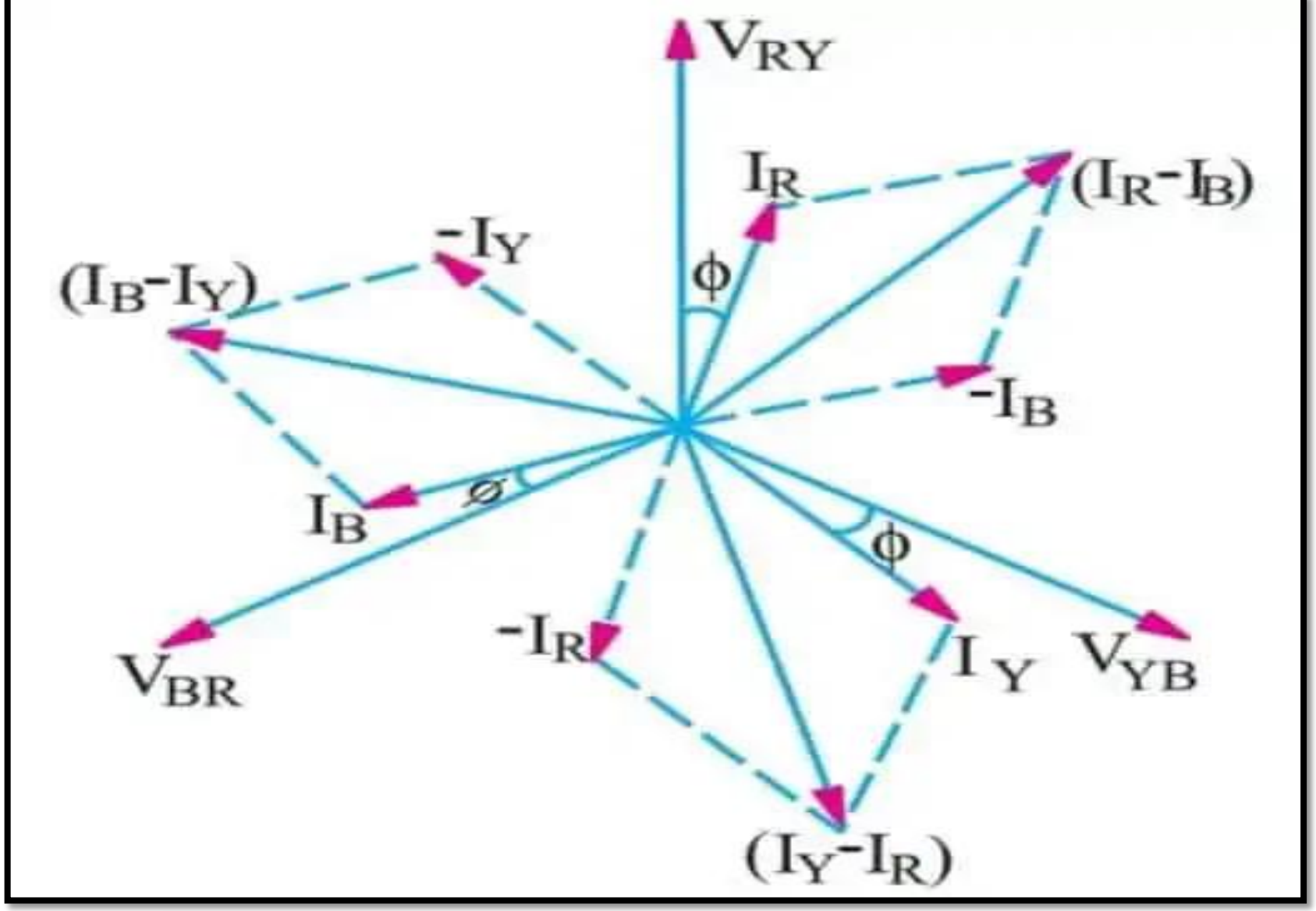
$V_L = V_P$

Line current = √3 * Phase current

$I_L = \sqrt{3} * I_P$



Delta Connection (Δ): 3 Phase Power, Voltage & Current Values



Line & Phase Current and Line & Phase Voltage in Delta (Δ) Connection

Compare Star and Delta connection (self study - 4 to 6 points)

Formulae

Most 3-phase generators produce 120/208V or 277/480V, giving them the power to run large-scale operations and equipment.

The three-phase power is mainly used for generation, transmission and distribution of electrical p

power, it is more economical as compared to single-phase power.

In 3 phase circuits ,the power is defined as the sum of various powers in a three-phase system. i.e.

generation-of-3-phase-power-in-3-phase-circuit-

$$P = 3V_{ph} I_{ph} Cos\varphi$$

Power in star connections in a 3 phase circuits is given as-

$$P = 3 \frac{V_L}{\sqrt{3}} I_L Cos\varphi \quad \ldots\ldots\ldots (1)$$

The phase voltage and line voltage in the star connection is represented as shown below:

generation-of-3-phase-power-

$$V_{ph} = \frac{V_L}{\sqrt{3}} \text{ and } I_{ph} = I_L$$

Therefore, equation (1) can be written as:

$$P = \sqrt{3}\, V_L I_L Cos\varphi \quad \ldots\ldots\ldots (2)$$

Power in delta connections in 3 phase circuits is given by the equation shown below:

$$P = 3\,V_L \frac{I_L}{\sqrt{3}} \cos\varphi \quad \ldots\ldots\ldots (3)$$

In delta connections, the relation between phase and line voltage and phase and line current is given as:-

$$V_{ph} = V_L \quad \text{and} \quad I_{ph} = \frac{I_L}{\sqrt{3}}$$

Thus, the total power in a 3 phase balanced load system, irrespective of their connections, whether the system is star connected or delta connected, the power is given by the relation:
$\sqrt{3}$ *VL*IL*Cosϕ, (active power)
Its units are kilowatt (kW) or Watt (W).

Apparent Power is given as:

$$P_a = \sqrt{3}\,V_L I_L$$

The unit of apparent power is kilovolt-ampere (kVA) or volt-ampere (VA).

Reactive Power is given by the equation:

$$P_r = \sqrt{3}\,V_L I_L \sin\varphi$$

Its units are kilovolt-ampere reactive (kVAR) or volt-ampere reactive (VAR).

EXERCISES 2.3

1. For the parallel circuit shown in Fig. 2.171, find the branch currents and the total current. [*Ans.* (10∠53.13) A; (5∠0) A; (13.60∠36.03) A]

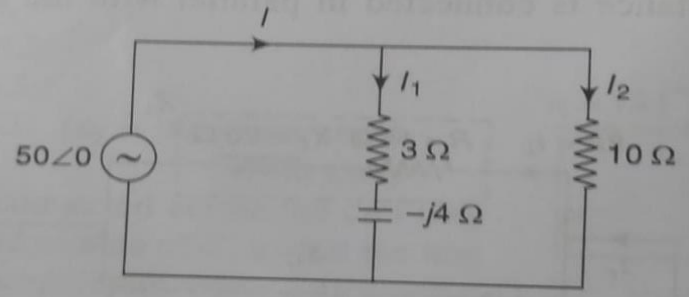


Fig. 2.171

2. Two impedances $\overline{Z}_1 = (12 + j15)\ \Omega$ and $\overline{Z}_2 = (8 - j4)\ \Omega$ are connected in parallel. If potential difference across the impedances is $(230 + j0)$ V, calculate: (i) total current and current supplied to each branch, (ii) total power and power consumed by each branch, and (iii) overall pf and pf of each branch.

[*Ans.* (i) (30.59∠4.04) A, (11.97∠−51.34) A, (25.73∠26.57) A, (ii) 7018 W, 1719.37 W, 5296 W, (iii) 0.998 lead, 0.625 lag, 0.894 lead]

3. A circuit consisting of branches *A* and *B* are connected in parallel across 230 V, 50 Hz supply.
Branch *A* – A resistance of 7 Ω is in series with 0.0125 H inductor.
Branch *B* – A resistance of 8 Ω is in series with 1000 μF capacitor.
Find the current in each branch and total current. Draw the phasor diagram.

[*Ans.* (28.64∠−29.31) A; (26.71∠21.68) A; (50∠−4.77) A]

4. Find the reading of the ammeter, when the voltmeter across 3 Ω resistor in the circuit shown in Fig. 2.172 reads 45 V.

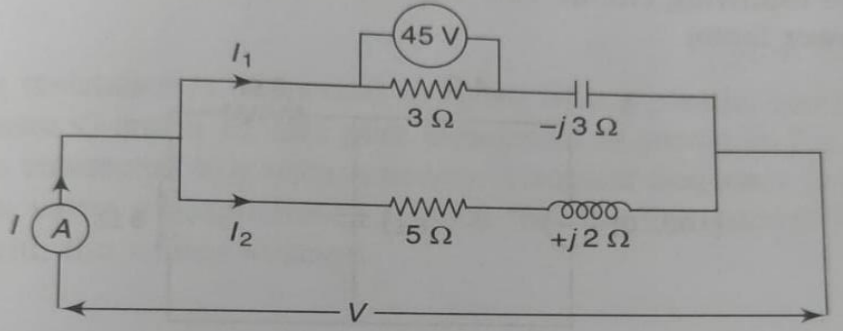


Fig. 2.172

[*Ans.* 22.43 A]

5. A single-phase circuit consists of three parallel branches, the currents in the respective branches are represented by

$$i_1 = 20 \sin 314t\ \text{A},\ i_2 = 30 \sin\left(314t - \frac{\pi}{4}\right)\ \text{A},\ i_3 = 18 \sin\left(314t + \frac{\pi}{4}\right)\ \text{A}$$

Express the total current in the similar form, if the supply voltage is 200sin 314*t* V. Find impedance, resistance and reactance of the circuit.

[*Ans.* 54.6 sin (314*t* − 8.94); (3.66 ∠8.94) Ω; 3.62 Ω; 0.569 Ω]

6. Two impedances $\overline{Z}_1 = (10 + j5)\ \Omega$ and $\overline{Z}_2 = (25 - j10)\ \Omega$ are in parallel. If this parallel circuit is connected across 100 V, find (i) total admittance, (ii) line current, (iii) power consumed, (iv) equivalent impedance of the circuit.

[*Ans.* (0.1175∠−12.9) ℧; (11.75∠−12.9) A; 1145 W; (8.51∠12.9) Ω]

7. A series circuit having a resistance of 6 Ω and an inductive reactance of 8 Ω is connected across a 200 V, 50 Hz ac supply in parallel with another series circuit having a resistance of 8 Ω capacitive reactance of 6 Ω.
   (i) Calculate the total admittance, total conductance and total susceptance of the circuit
   (ii) Calculate the total current and total power factor.
   (iii) Calculate the value of pure reactance to be connected in parallel with above combination to make the total pf unity.

[*Ans.* 0.14 ℧; 0.14 ℧; 0.02 ℧; (28∠−8.13) A; 0.989 lagging; $X_C = 50\ \Omega$]

FECO_BEEE_UNIT 2-Part 2_Nirmala K

8. An impedance of $(2 + j6)\ \Omega$ is connected in series with two impedances of $(10 + j4)\ \Omega$ and $(12 - j8)\ \Omega$ which are in parallel. Calculate the supply current and power factor if the circuit is connected to 200 V.

[*Ans.* $(18.74\angle -33)$ A; 0.839 lagging]

9. A coil of 20 Ω resistance has an inductance of 0.2 H is connected in parallel with a condenser of 100 μF capacitance. Calculate the frequency at which this circuit will behave as a non-inductive resistance. Find also the value of dynamic resistance. [*Ans.* 31.83 Hz; 100 Ω]

10. An equipment consumes 2 kW when connected across a 110 V, 100 Hz supply and takes a lagging current of 25 A. If a capacitor is connected in parallel with equipment to make the PF unity, find its capacitance.

[*Ans.* 248.29 μF]

11. For the following circuit, find (i) the current delivered by the source and (ii) power factor

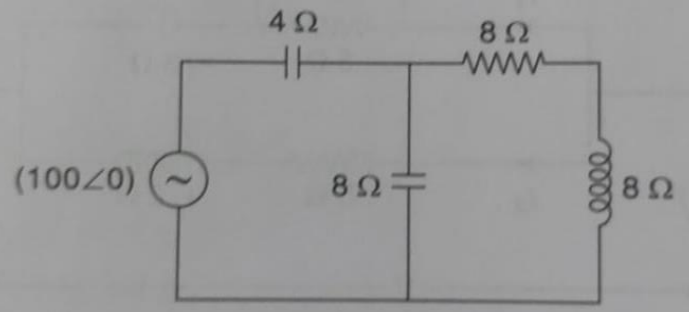


Fig. 2.173

[*Ans.* $(6.93\angle 56.31)$ A; 0.555 leading]

12. Calculate the current, power, and power factor for each impedance shown in Fig. 2.174. Also calculate the total current, power, and power factor of the whole combination.

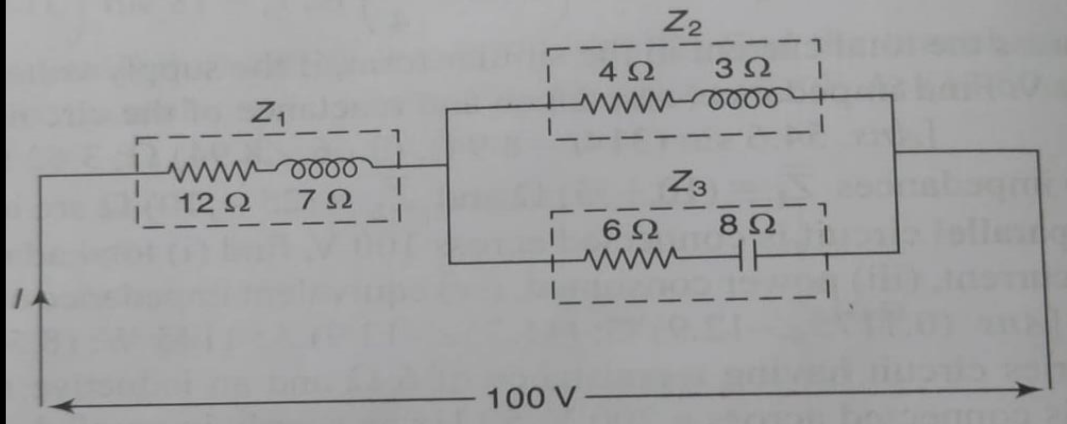


Fig. 2.174

[*Ans.* For Z_1: 5.51 A; 364.3 W; 0.864 lag;
For Z_2: 4.93 A; 97.21 W; 0.8 lag;
For Z_3: 2.46 A; 36.31 W; 0.6 lead;
For circuit: 5.51 A; 497.9 W; 0.903 lag

13. The circuit shown in Fig. 2.175 operates at a frequency of 50 Hz. Determine the value of C such that the input voltage V and the input current I are in the same phase.

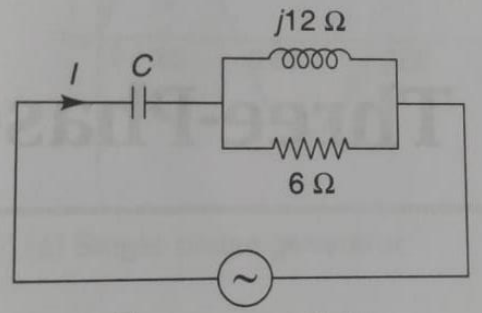


Fig. 2.175

[*Ans.* 1326.29 μF]

Part 2_Nirmala k

14. A 20 Ω resistance is connected in series with a parallel combination of a capacitance C and a 15 mH pure inductance as shown in Fig. 2.176. This circuit is connected to a voltage source of angular frequency $\omega = 1000$ rad/s. Find the value of capacitance C such that the line current is 45° out of phase with the source voltage.

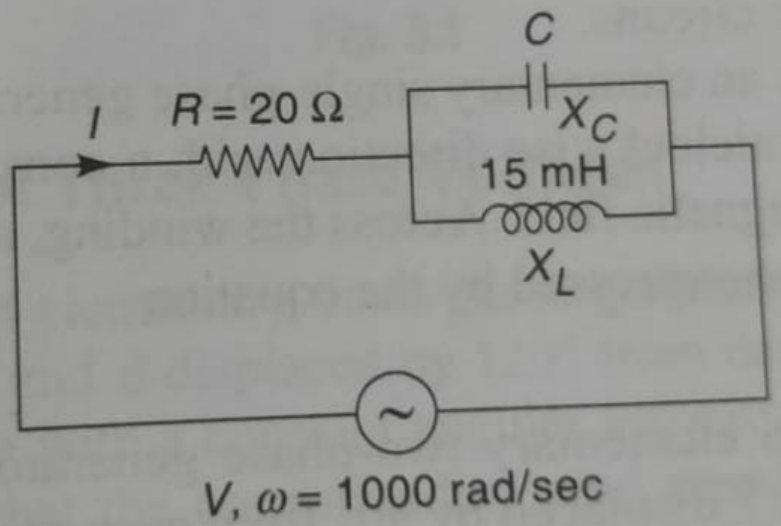


Fig. 2.176

[Ans. 16.67 μF]